

(12) **United States Patent**
Li et al.

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(54) **MICRO-LED STRUCTURE AND MICRO-LED CHIP INCLUDING SAME**

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(73) Assignee: **Jade Bird Display (Shanghai) Limited**, Shanghai (CN)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 713 days.

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Related U.S. Application Data

(60) Provisional application No. 63/131,128, filed on Dec. 28, 2020.

(51) **Int. Cl.**
H10H 20/812 (2025.01)
H01L 25/075 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **H10H 20/812** (2025.01); **H01L 25/0753** (2013.01); **H10H 20/81** (2025.01);
(Continued)

(58) **Field of Classification Search**
None
See application file for complete search history.

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Primary Examiner — Yara B Green

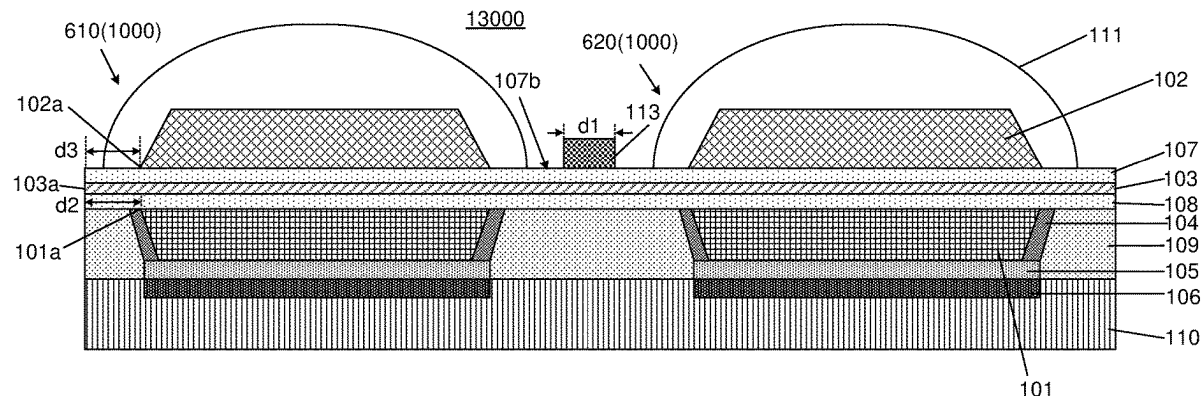
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(74) *Attorney, Agent, or Firm* — Finnegan, Henderson, Farabow, Garrett & Dunner, LLP

(57) **ABSTRACT**

A micro-LED chip includes multiple micro-LEDs. At least one micro-LED includes: a first type conductive layer; a second type conductive layer stacked on the first type conductive layer; and a light emitting layer formed between the first type conductive layer and the second type conductive layer. The light emitting layer extends along a horizontal level from a top edge of the first type conductive layer and from a bottom edge of the second type conductive layer, such that an edge of the light emitting layer does not contact the top edge of the first type conductive layer and the bottom edge of the second type conductive layer, and the bottom edge of the second type conductive layer is aligned with the top edge of the first type conductive layer. The micro-LED chip further includes a metal layer formed on the light emitting layer between adjacent micro-LEDs.

21 Claims, 39 Drawing Sheets



(51) **Int. Cl.**

H10H 20/81 (2025.01)
H10H 20/811 (2025.01)
H10H 20/814 (2025.01)
H10H 20/819 (2025.01)
H10H 20/821 (2025.01)
H10H 20/83 (2025.01)
H10H 20/831 (2025.01)
H10H 20/841 (2025.01)
H10H 20/851 (2025.01)
H10H 20/855 (2025.01)
H10H 20/856 (2025.01)
H10H 20/857 (2025.01)
H10H 29/10 (2025.01)
H10H 29/14 (2025.01)

(52) **U.S. Cl.**

CPC *H10H 20/811* (2025.01); *H10H 20/814*
(2025.01); *H10H 20/819* (2025.01); *H10H*
20/821 (2025.01); *H10H 20/831* (2025.01);
H10H 20/841 (2025.01); *H10H 20/851*
(2025.01); *H10H 20/855* (2025.01); *H10H*
20/856 (2025.01); *H10H 20/857* (2025.01);

H10H 29/10 (2025.01); *H10H 29/14*
(2025.01); *H10H 20/83* (2025.01)

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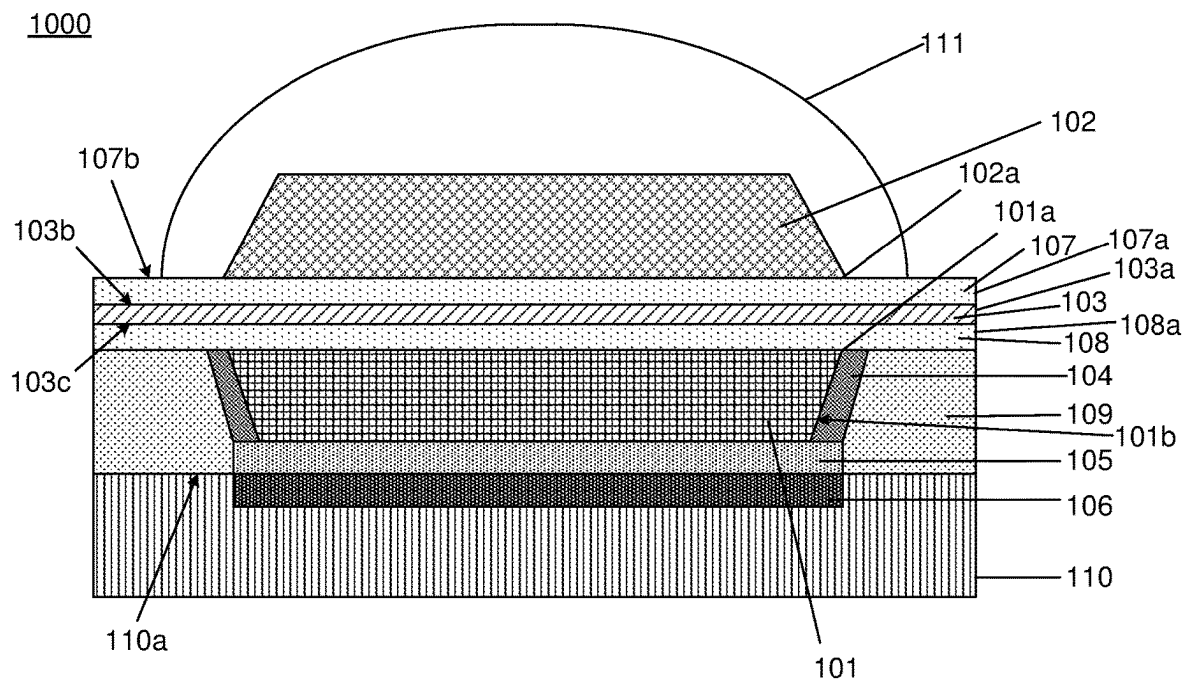


FIG. 1A

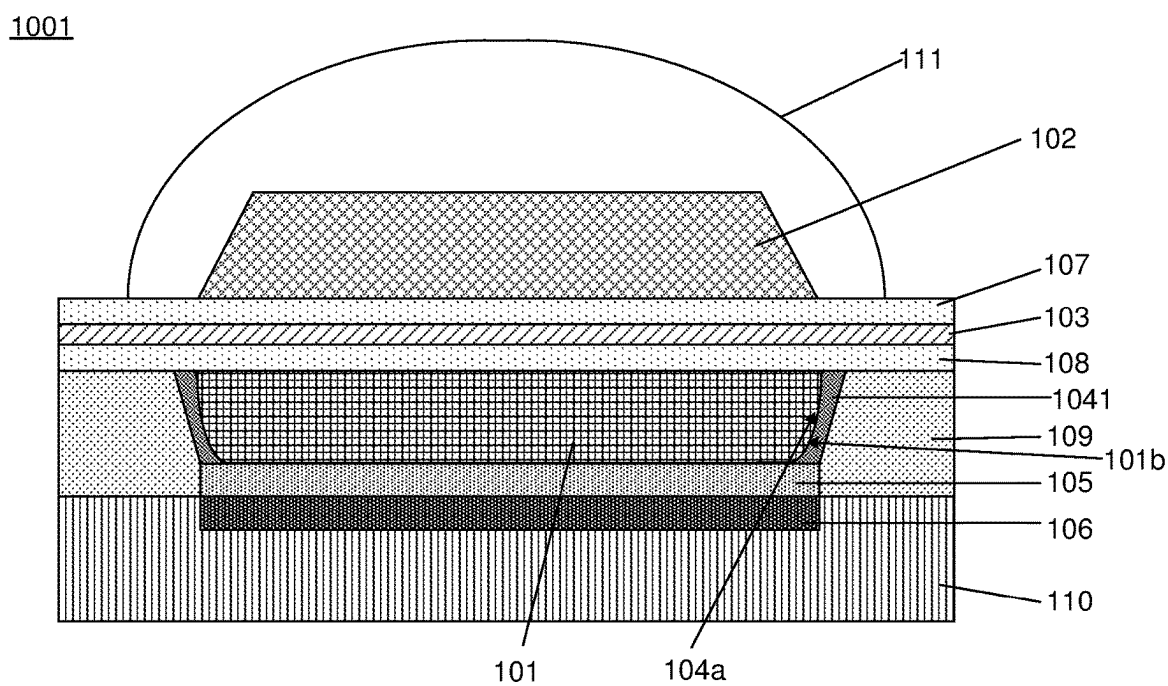


FIG. 1B

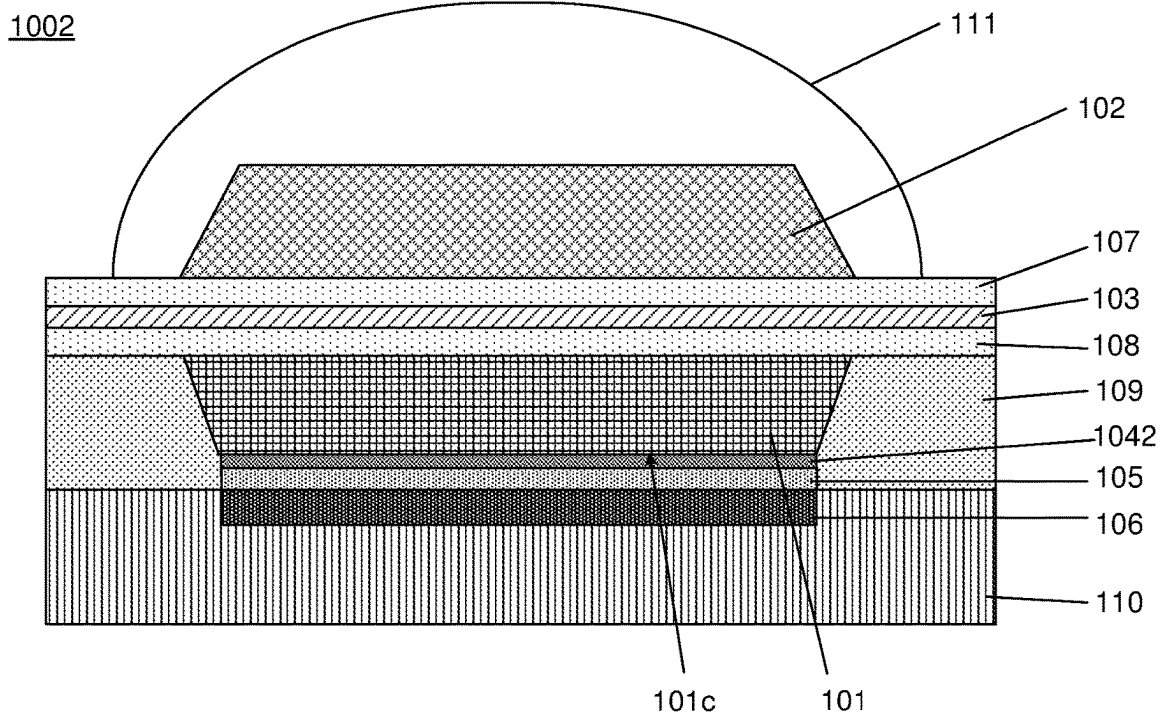


FIG. 1C

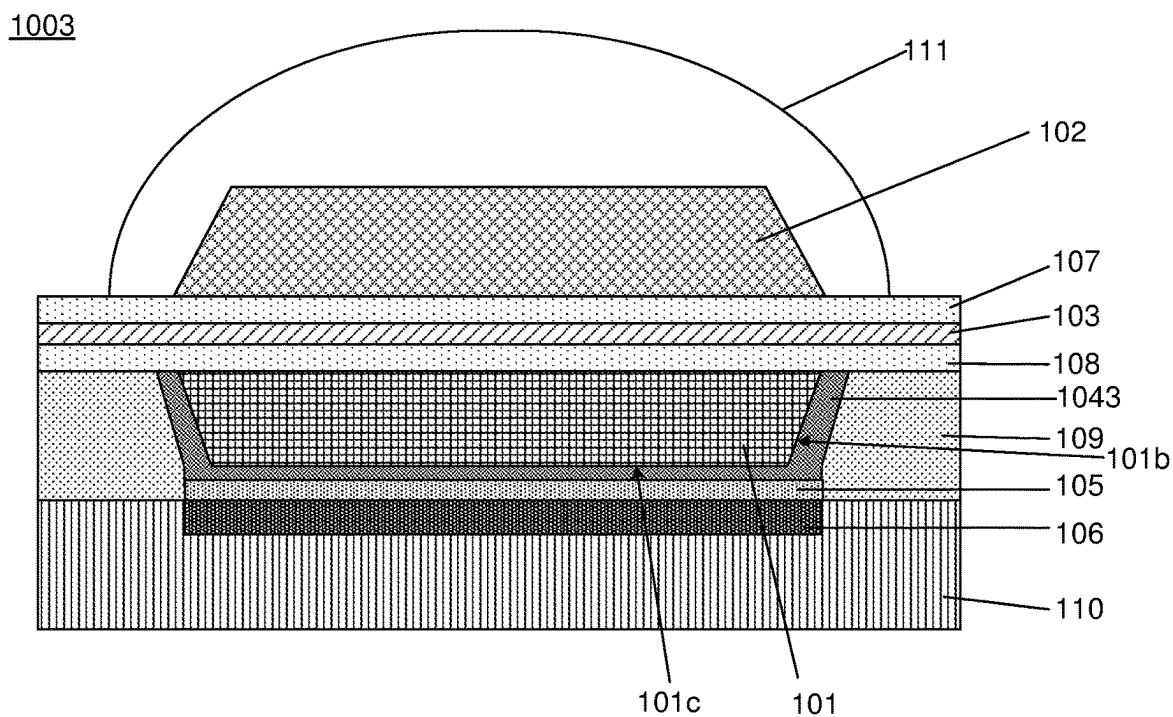


FIG. 1D

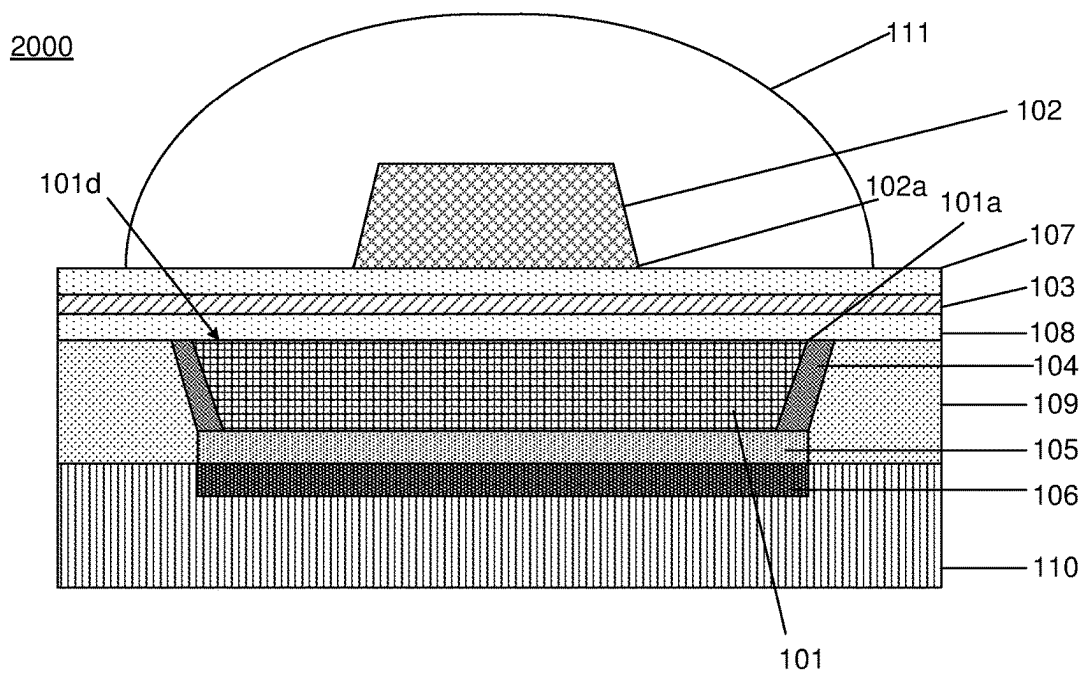


FIG. 2A

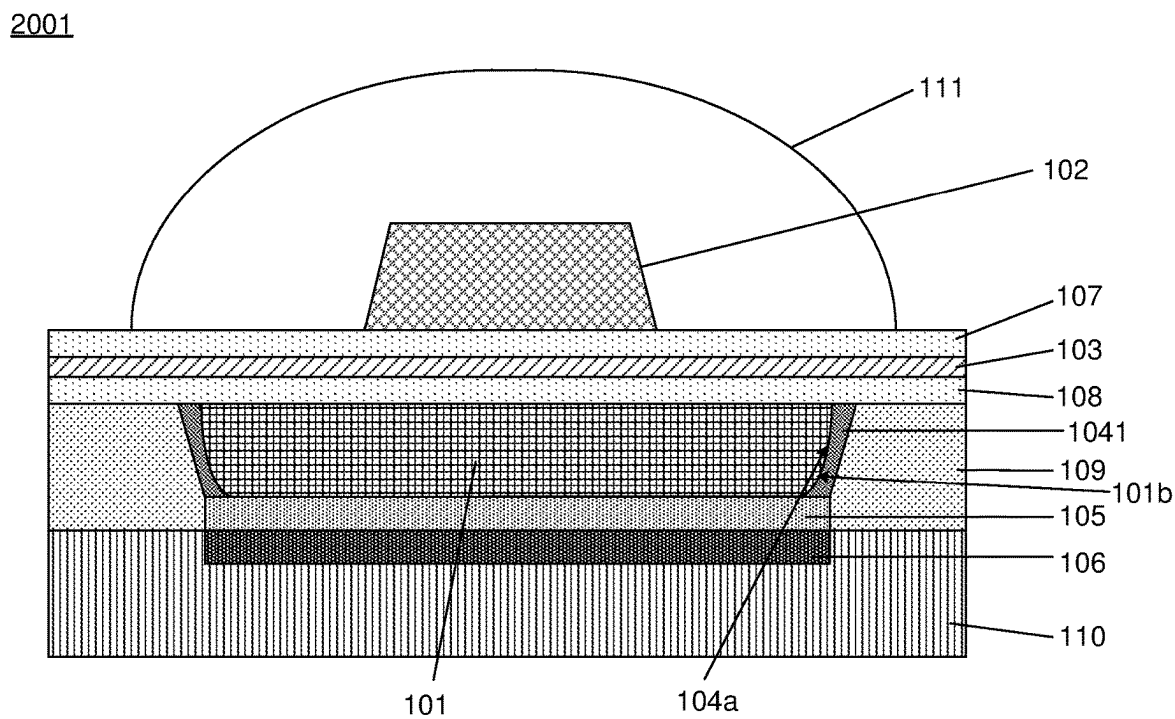


FIG. 2B

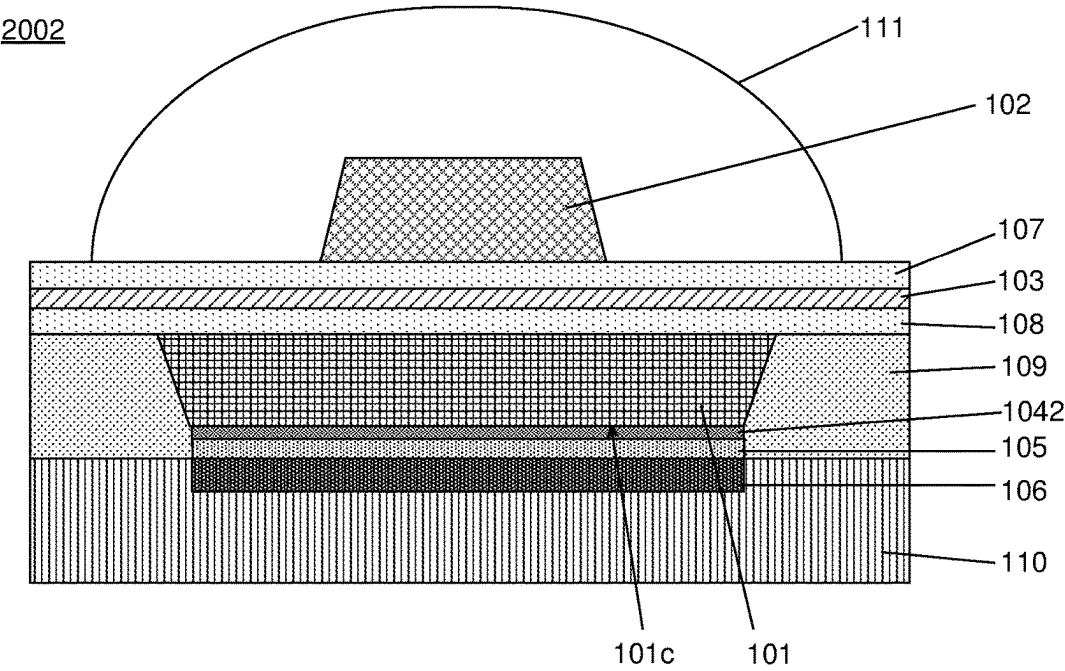


FIG. 2C

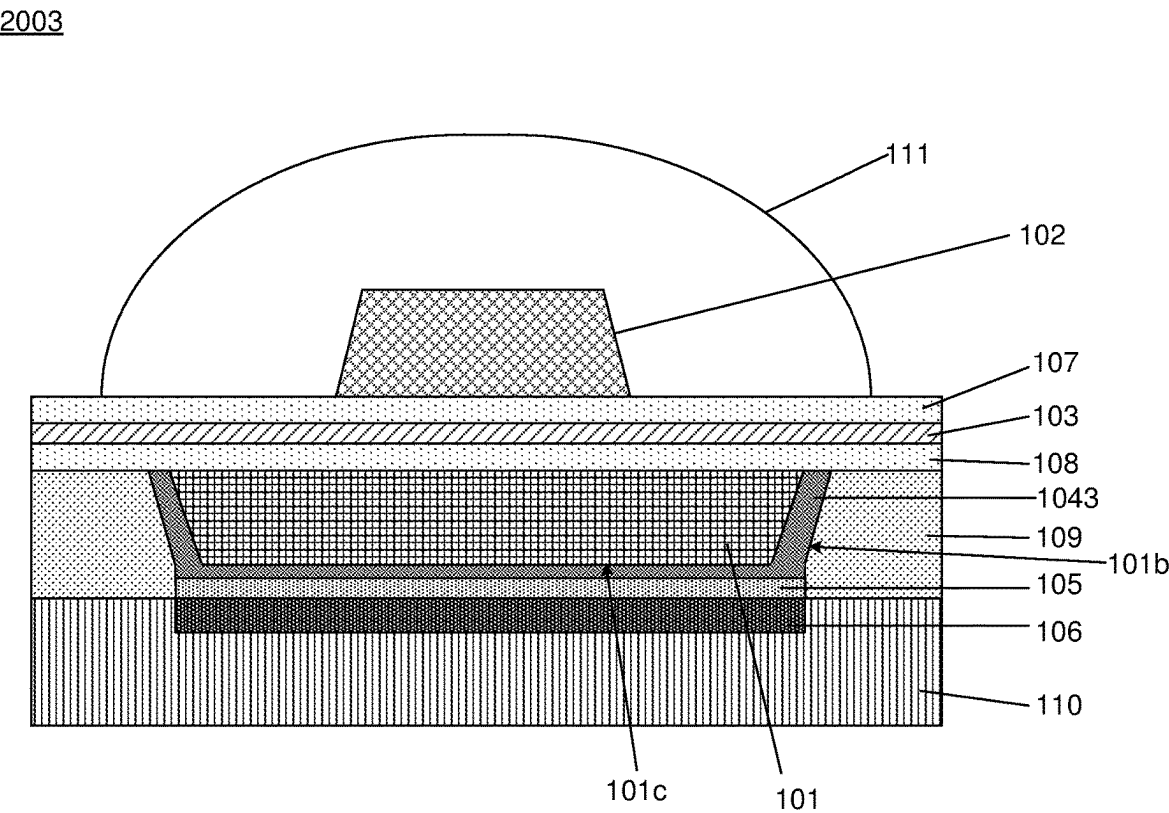


FIG. 2D

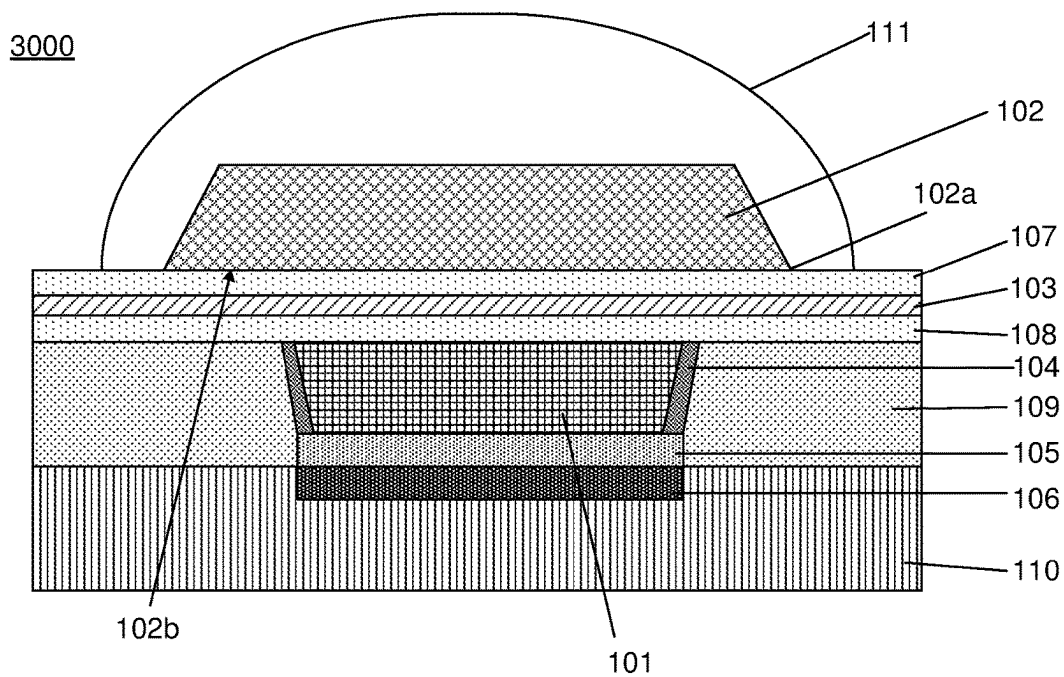


FIG. 3A

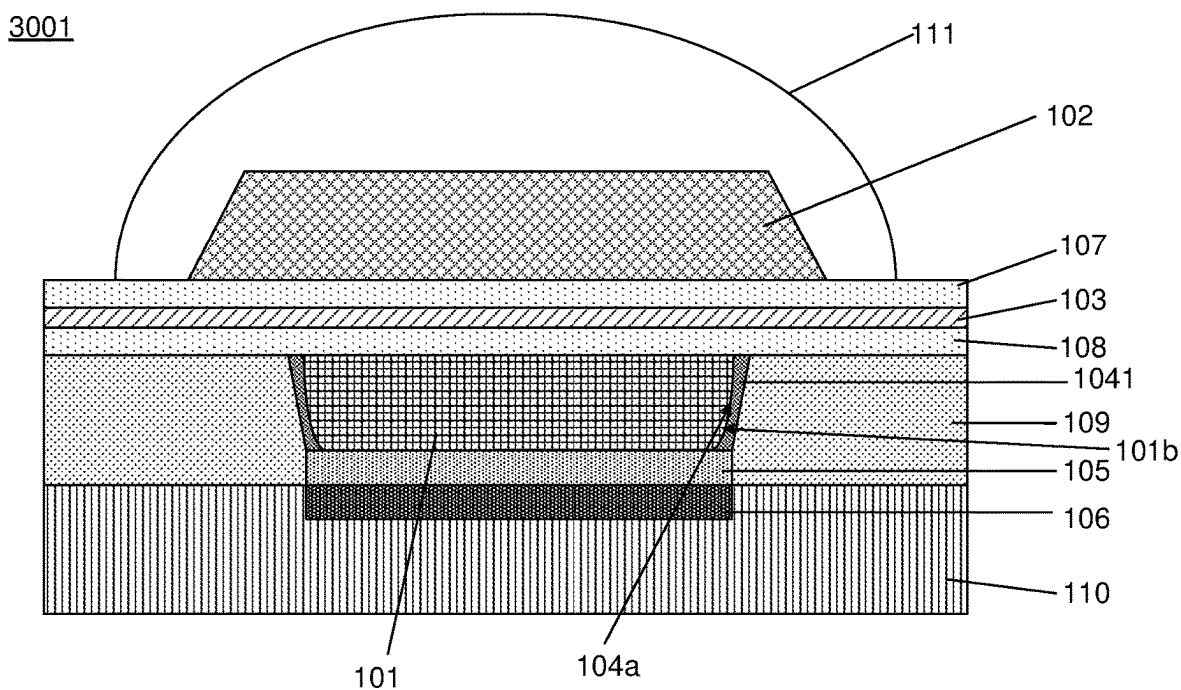


FIG. 3B

3002

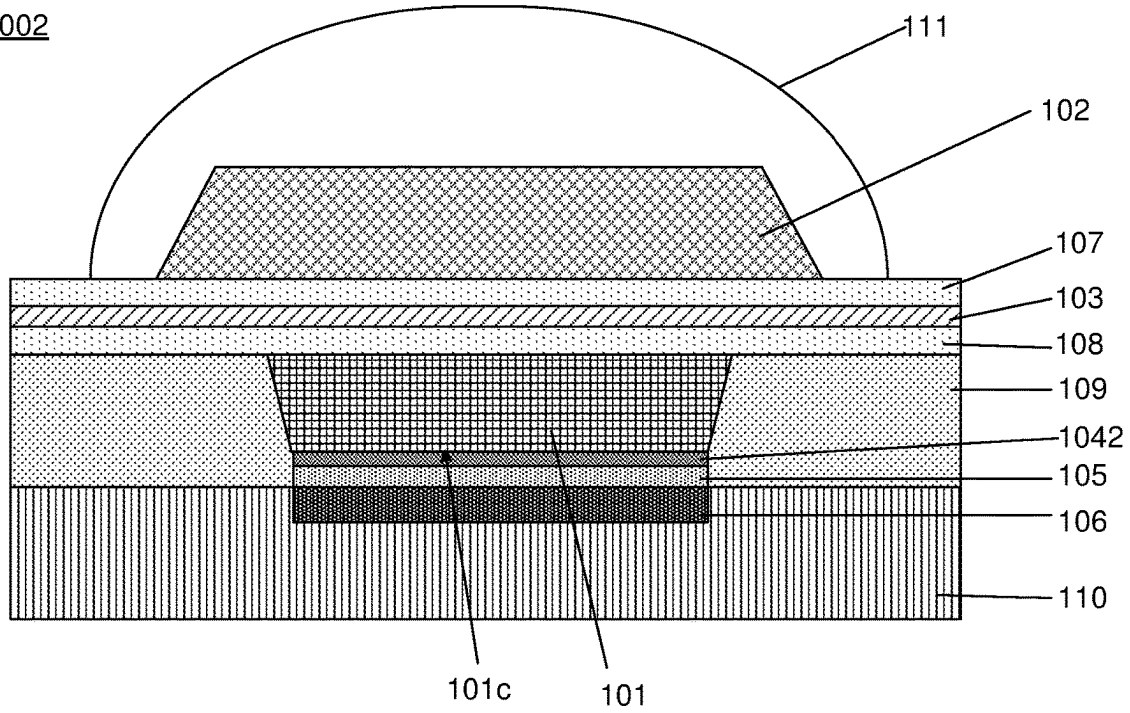


FIG. 3C

3003

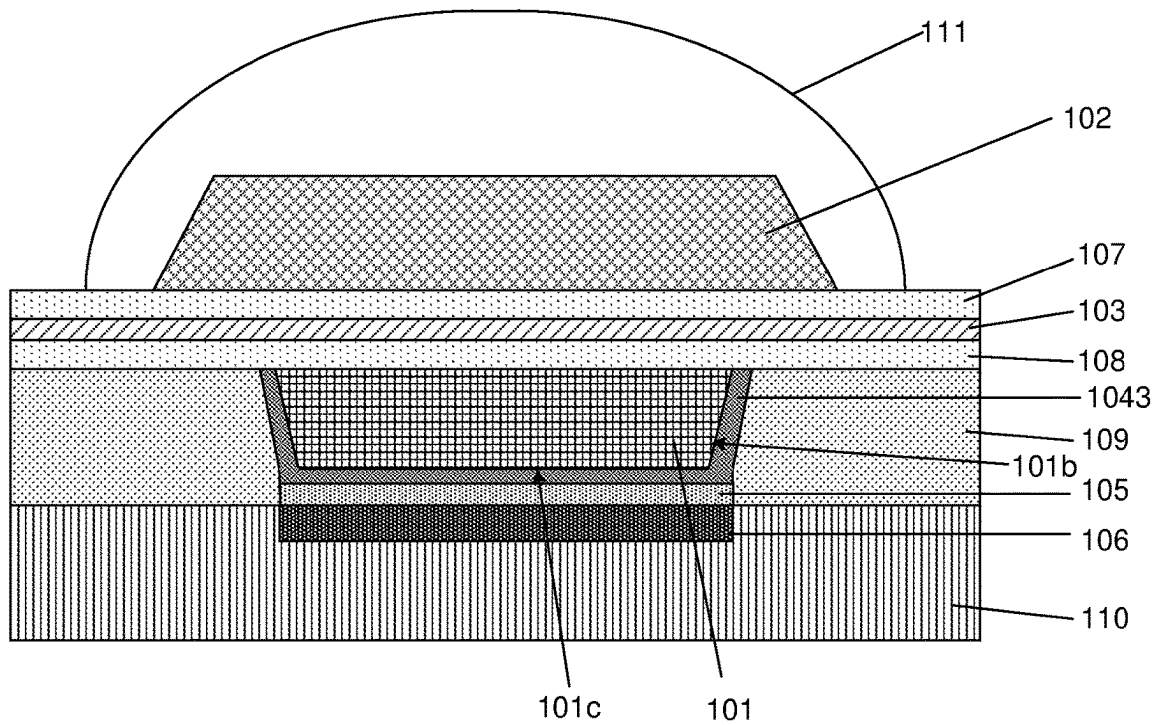


FIG. 3D

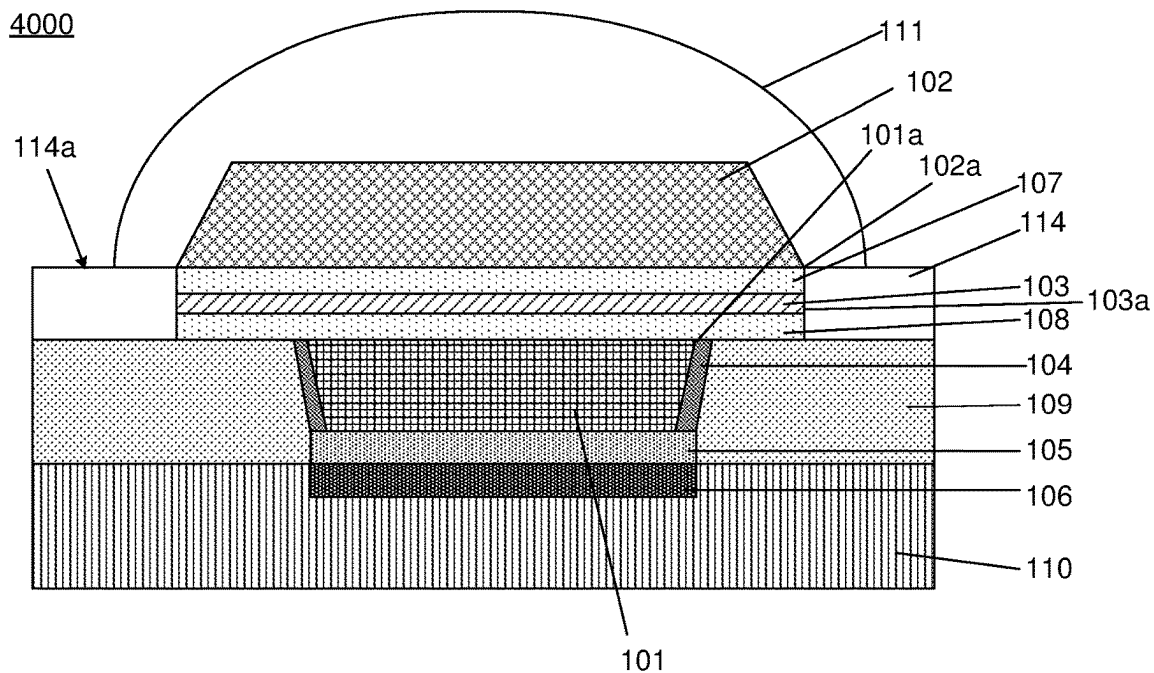


FIG. 4A

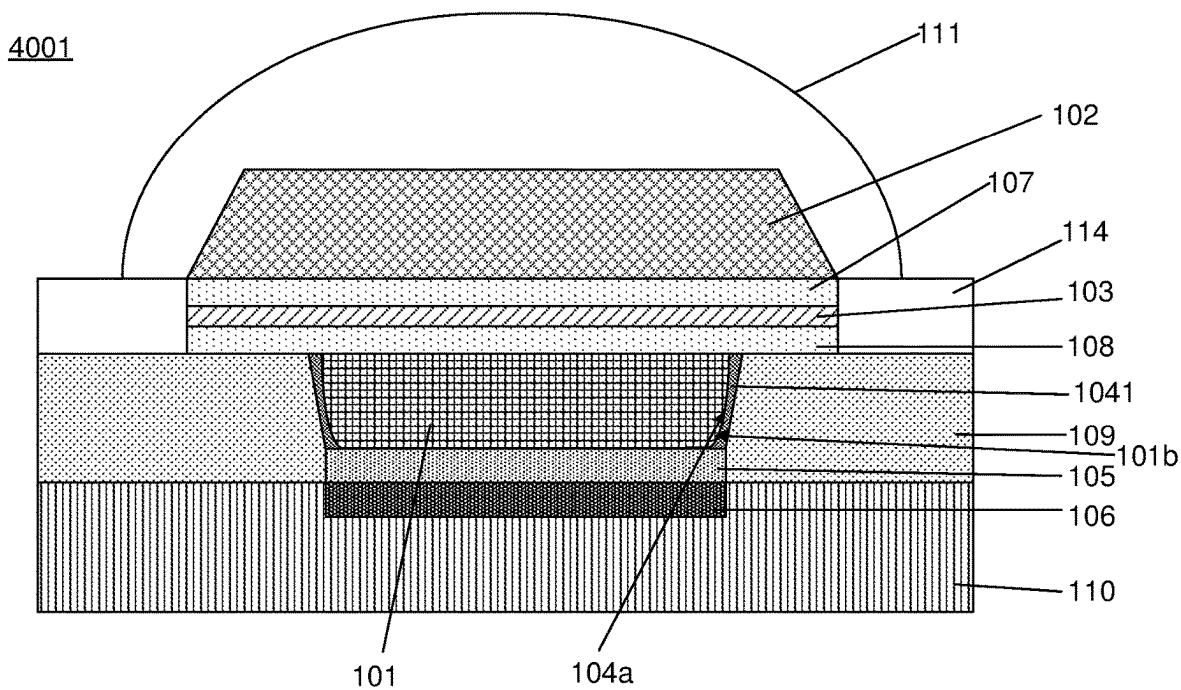


FIG. 4B

4002

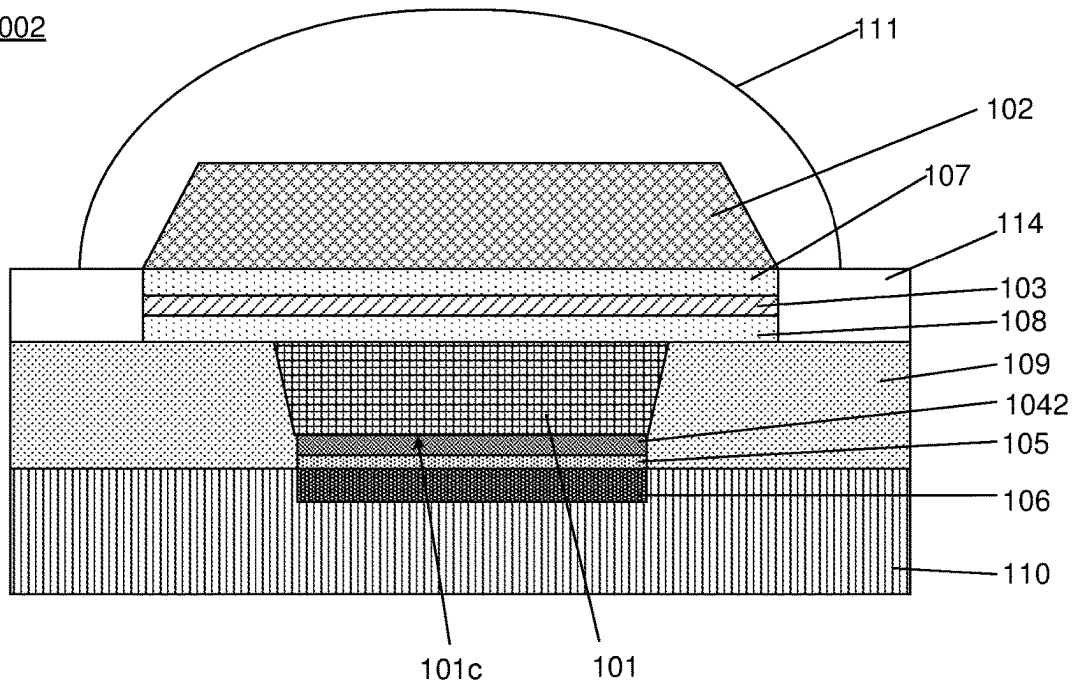


FIG. 4C

4003

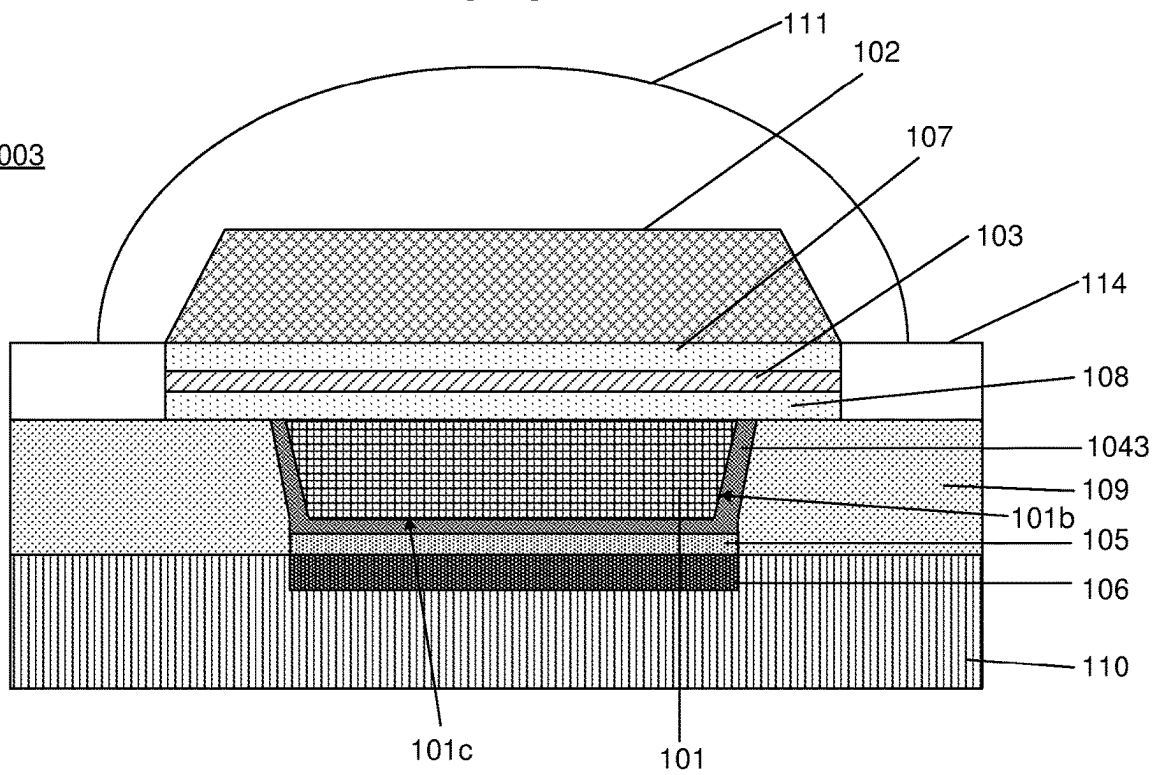


FIG. 4D

5000

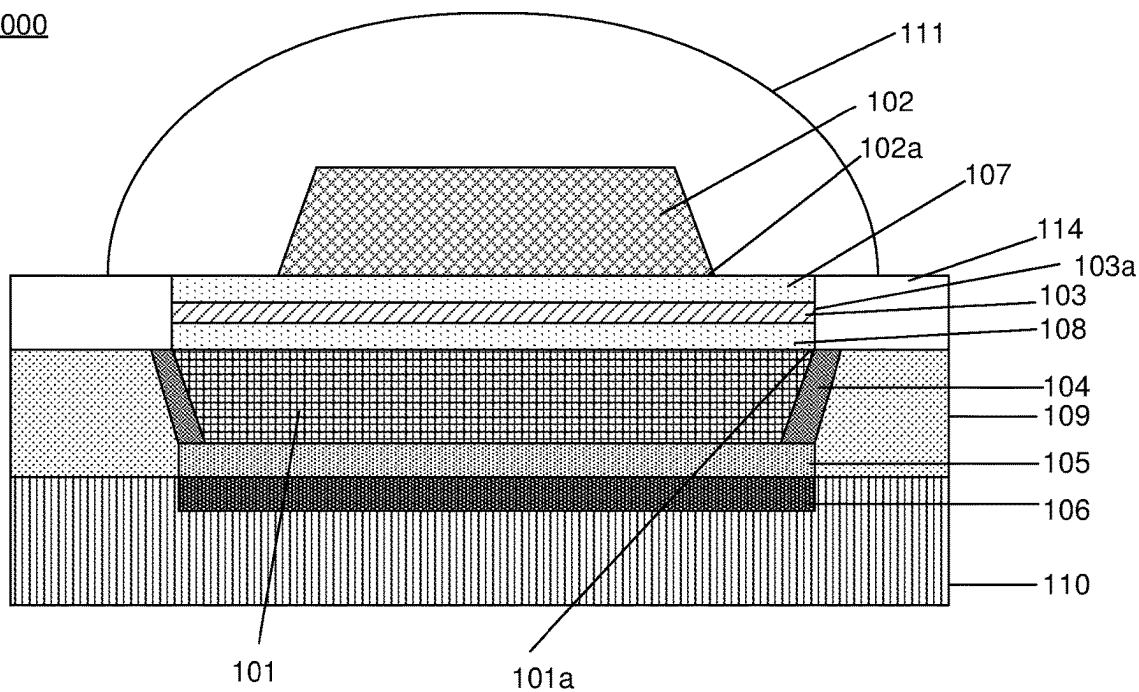


FIG. 5A

5001

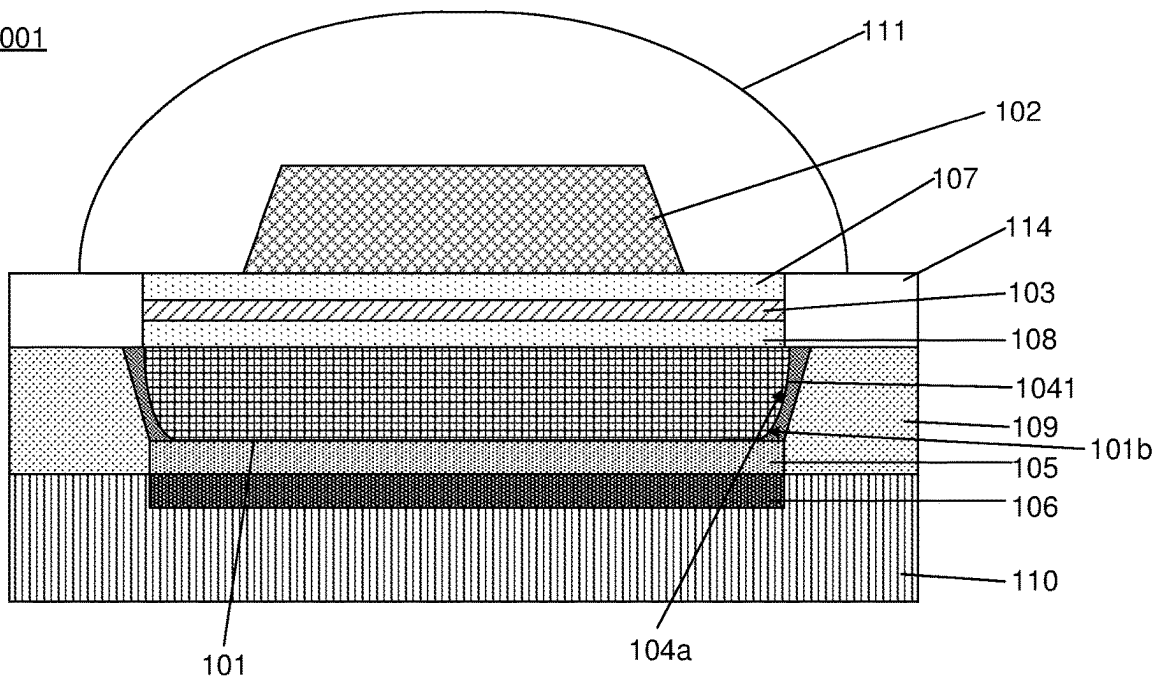


FIG. 5B

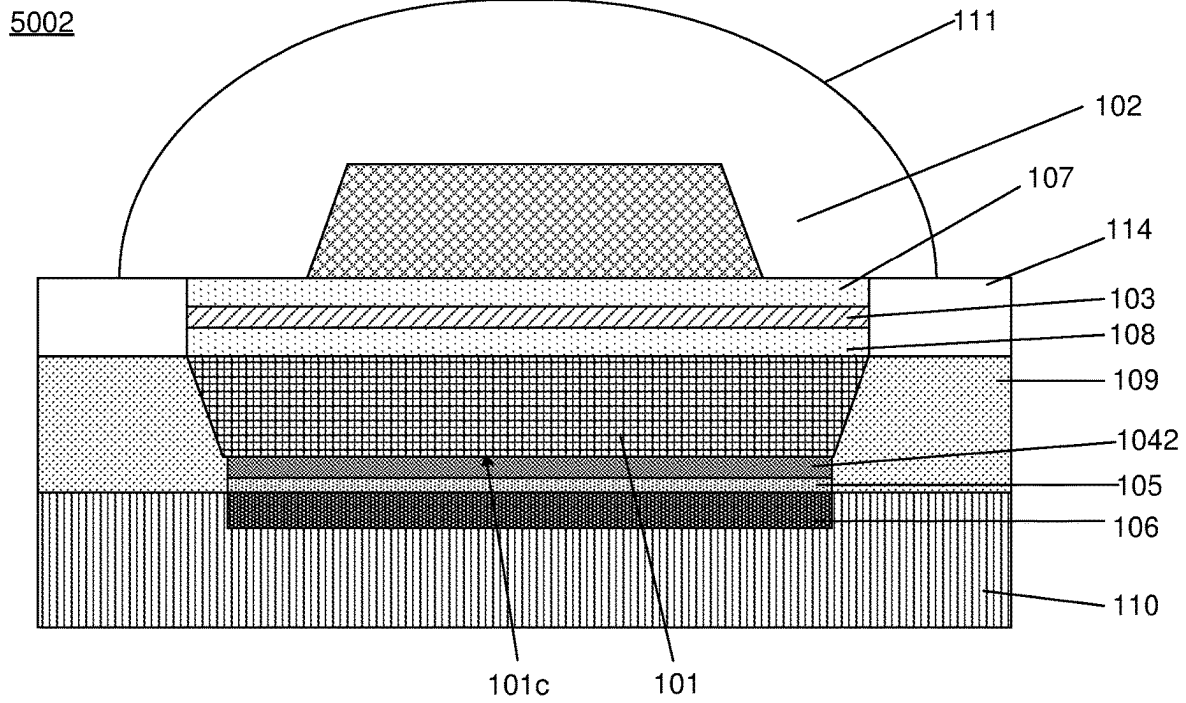


FIG. 5C

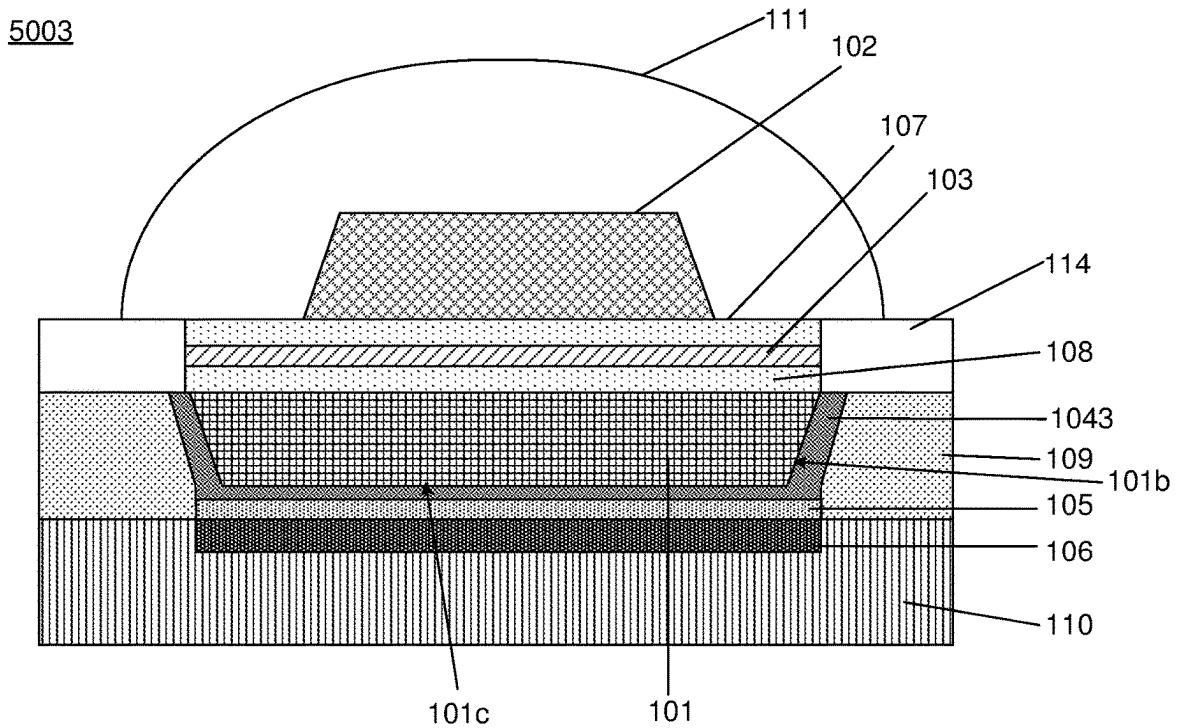


FIG. 5D

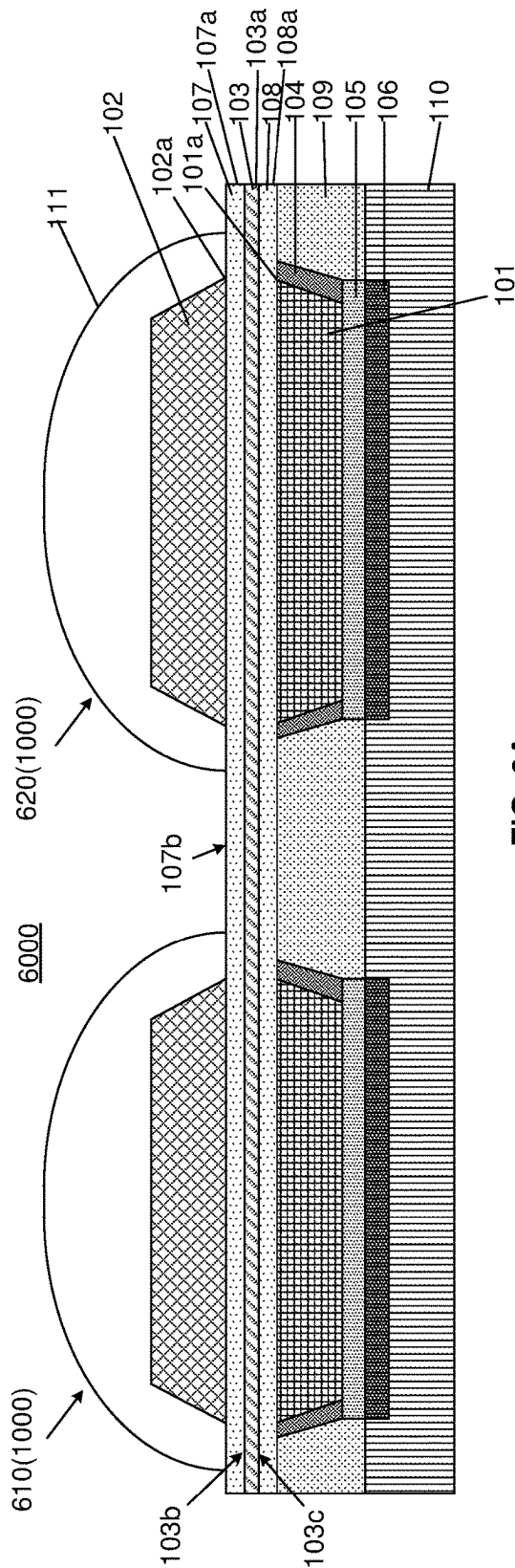


FIG. 6A

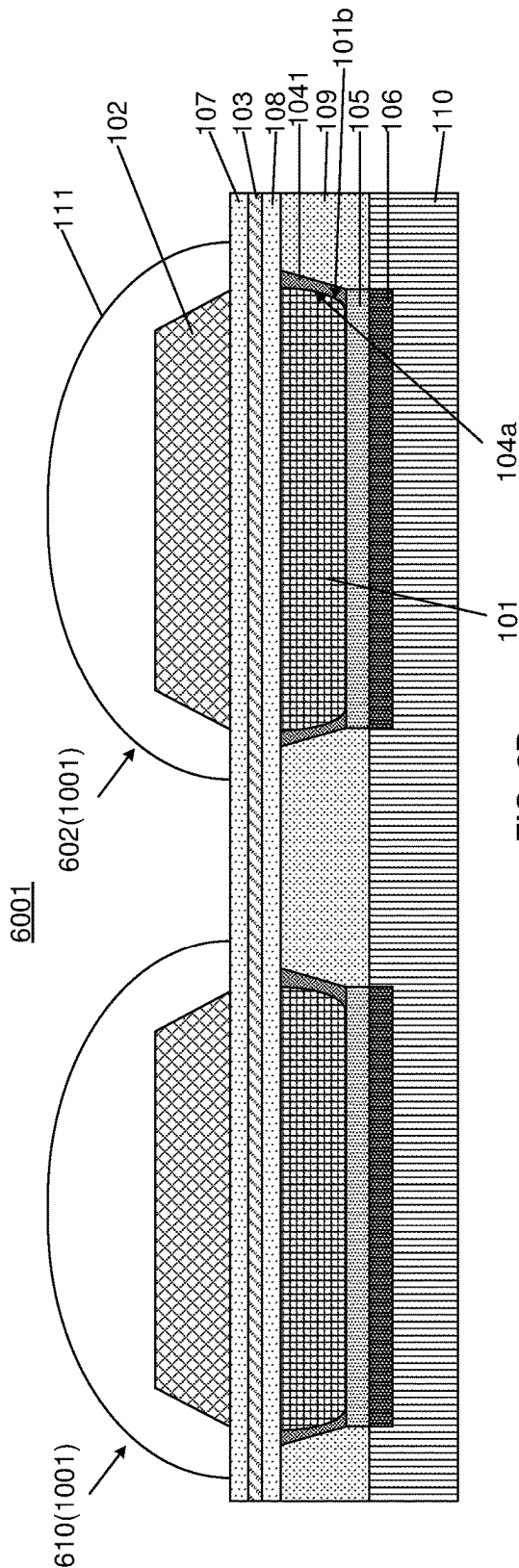


FIG. 6B

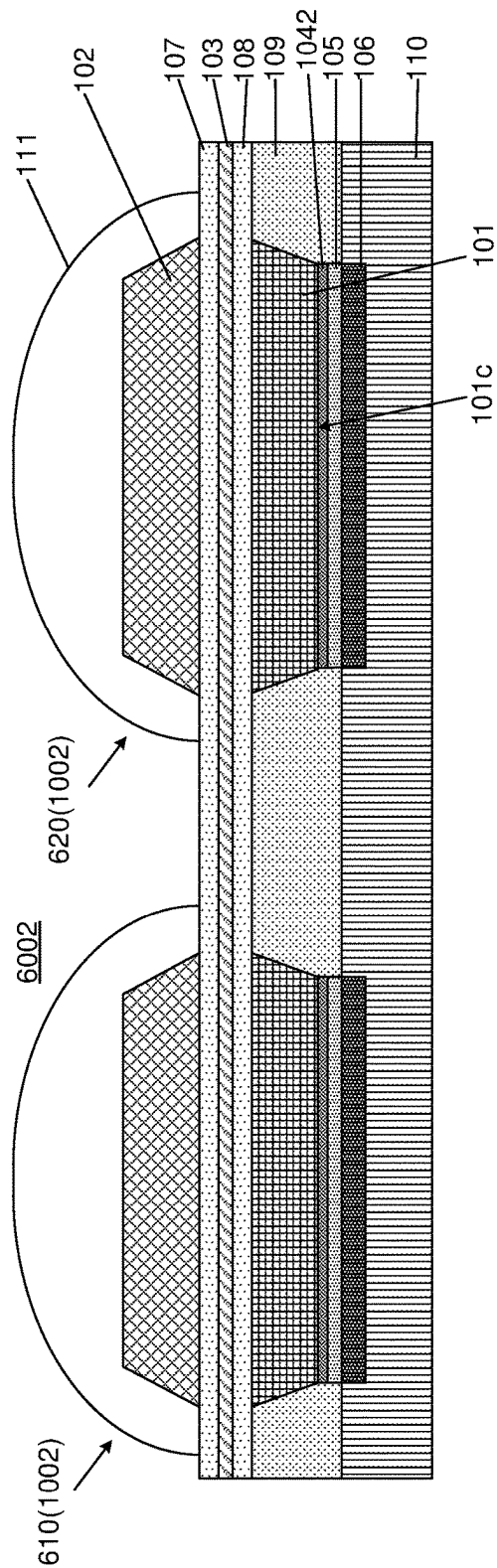


FIG. 6C

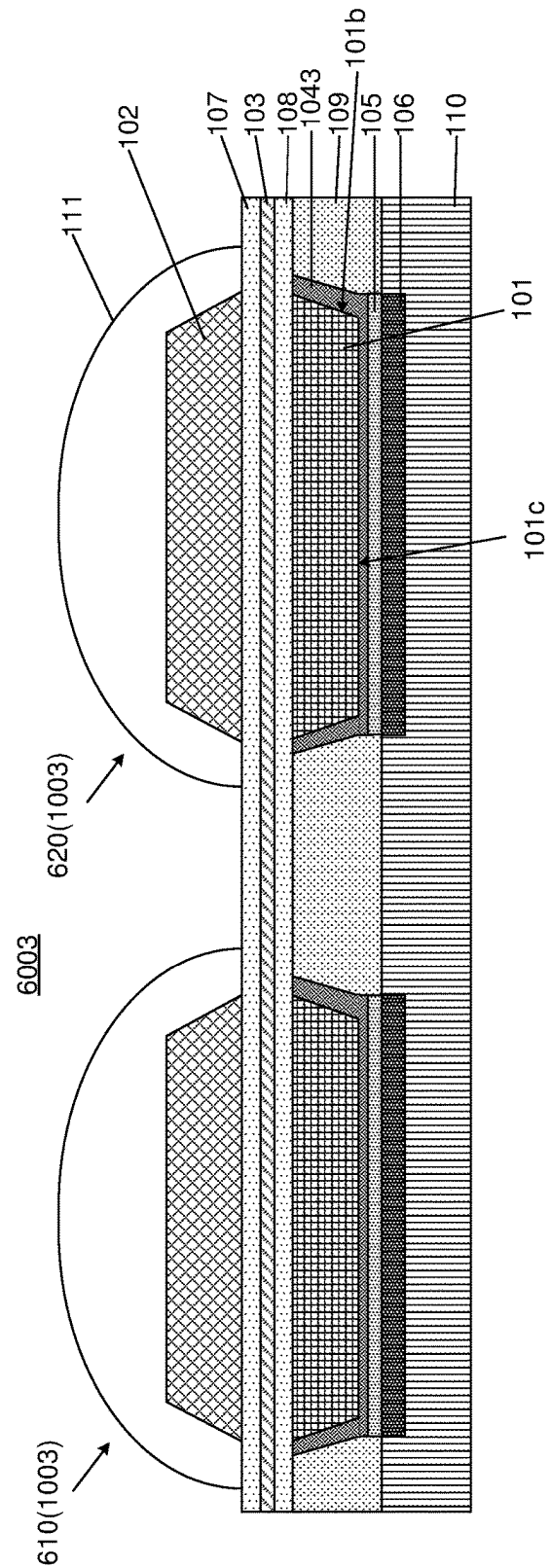


FIG. 6D

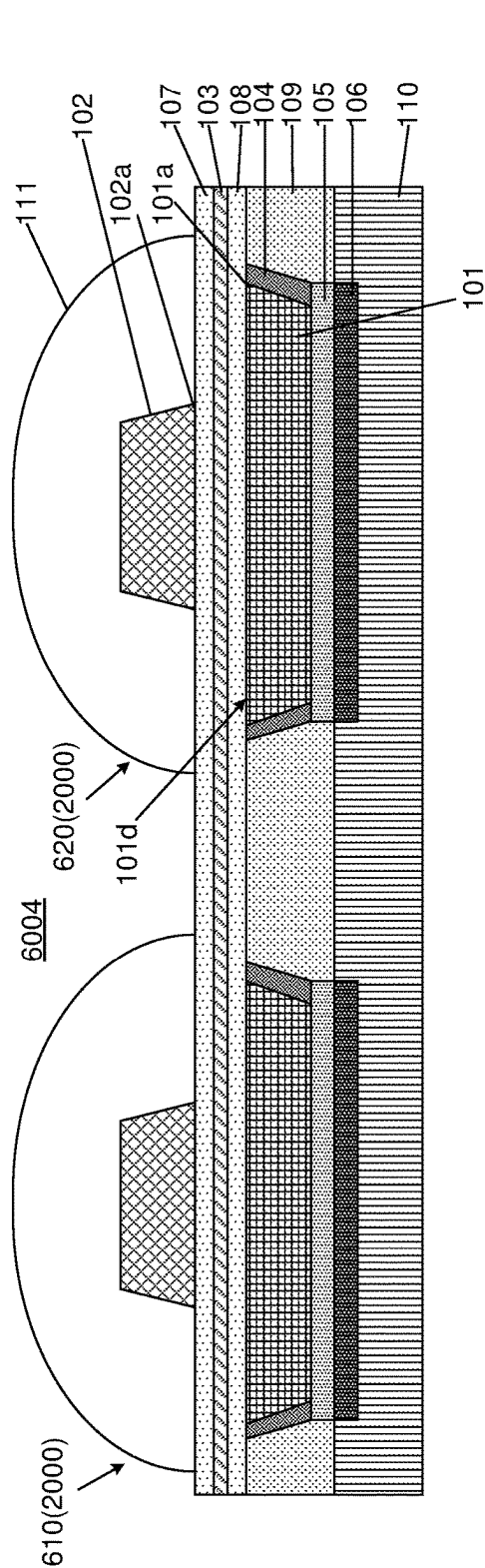


FIG. 6E

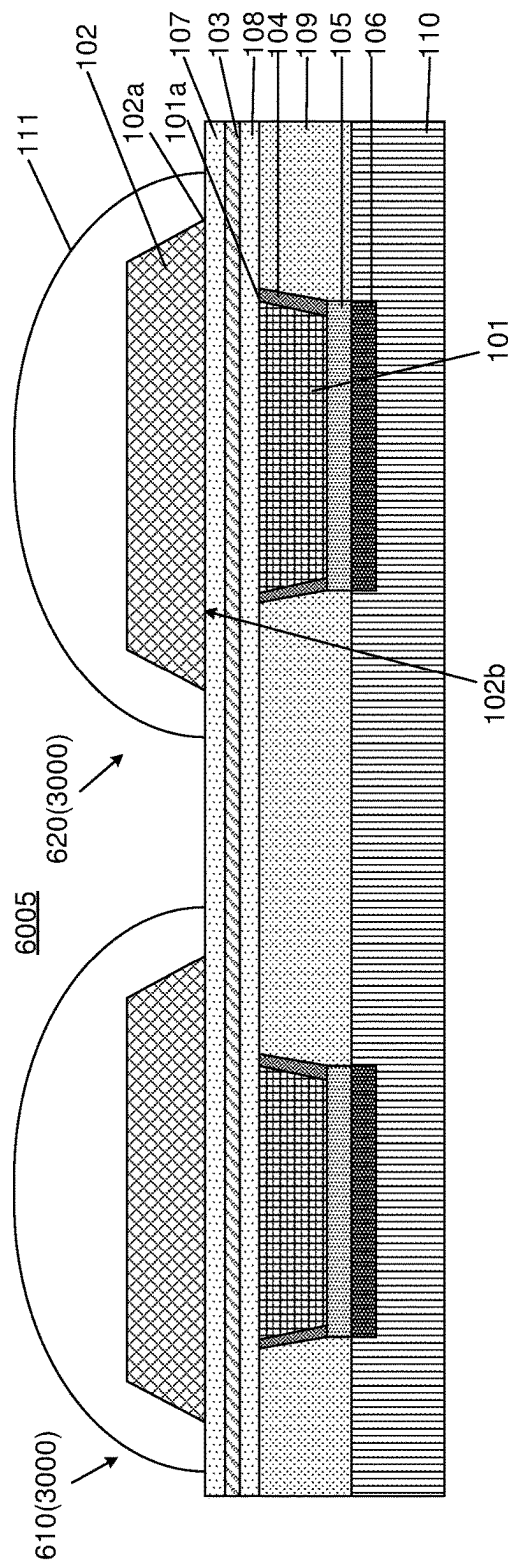


FIG. 6F

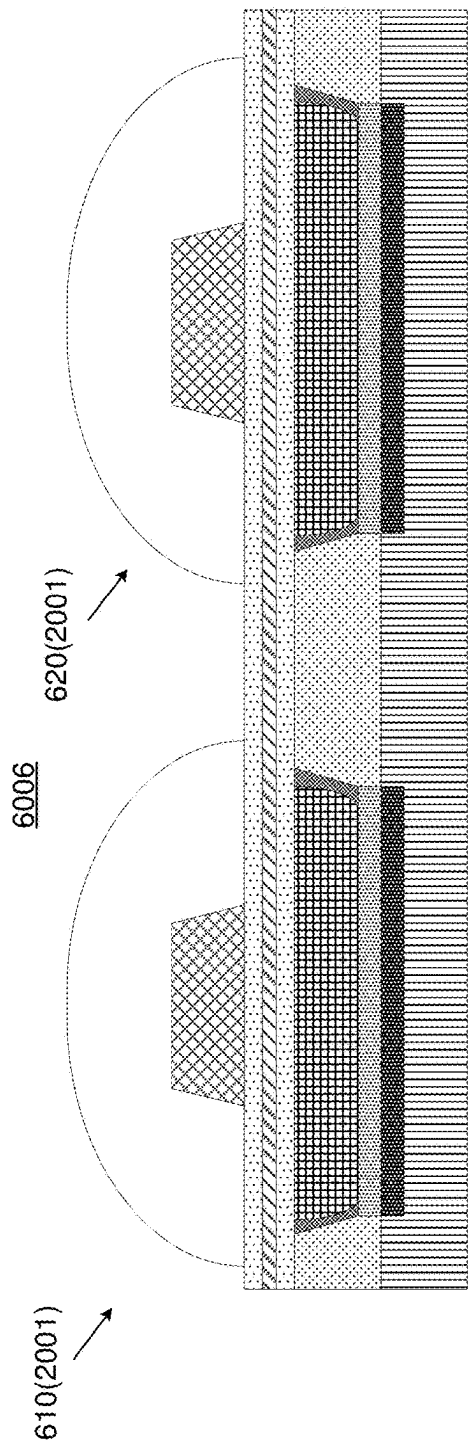


FIG. 6G

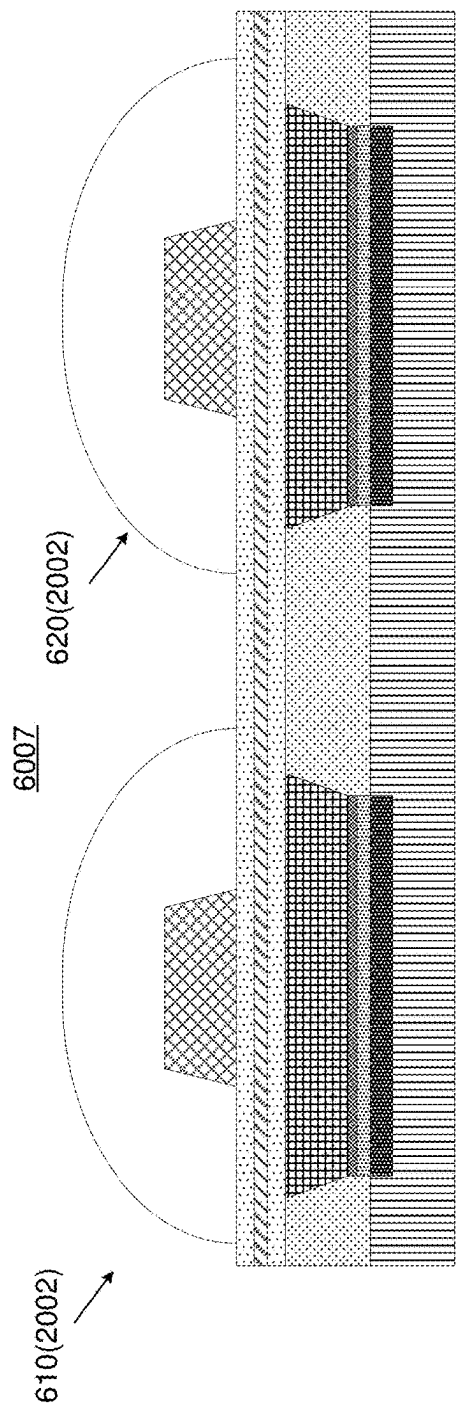


FIG. 6H

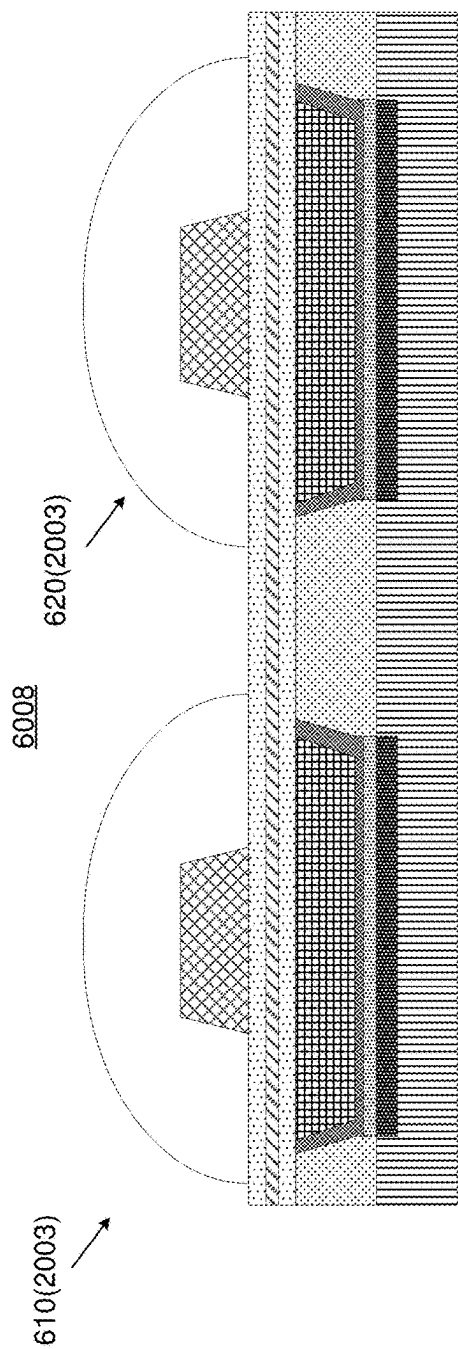


FIG. 6I

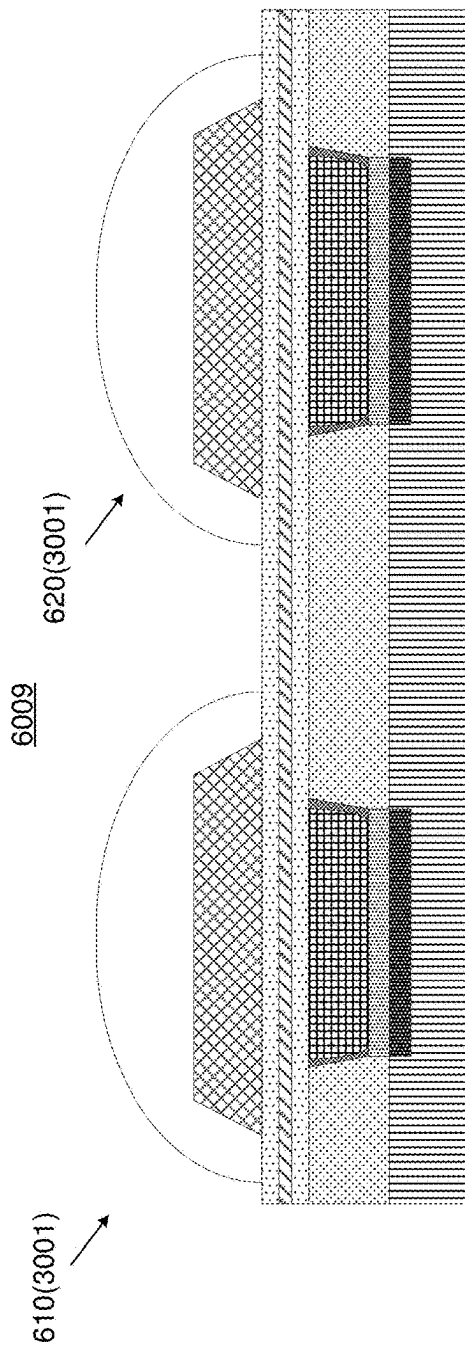


FIG. 6J

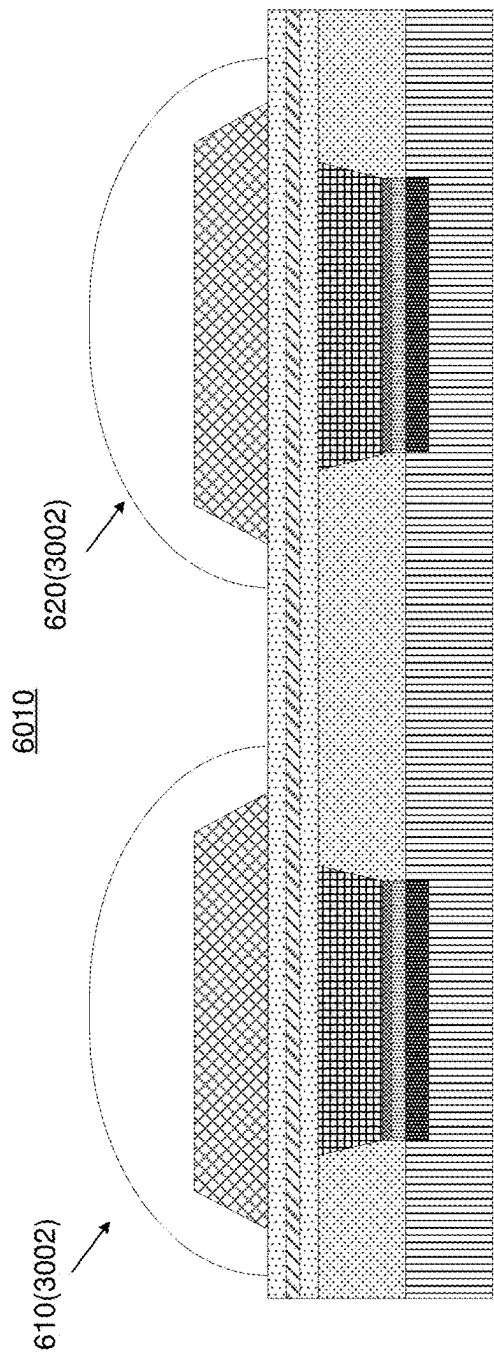


FIG. 6K

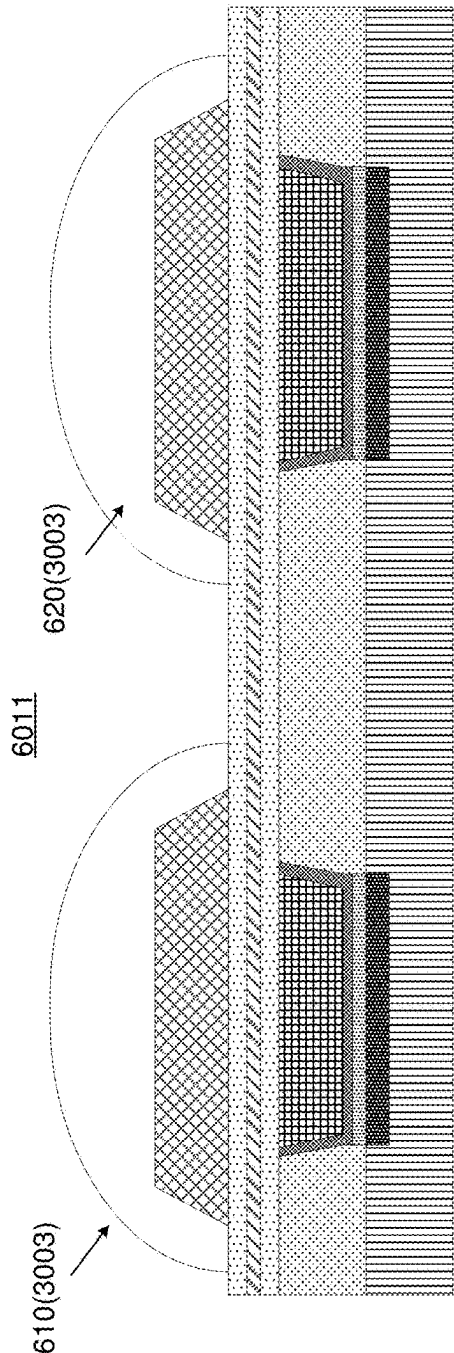


FIG. 6L

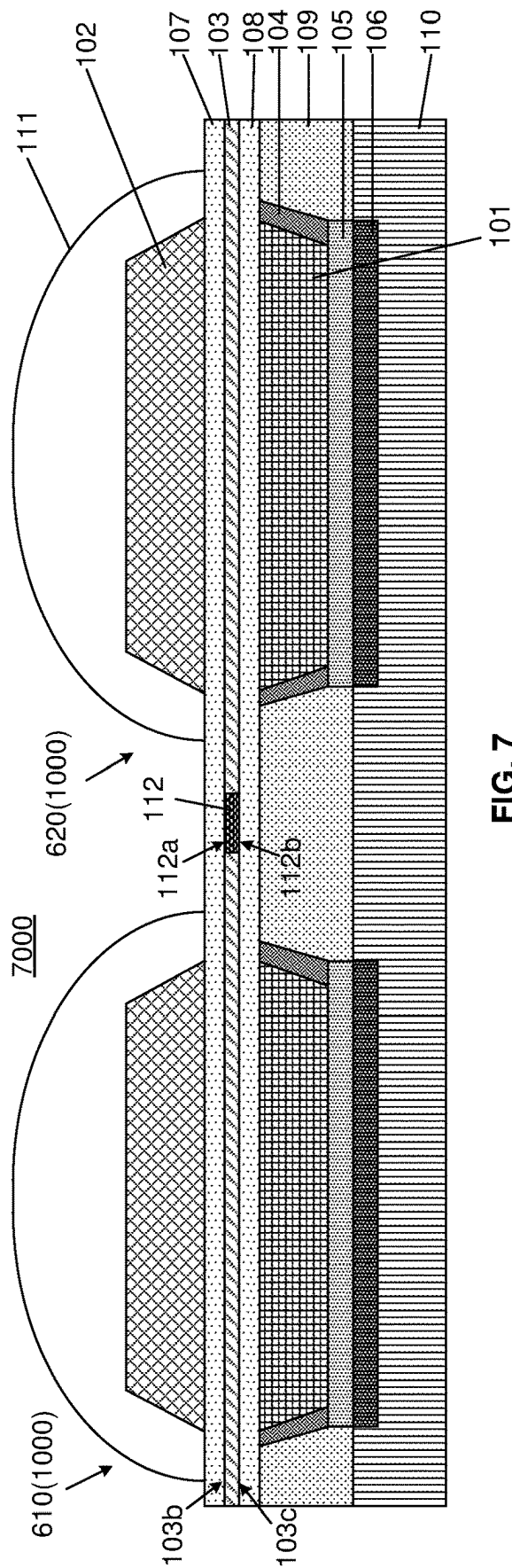


FIG. 7

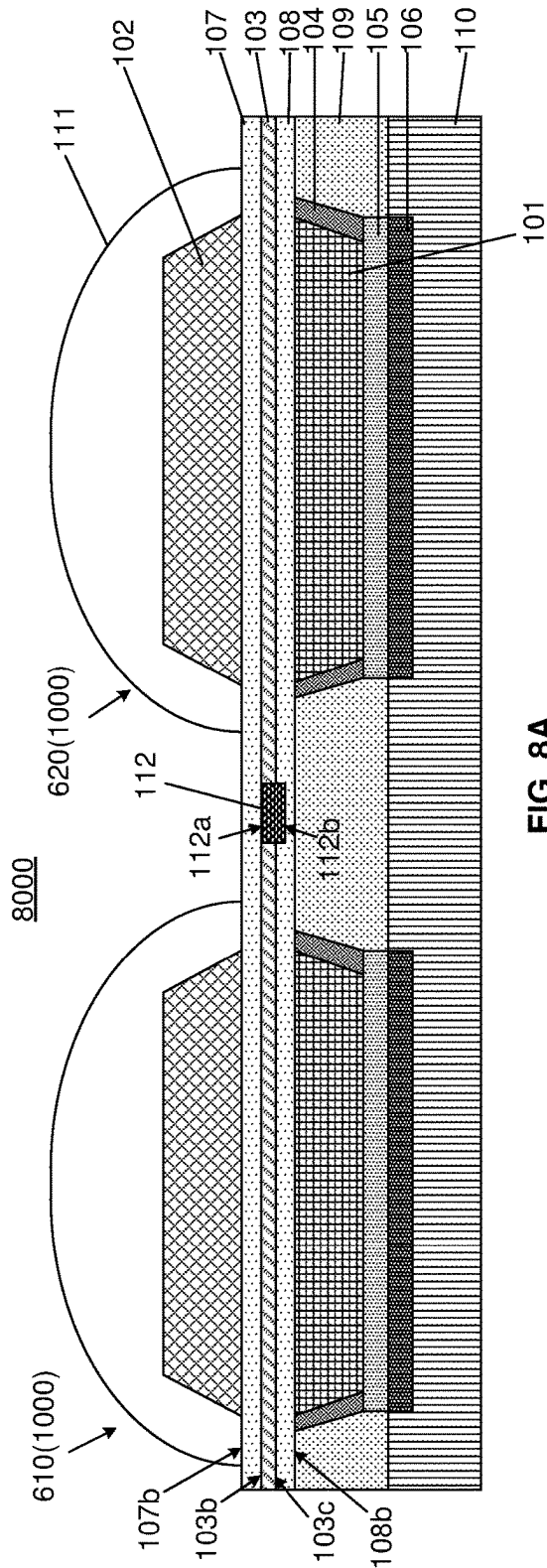


FIG. 8A

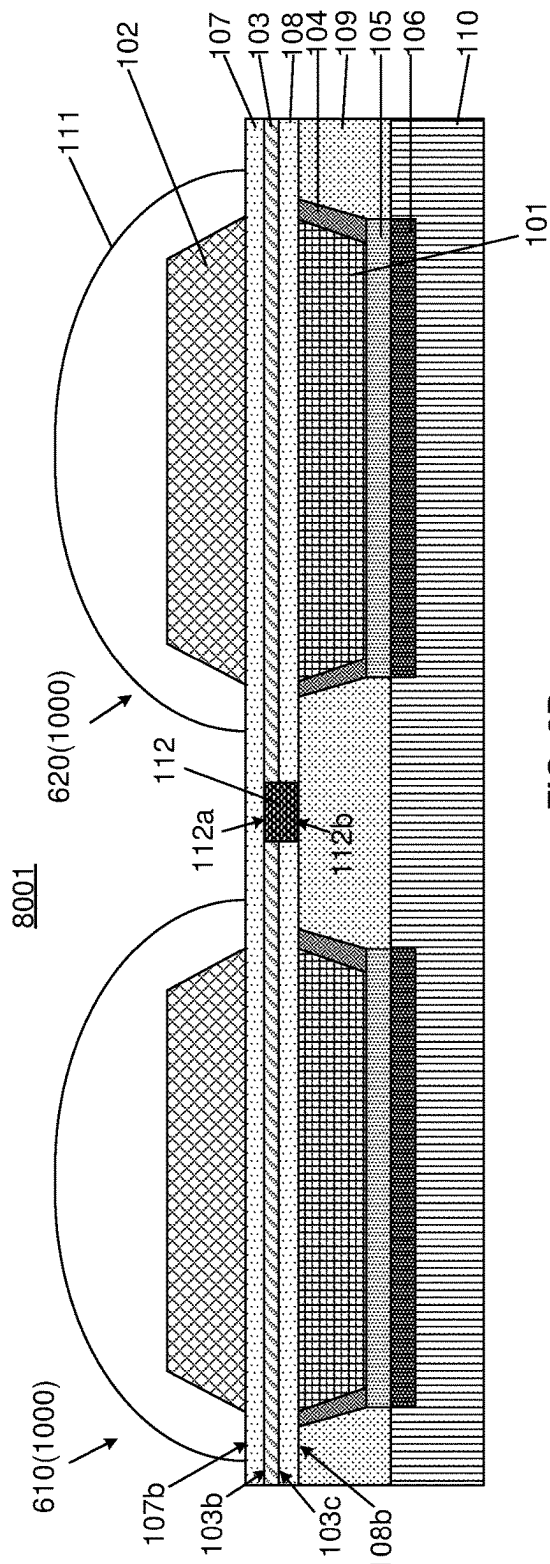


FIG. 8B

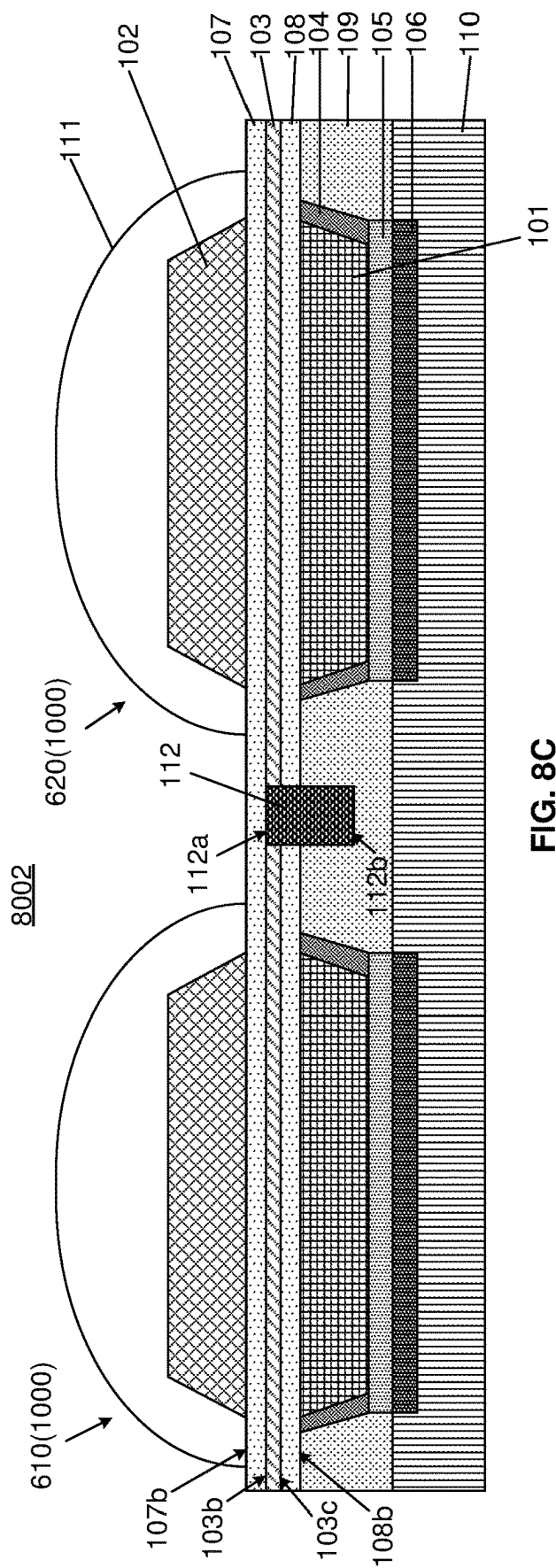


FIG. 8C

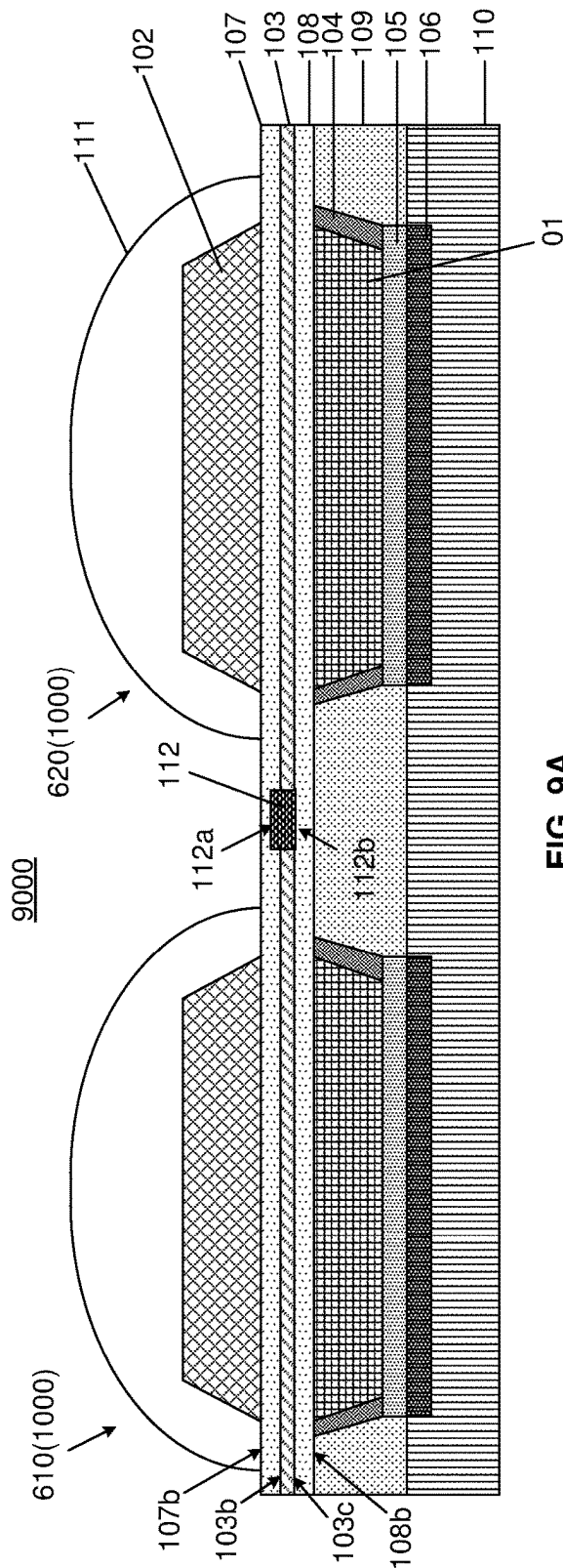


FIG. 9A

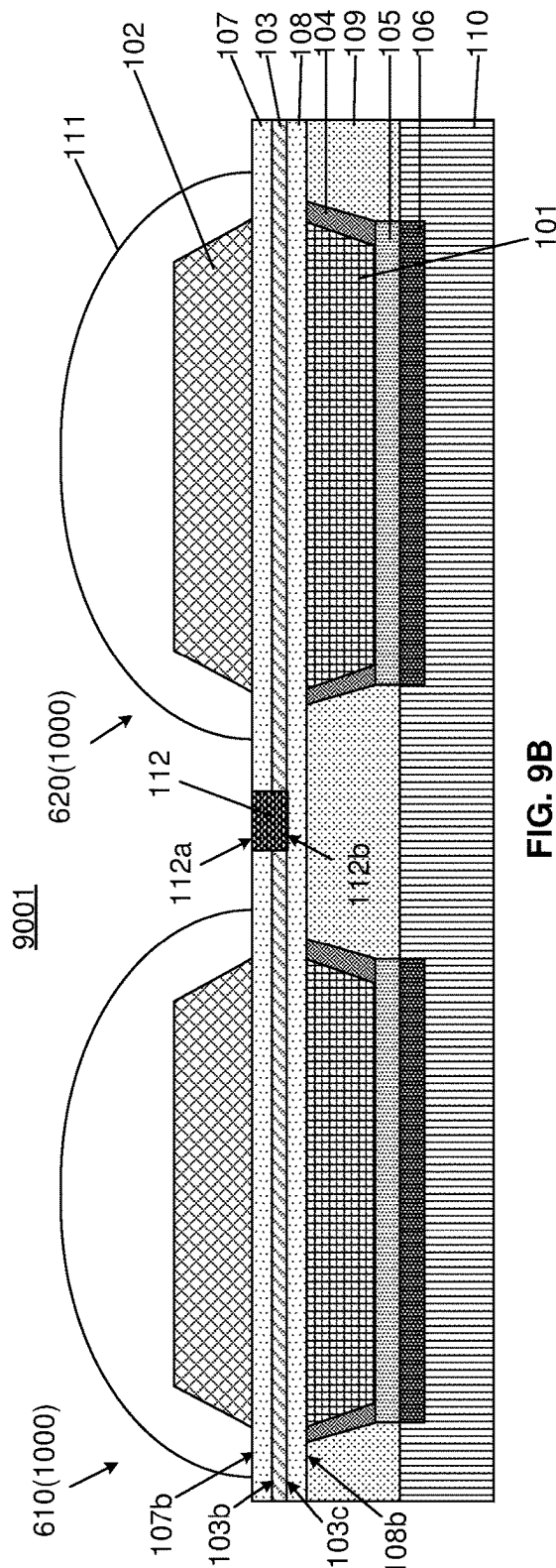


FIG. 9B

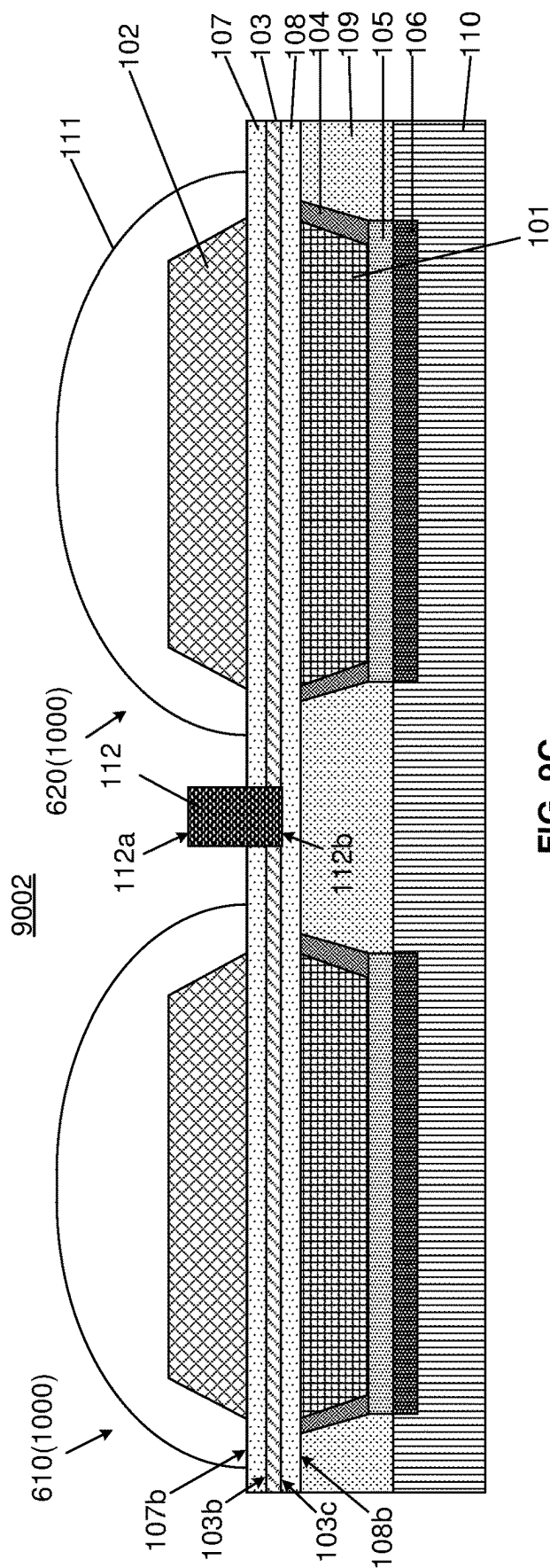
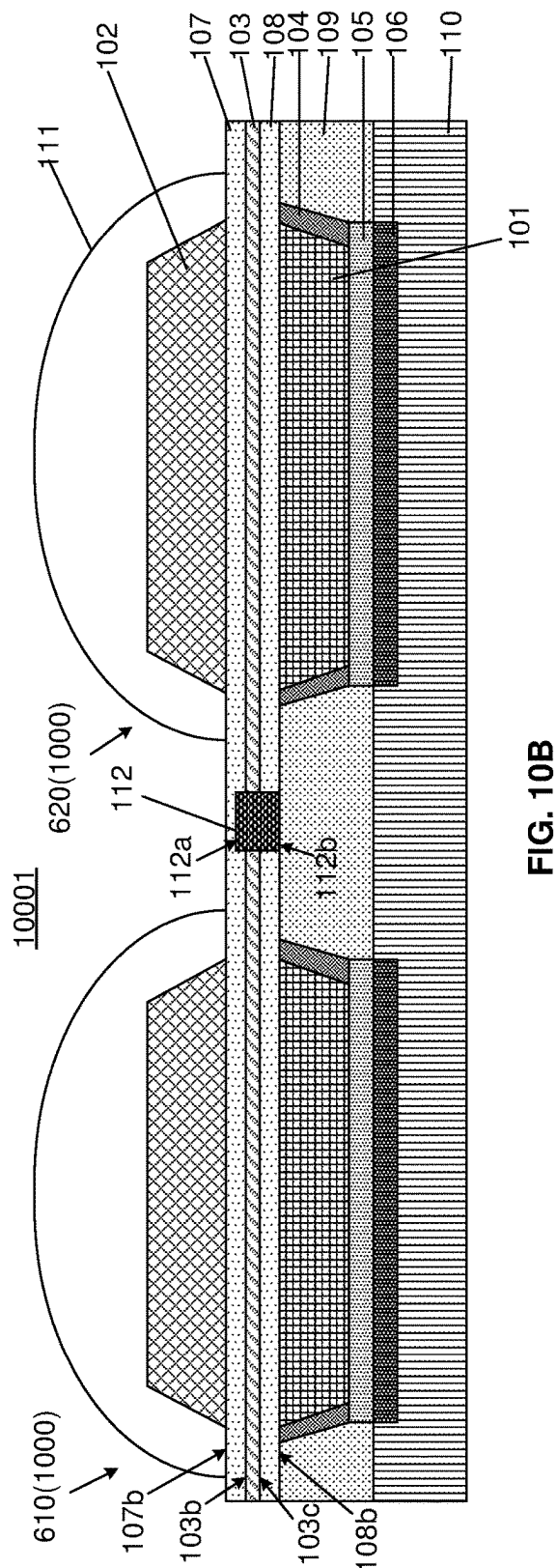
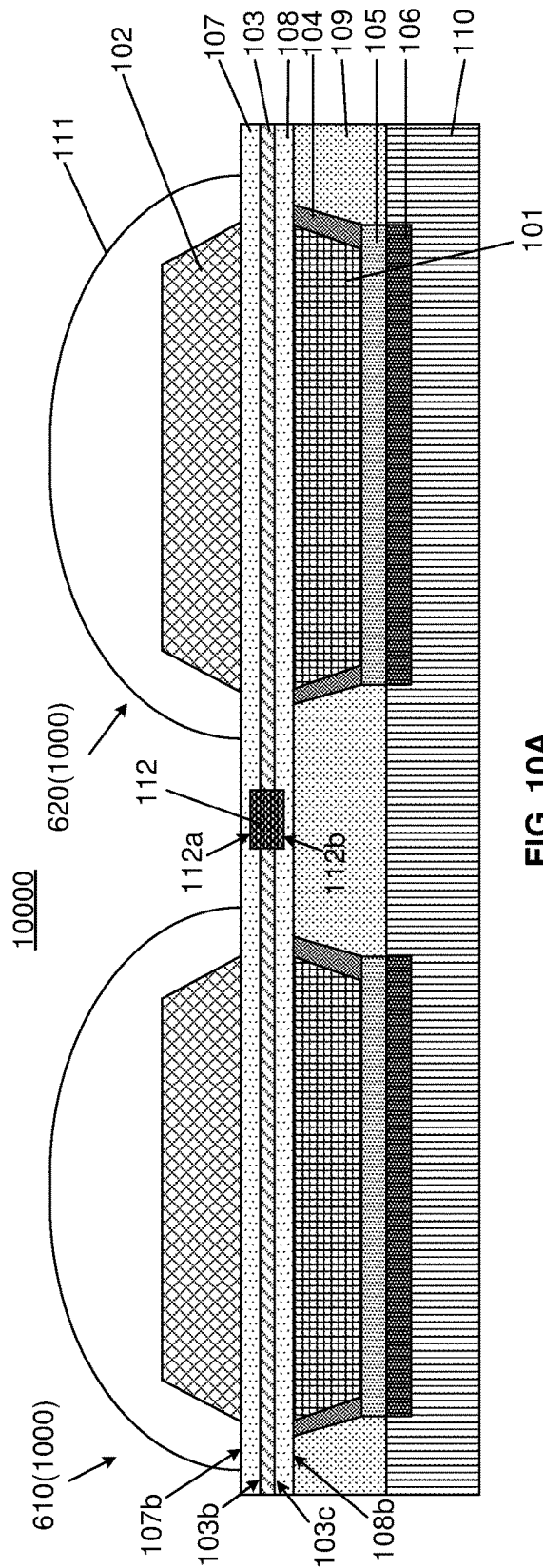


FIG. 9C



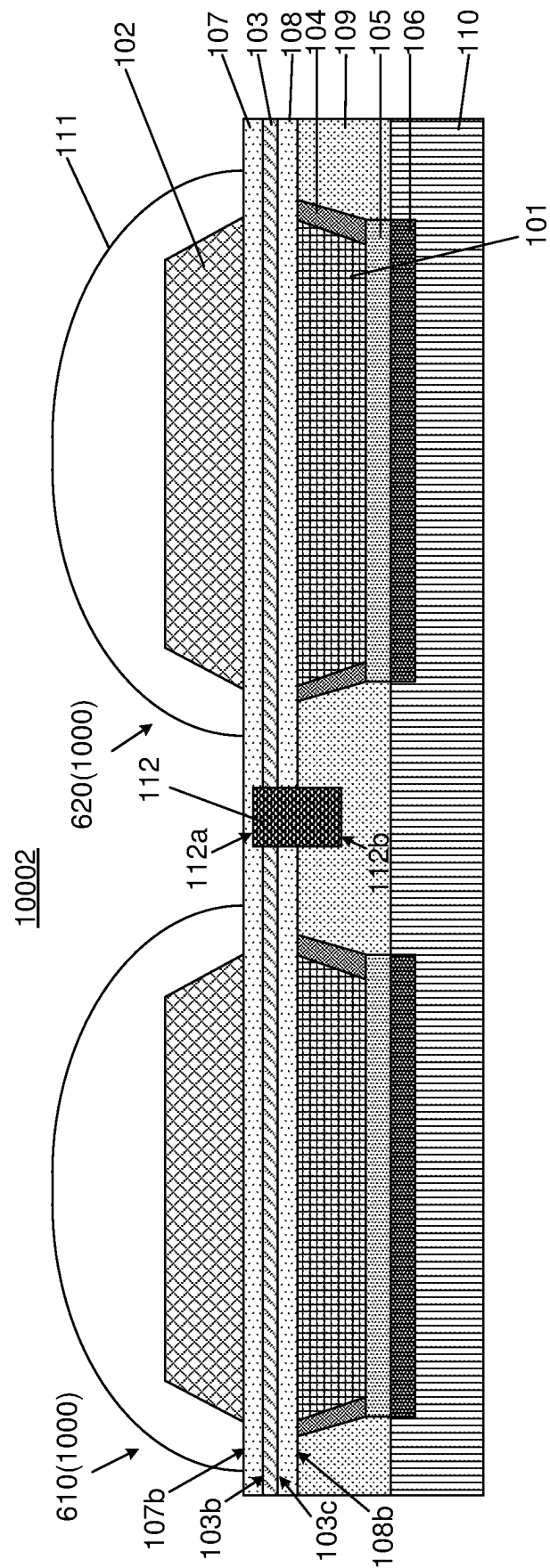


FIG. 10C

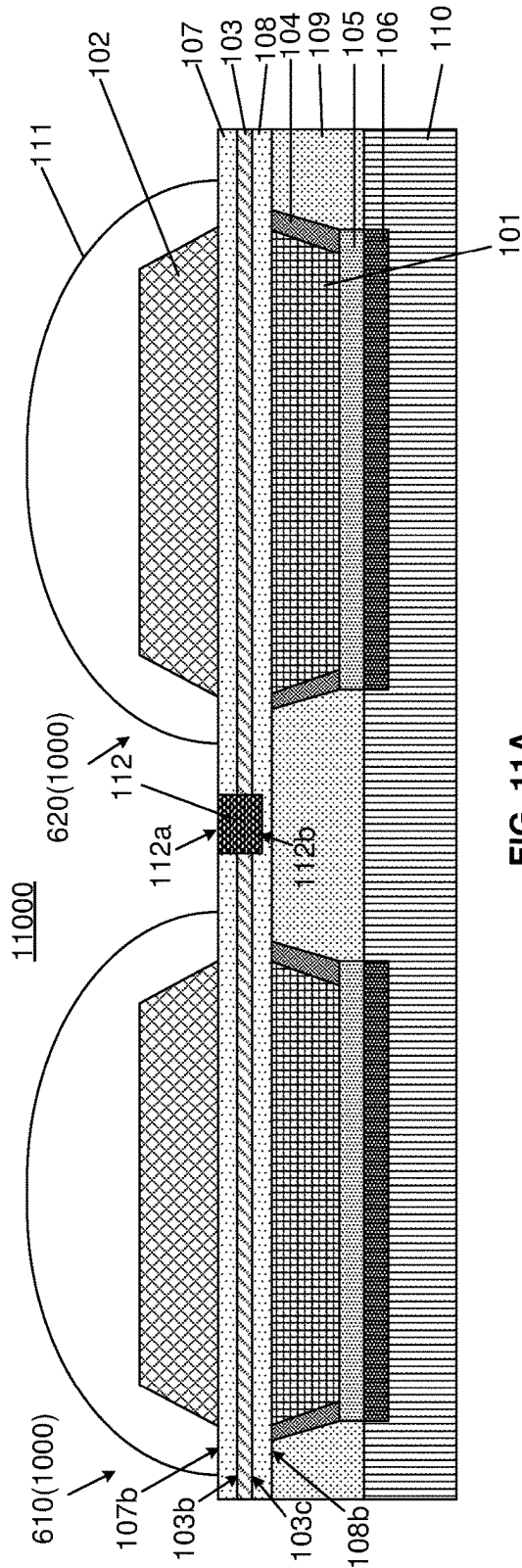


FIG. 11A

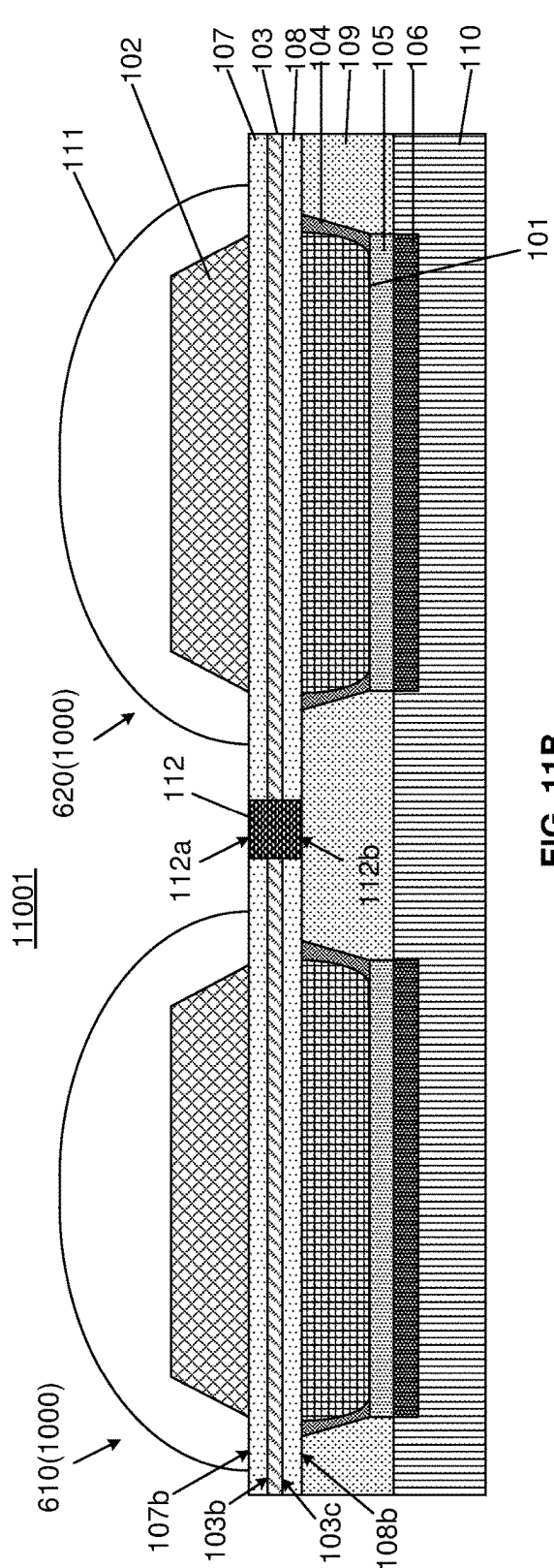
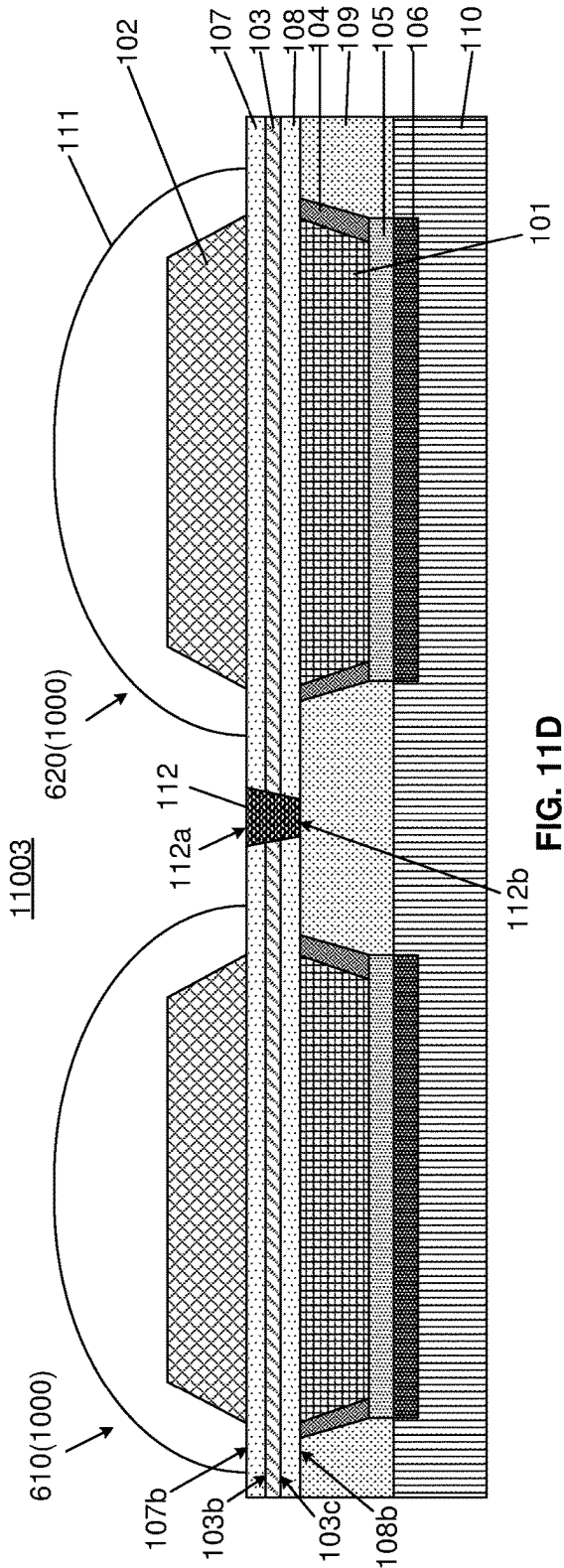
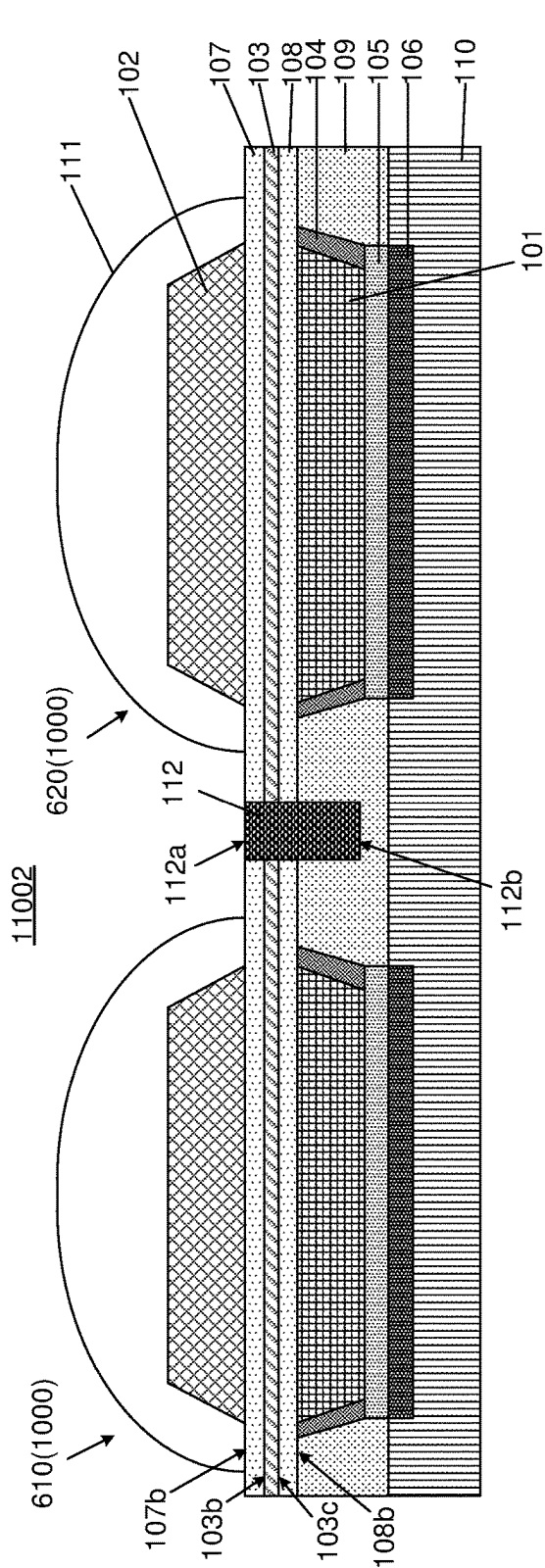
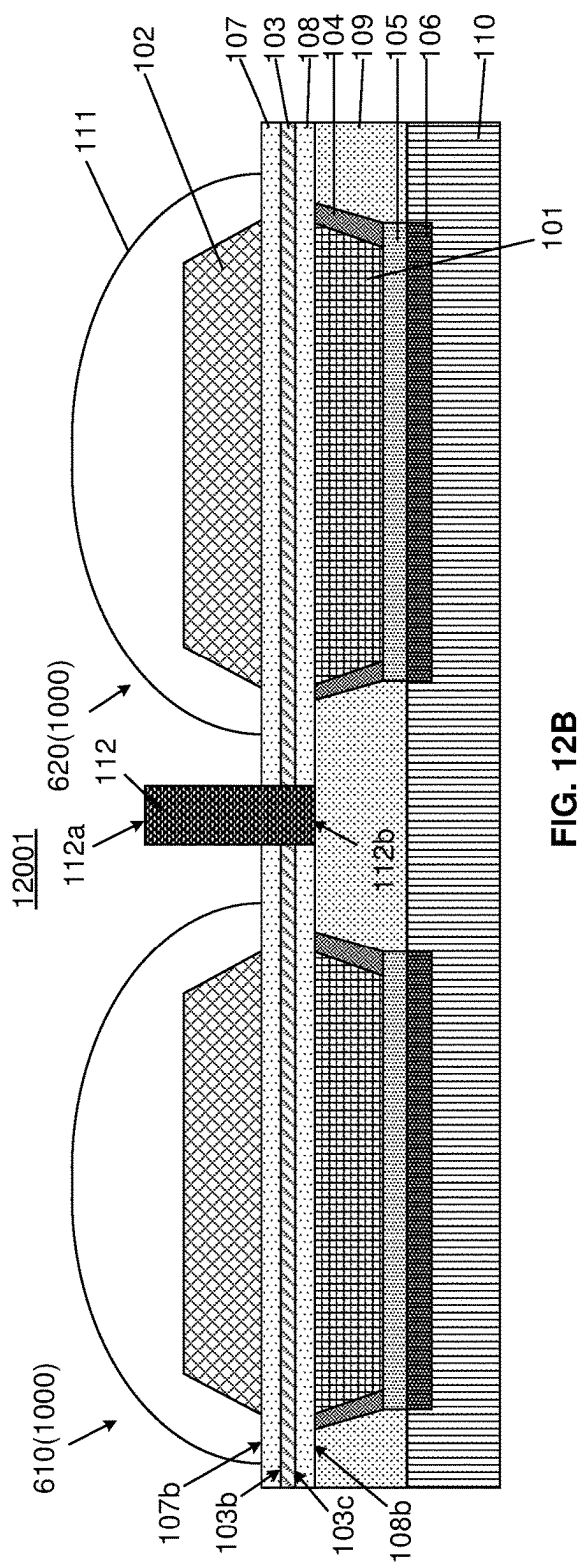
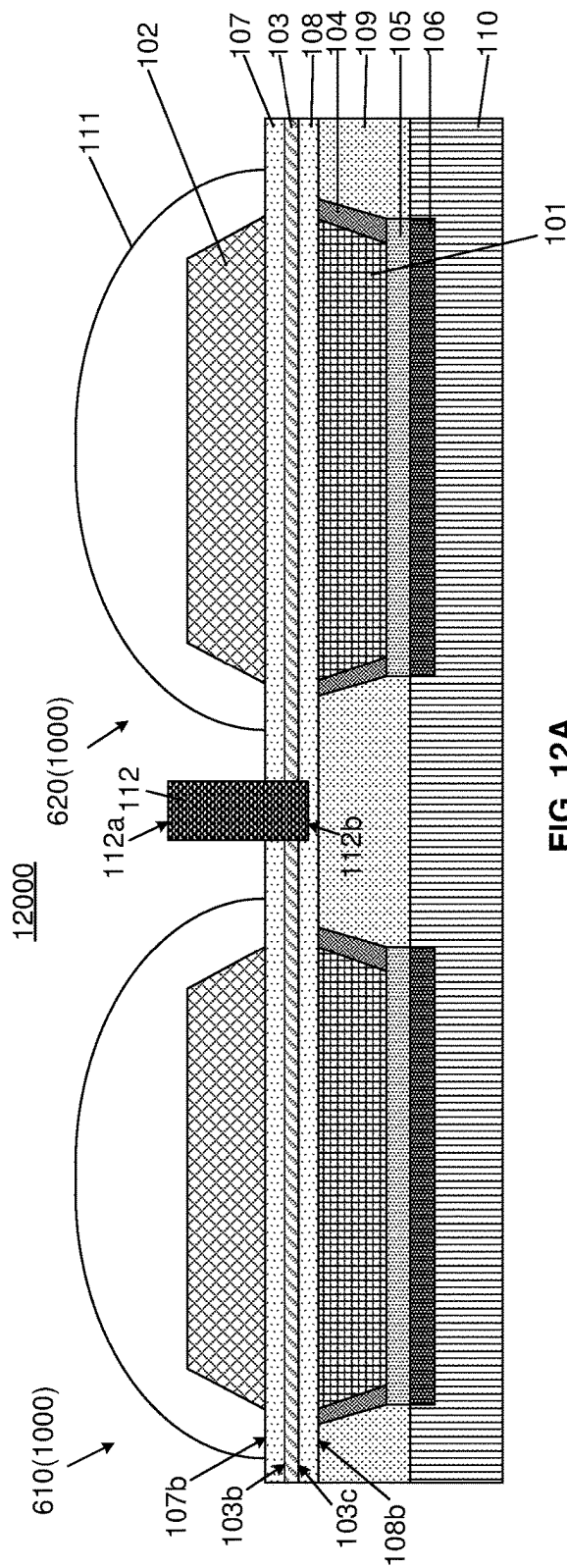
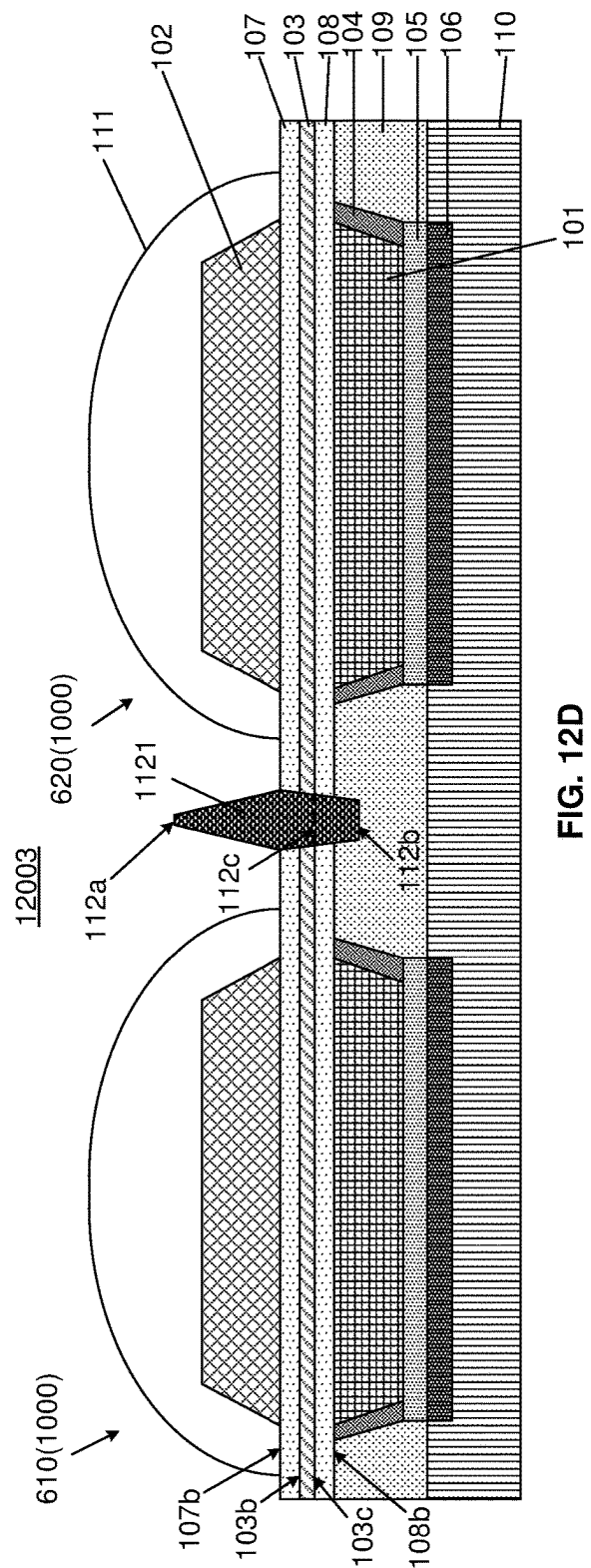
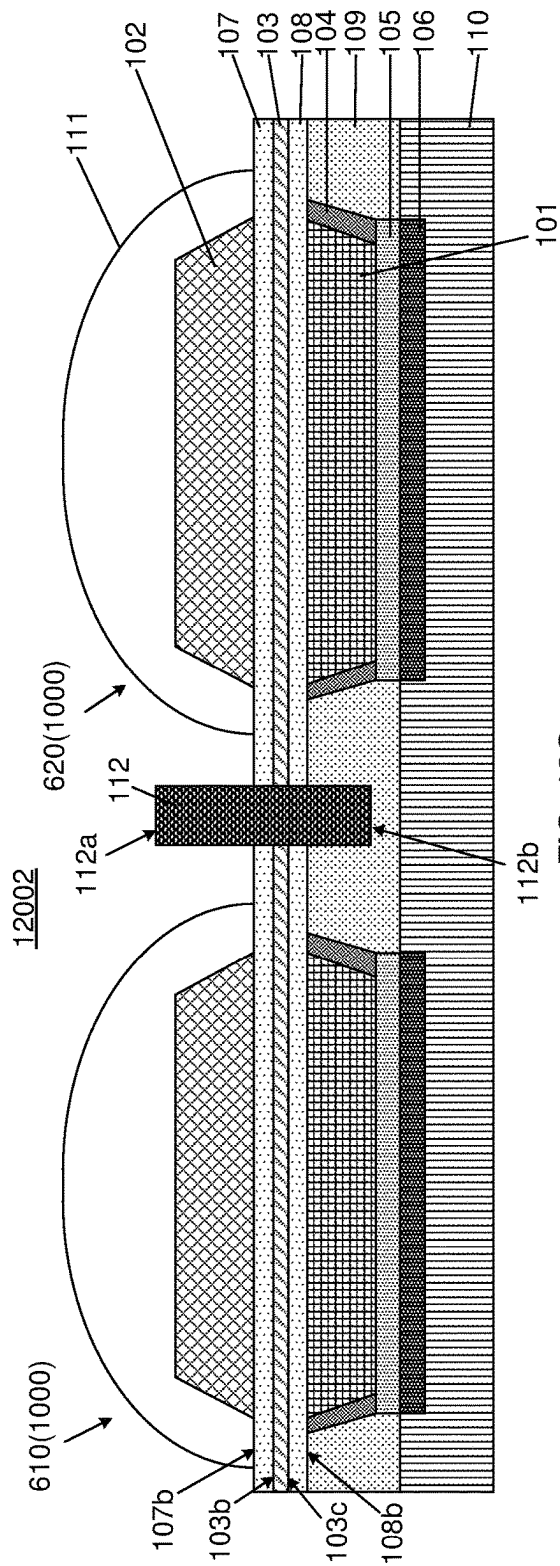


FIG. 11B







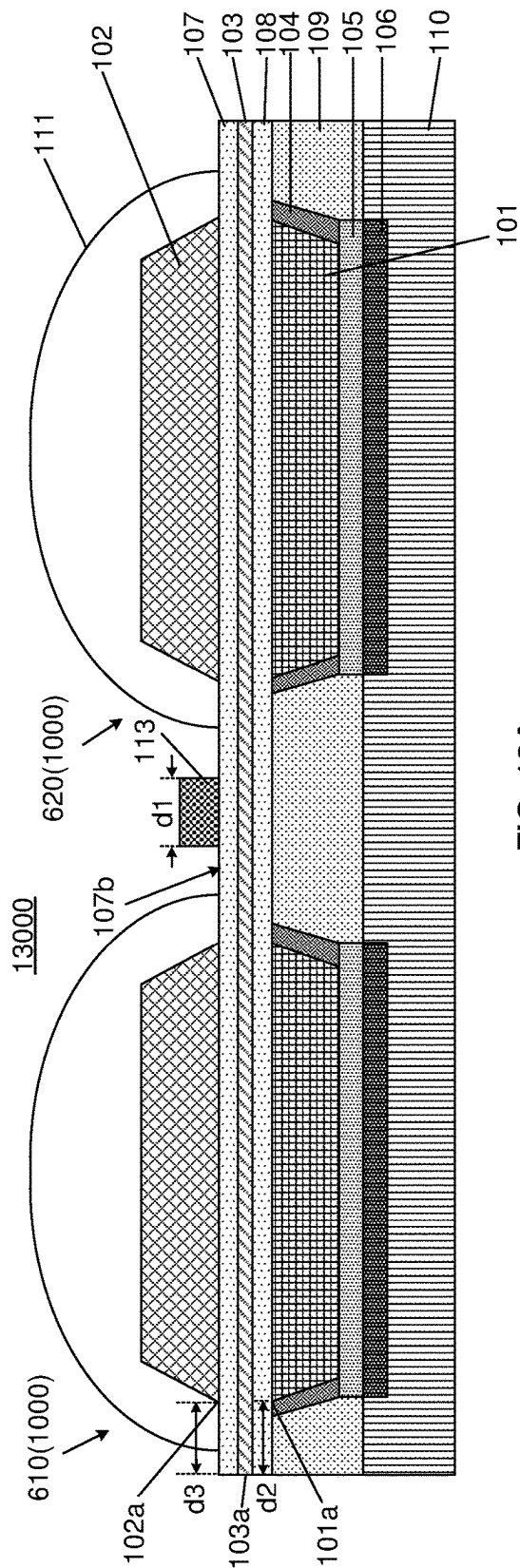


FIG. 13A

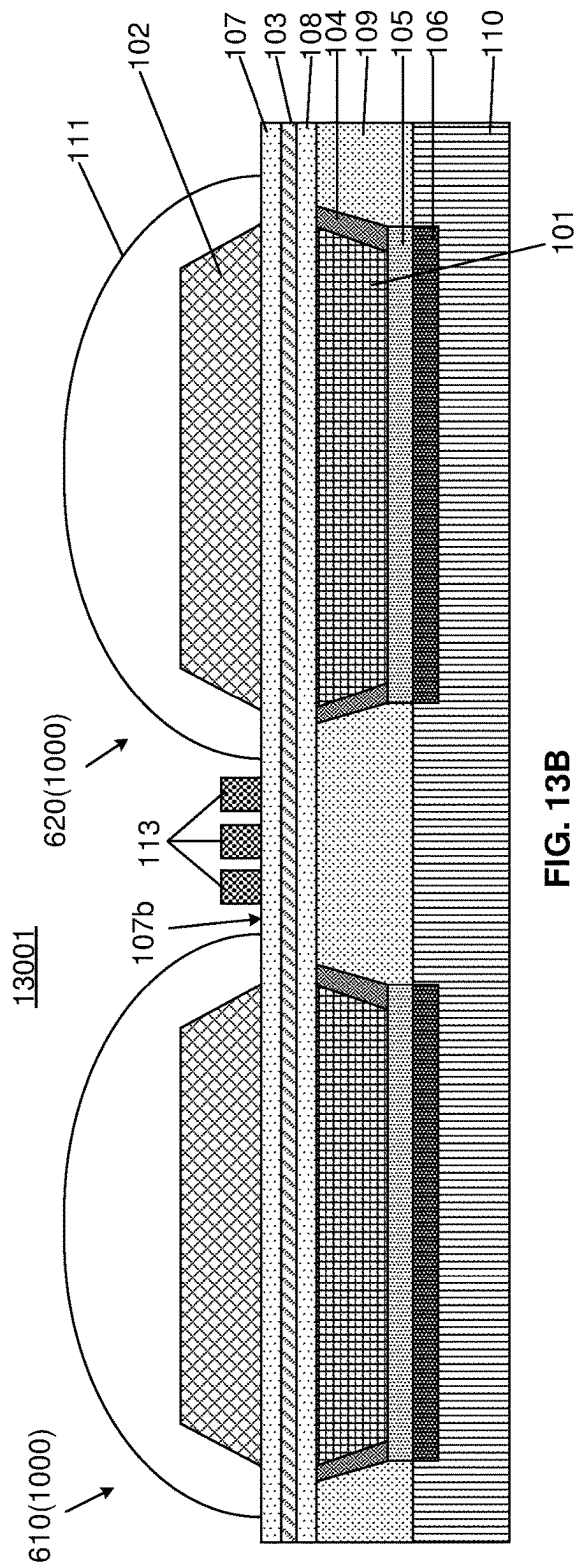


FIG. 13B

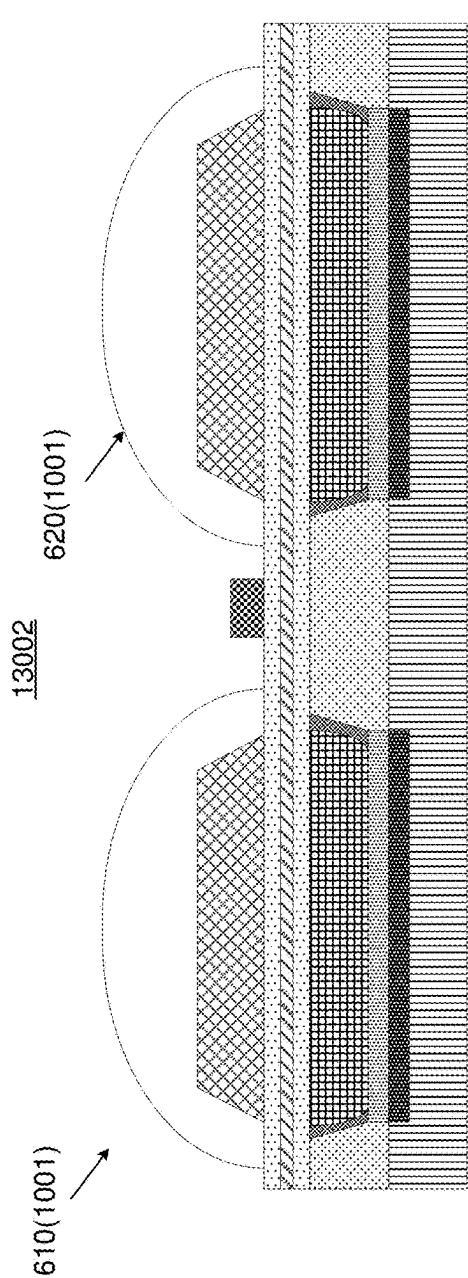


FIG. 13C

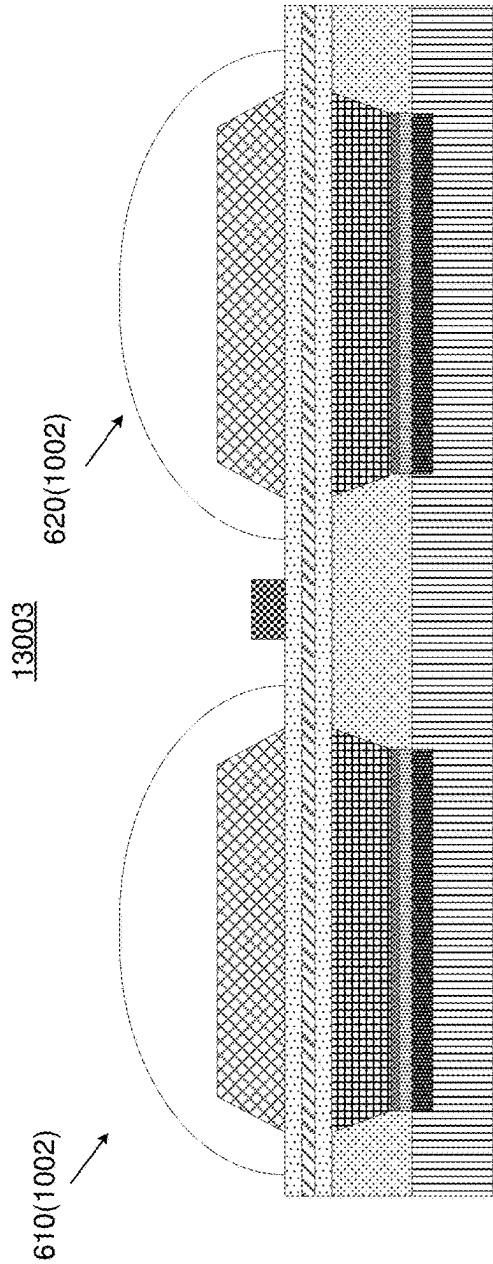


FIG. 13D

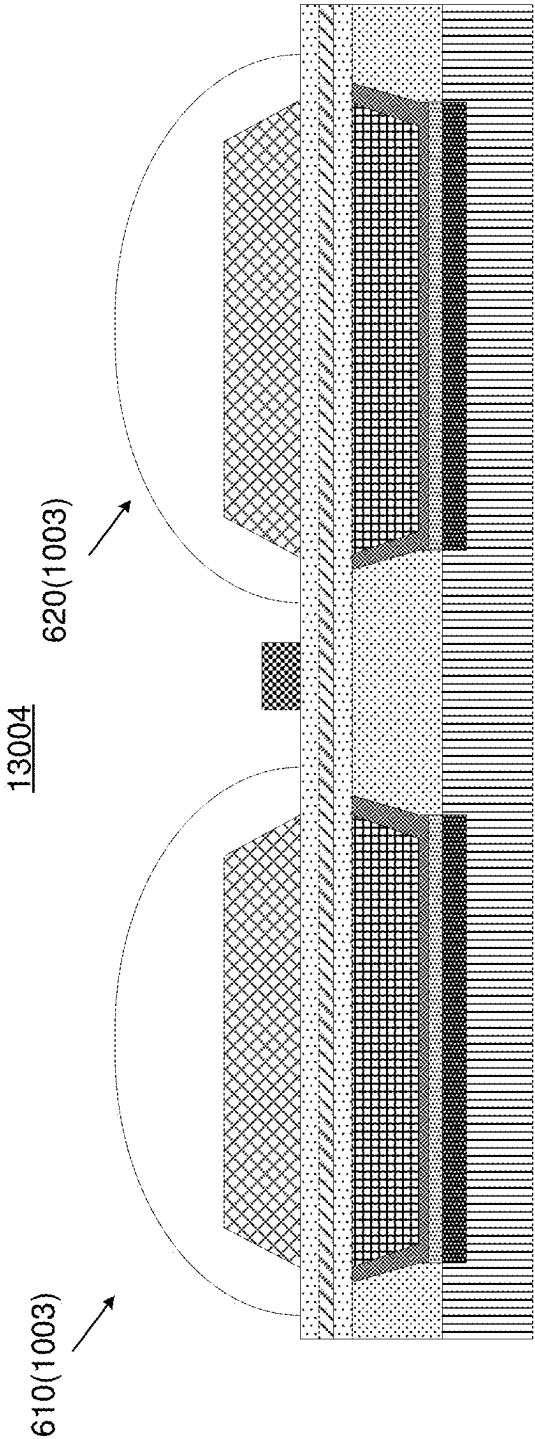


FIG. 13E

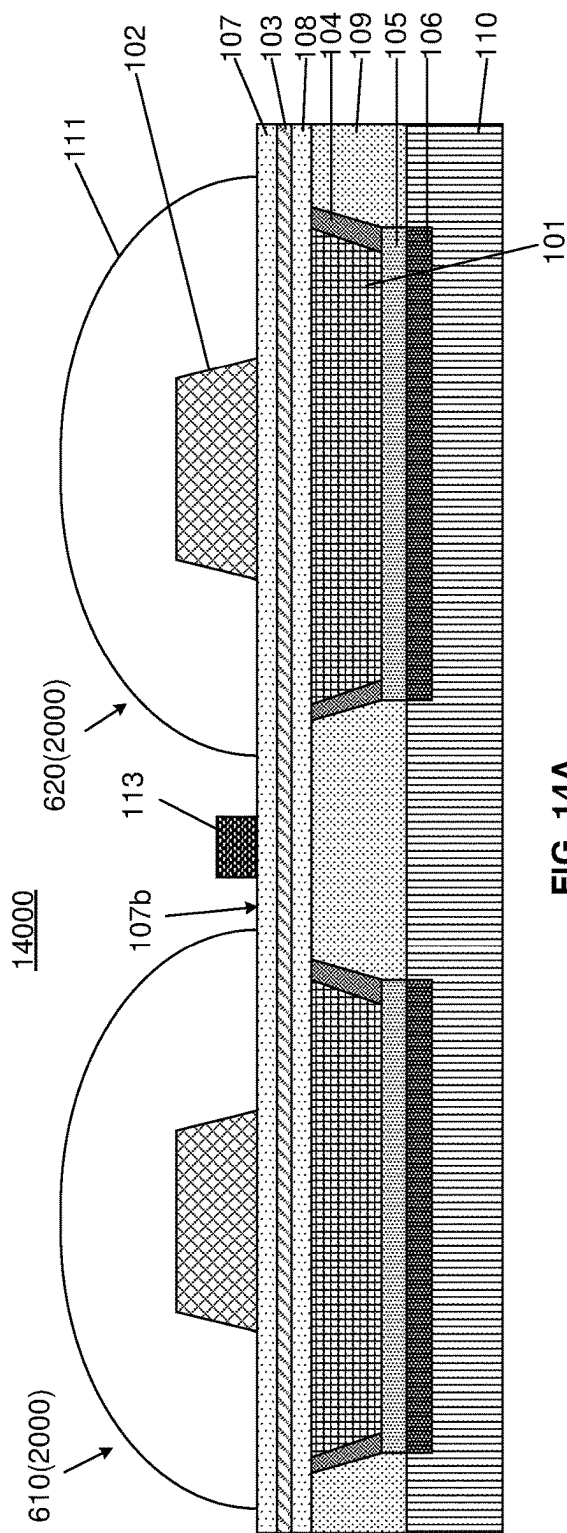


FIG. 14A

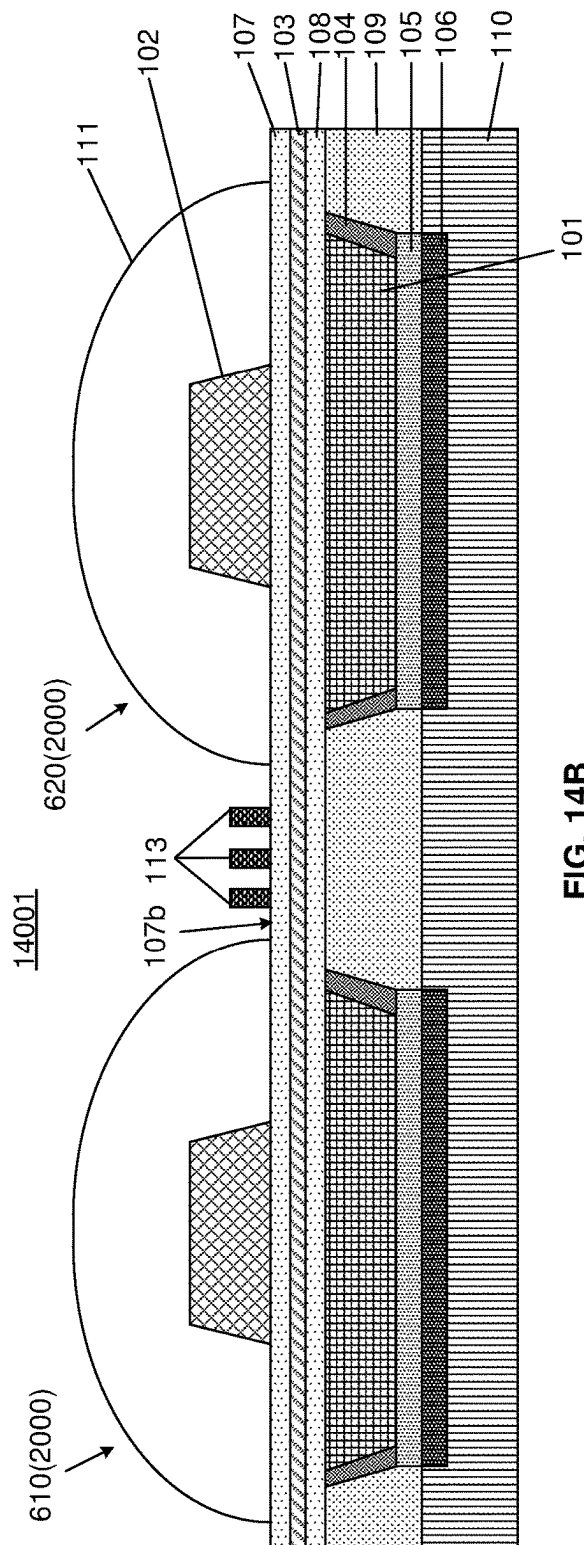


FIG. 14B

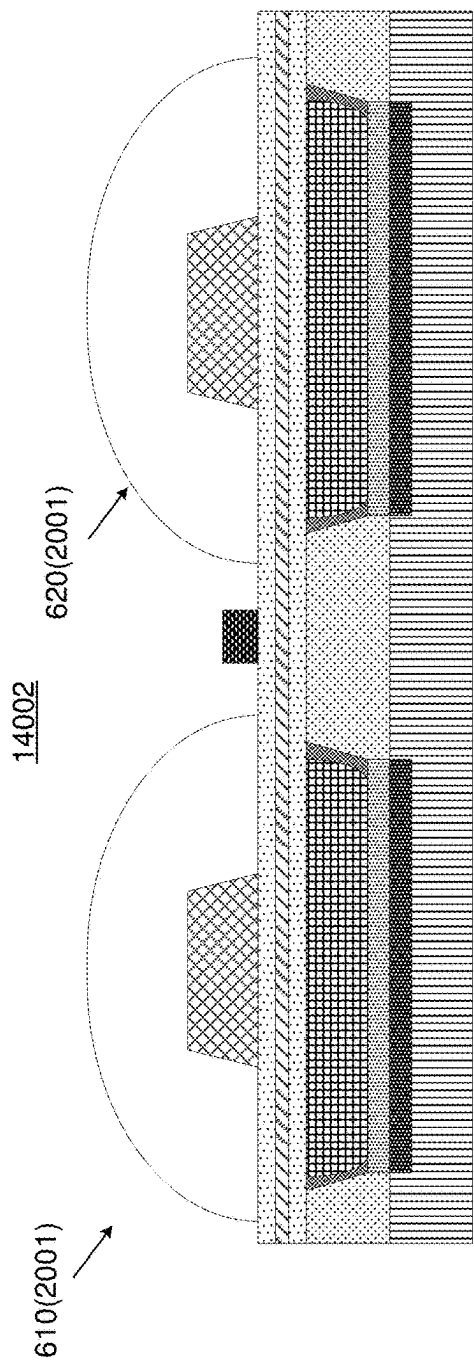


FIG. 14C

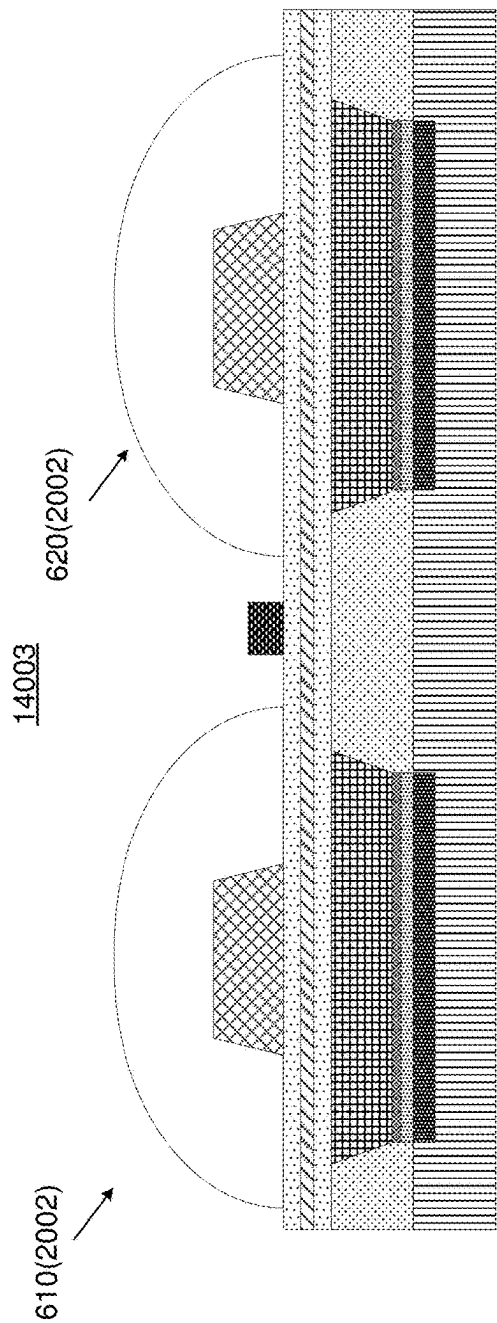


FIG. 14D

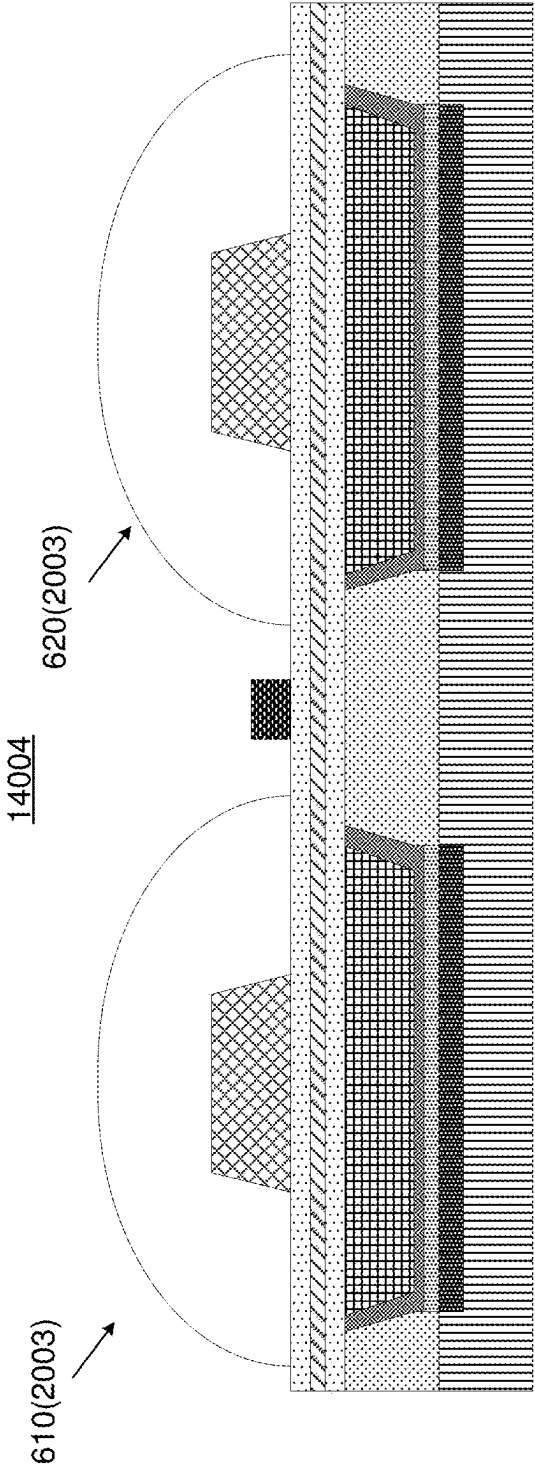


FIG. 14E

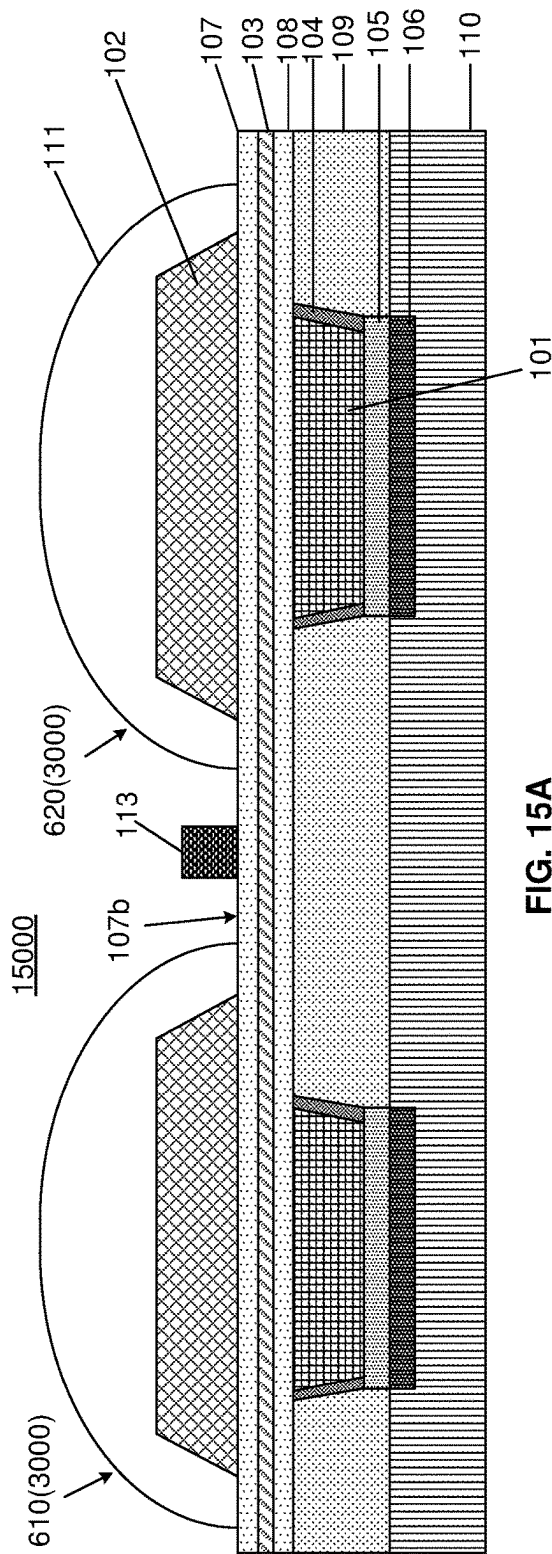


FIG. 15A

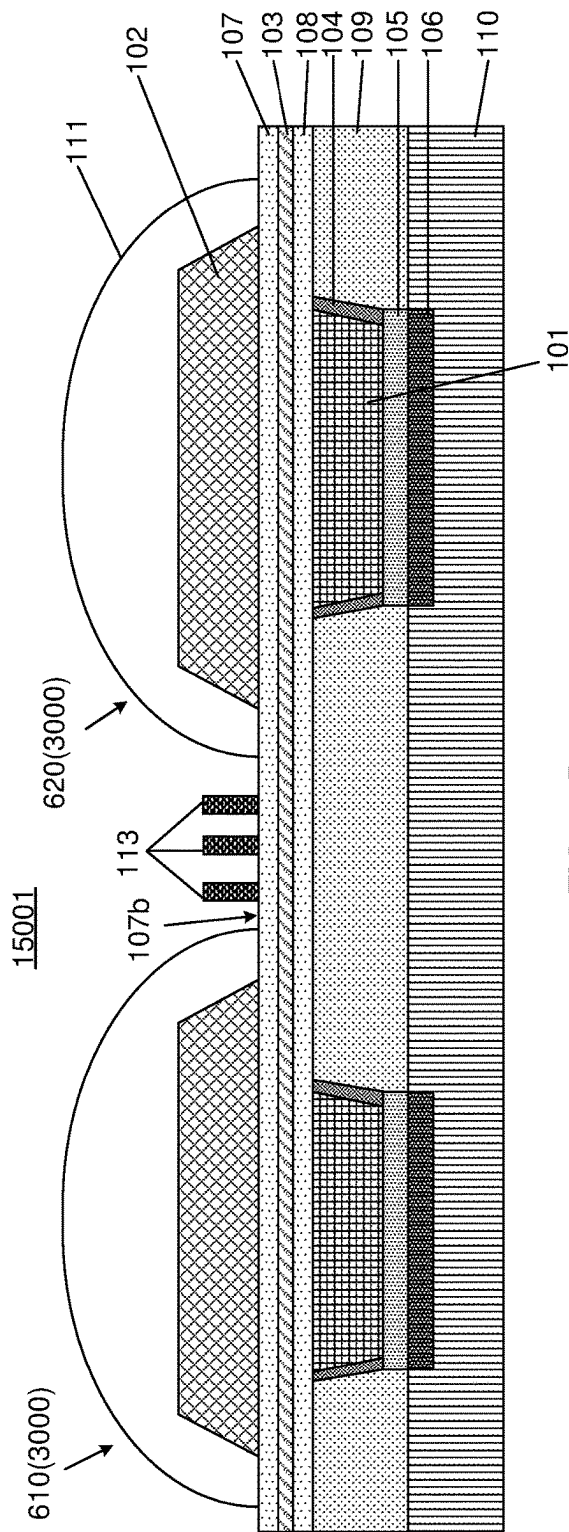


FIG. 15B

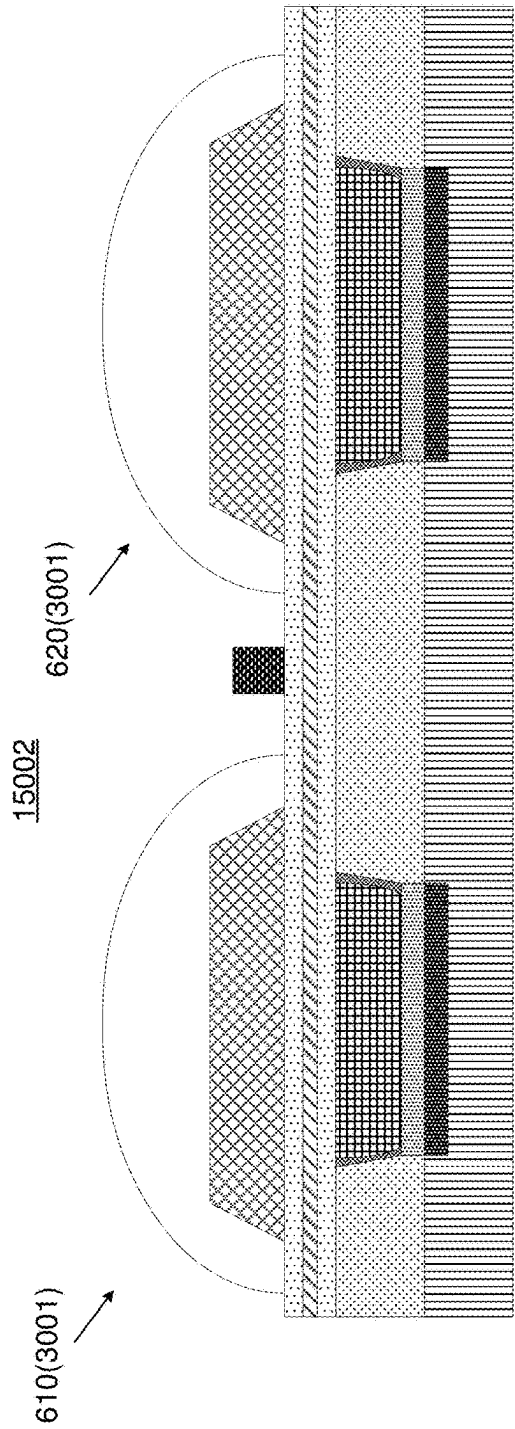


FIG. 15C

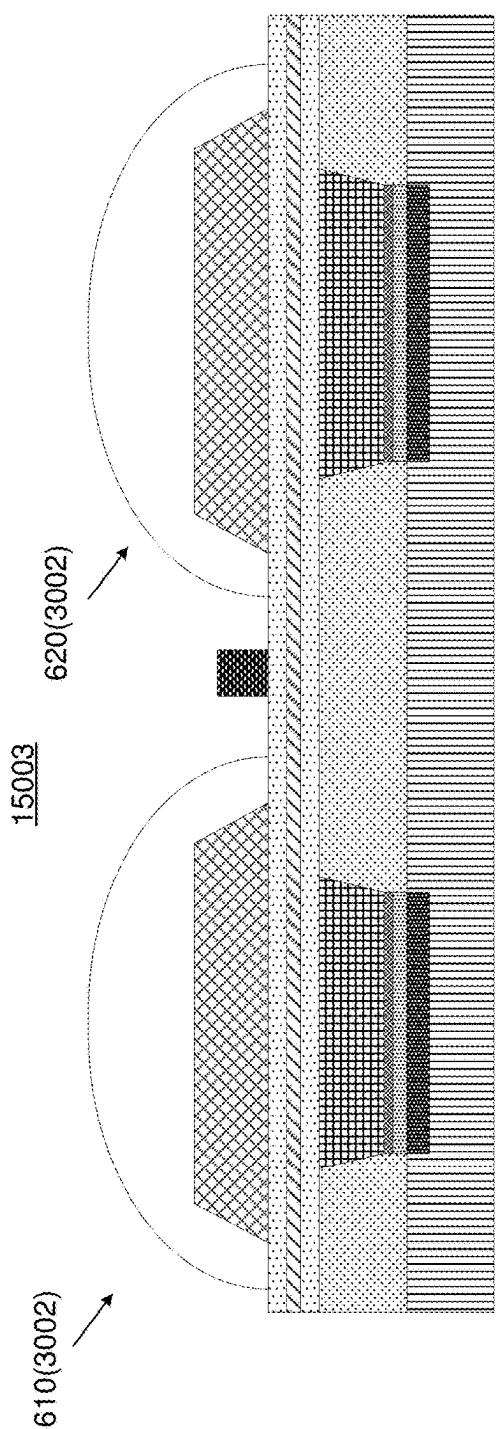


FIG. 15D

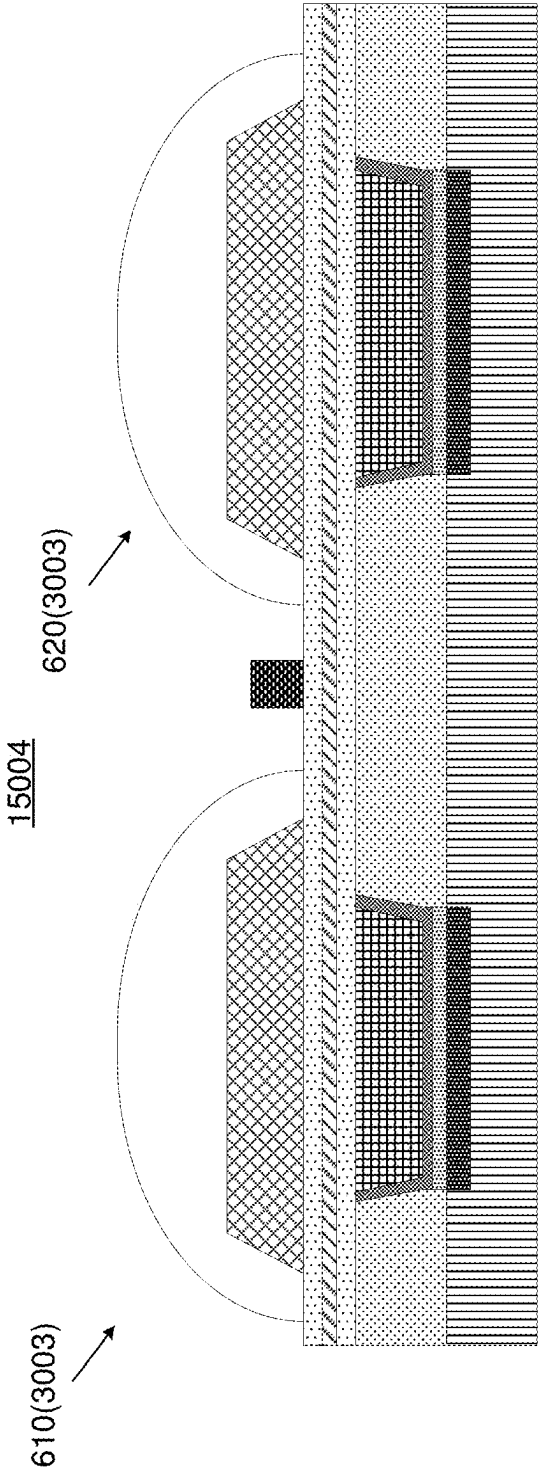


FIG. 15E

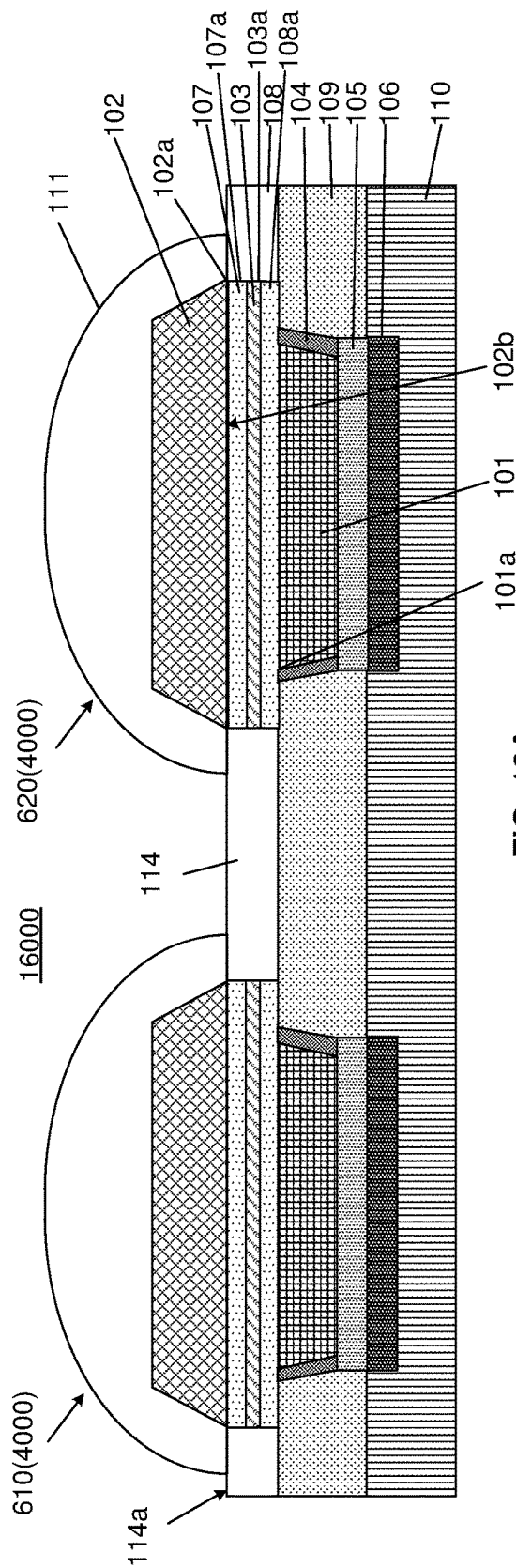


FIG. 16A

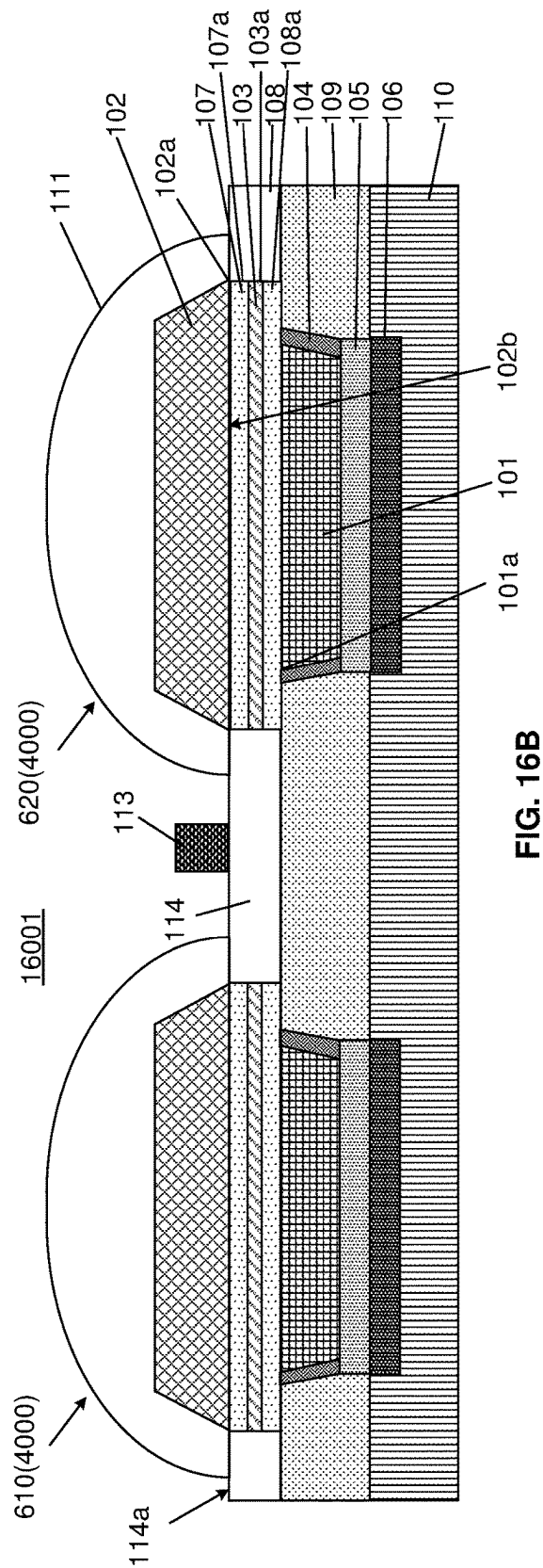


FIG. 16B

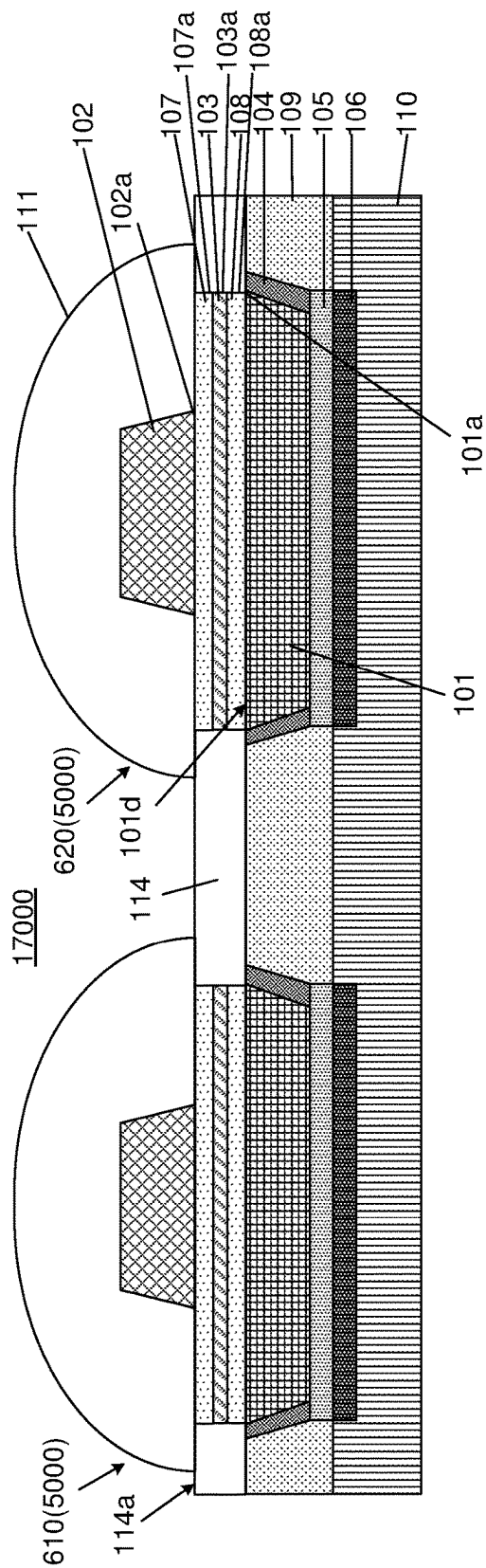


FIG. 17A

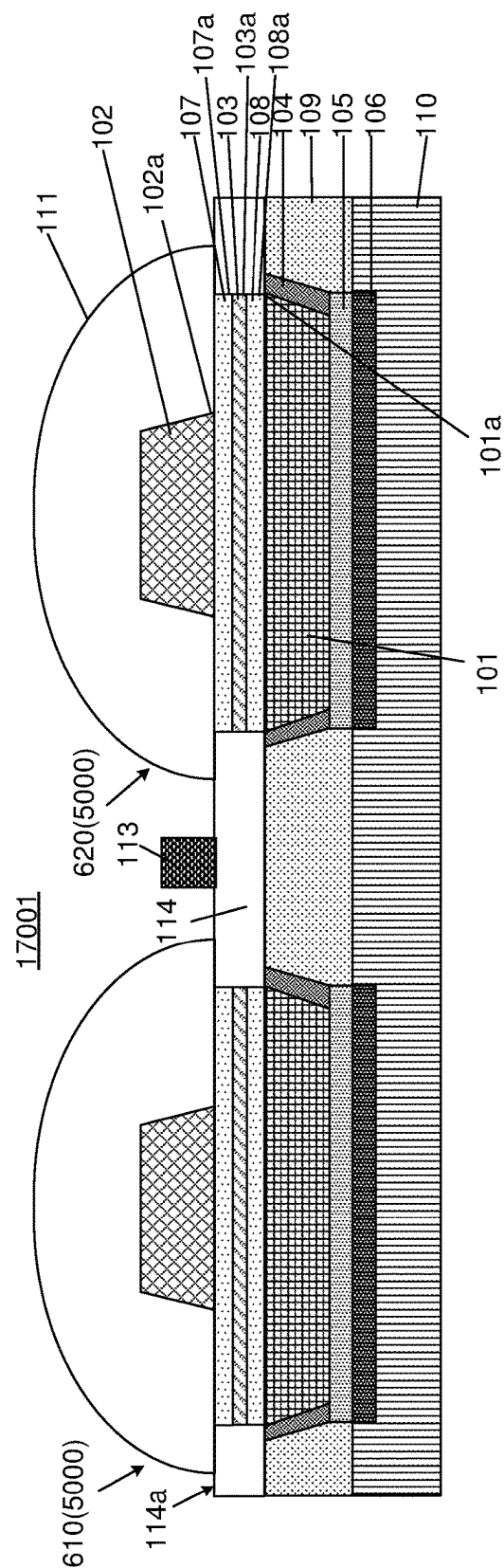


FIG. 17B

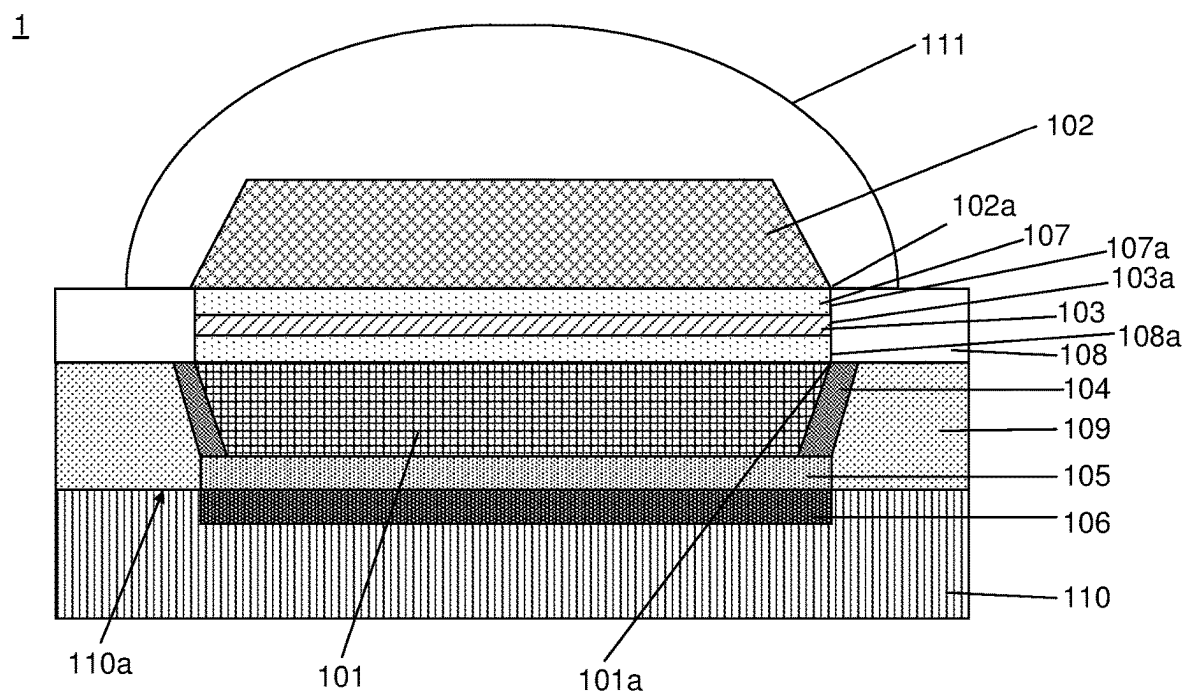


FIG. 18

1

MICRO-LED STRUCTURE AND MICRO-LED CHIP INCLUDING SAME

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims the benefit of priority to U.S. Provisional Application No. 63/131,128, filed on Dec. 28, 2020, the entire contents of which are incorporated herein by reference.

FIELD

The present disclosure relates to a micro-LED structure and a micro-LED chip including the micro-LED structure.

BACKGROUND

A micro-light emitting diode (micro-LED) is a device that emits light using an electric signal and has a size on the order of micrometers or even smaller. The micro-LED can be driven at a low voltage such that it is widely implemented in small-sized optical elements. In recent years, the micro-LED has been developed as an illuminating light source by increasing its efficiency.

SUMMARY

According to one aspect of embodiments of the present disclosure, a micro-LED structure includes: a first type conductive layer; a second type conductive layer stacked on the first type conductive layer; and a light emitting layer formed between the first type conductive layer and the second type conductive layer. The light emitting layer extends along a horizontal level away from a top edge of the first type conductive layer and a bottom edge of the second type conductive layer, such that an edge of the light emitting layer does not contact the top edge of the first type conductive layer and the bottom edge of the second type conductive layer. The bottom edge of the second type conductive layer is aligned with the top edge of the first type conductive layer.

According to another aspect of embodiments of the present disclosure, a micro-LED structure includes: a first type conductive layer; a second type conductive layer stacked on the first type conductive layer; and a light emitting layer formed between the first type conductive layer and the second type conductive layer. The light emitting layer extrudes along a horizontal level away from a top edge of the first type conductive layer and a bottom edge of the second type conductive layer, such that an edge of the light emitting layer does not contact the top edge of the first type conductive layer and the bottom edge of the second type conductive layer. A profile of the second type conductive layer perpendicularly projected on a top surface of the first type conductive layer is surrounded by the top edge of the first type conductive layer.

According to another aspect of embodiments of the present disclosure, a micro-LED structure includes a first type conductive layer; a second type conductive layer stacked on the first type conductive layer; and a light emitting layer formed between the first type conductive layer and the second type conductive layer. The light emitting layer extends along a horizontal level away from a top edge of the first type conductive layer and a bottom edge of the second type conductive layer, such that an edge of the light emitting layer does not contact the top edge of the first type conductive layer and the bottom edge of the second type conductive

2

layer. A profile of the first type conductive layer perpendicularly projected on a bottom surface of the second type conductive layer is surrounded by the bottom edge of the second type conductive layer.

According to another aspect of embodiments of the present disclosure, a micro-LED structure includes a first type conductive layer; a second type conductive layer stacked on the first type conductive layer; and a light emitting layer formed between the first type conductive layer and the second type conductive layer. The light emitting layer extends along a horizontal level from an edge of the first type conductive layer. An edge of the light emitting layer is aligned with an edge of the second type conductive layer. The edge of the second type conductive layer extends along the horizontal level away from the edge of the first type conductive layer.

According to another aspect of embodiments of the present disclosure, a micro-LED structure includes: a first type conductive layer; a second type conductive layer stacked on the first type conductive layer; and a light emitting layer formed between the first type conductive layer and the second type conductive layer. The light emitting layer extends along a horizontal level from an edge of the second type conductive layer. An edge of the light emitting layer is aligned with an edge of the first type conductive layer. The edge of the first type conductive layer extends along the horizontal level away from the edge of the second type conductive layer.

According to another aspect of embodiments of the present disclosure, a micro-LED chip includes multiple micro-LEDs. At least one micro-LED of the multiple micro-LEDs includes: a first type conductive layer; a second type conductive layer stacked on the first type conductive layer; and a light emitting layer formed between the first type conductive layer and the second type conductive layer. The light emitting layer is continuously formed on the whole micro-LED chip, the multiple micro-LEDs sharing the light emitting layer.

According to another aspect of embodiments of the present disclosure, a micro-LED chip includes multiple micro-LEDs. At least one micro-LED of the multiple micro-LEDs includes: a first type conductive layer; a second type conductive layer stacked on the first type conductive layer; and a light emitting layer formed between the first type conductive layer and the second type conductive layer. The light emitting layer is continuously formed on the whole micro-LED chip, the multiple micro-LEDs sharing the light emitting layer. An isolation structure is formed between adjacent micro-LEDs, at least a portion of the isolation structure being formed in the light emitting layer.

According to another aspect of embodiments of the present disclosure, a micro-LED chip includes multiple micro-LEDs. At least one micro-LED of the multiple micro-LEDs includes: a first type conductive layer; a second type conductive layer stacked on the first type conductive layer; and a light emitting layer formed between the first type conductive layer and the second type conductive layer. The light emitting layer is continuously formed on the whole chip, the multiple micro-LEDs sharing the light emitting layer. An isolation structure is formed between adjacent micro-LEDs, at least a portion of the isolation structure being formed in the light emitting layer. A top surface of the isolation structure is aligned with a top of the light emitting layer, and a bottom surface of the isolation structure is under the light emitting layer.

According to another aspect of embodiments of the present disclosure, a micro-LED chip includes multiple micro-

According to another aspect of embodiments of the present disclosure, a micro-LED chip includes multiple micro-LEDs. At least one micro-LED of the multiple micro-LEDs includes: a first type conductive layer; a second type conductive layer stacked on the first type conductive layer; and a light emitting layer formed between the first type conductive

According to another aspect of embodiments of the present disclosure, a micro-LED chip includes multiple micro-LEDs. At least one micro-LED of the multiple micro-LEDs includes: a first type conductive layer; a second type conductive layer stacked on the first type conductive layer; and a light emitting layer formed between the first type conductive layer and the second type conductive layer. The light emitting layer is continuously formed on the whole micro-LED chip, the multiple micro-LEDs sharing the light emitting layer. The micro-LED chip further includes: a top spacer formed on a top surface of the light emitting layer; a bottom spacer formed on a bottom surface of the light emitting layer, wherein an edge of the top spacer is aligned

According to another aspect of embodiments of the present disclosure, a micro-LED chip includes multiple micro-LEDs. At least one micro-LED of the multiple micro-LEDs includes: a first type conductive layer; a second type conductive layer stacked on the first type conductive layer; and a light emitting layer formed between the first type conductive layer and the second type conductive layer. The light emitting layer is continuously formed on the whole micro-LED chip, the multiple micro-LEDs sharing the light emitting layer. The micro-LED chip further includes: a top spacer formed on a top surface of the light emitting layer; a bottom spacer formed on a bottom surface of the light emitting layer, wherein an edge of the top spacer is aligned with an edge of the light emitting layer, and an edge of the bottom spacer is aligned with the edge of the light emitting layer; and an isolation structure formed between adjacent micro-LEDs, wherein at least a portion of the isolation structure is formed in the light emitting layer, a top surface of the isolation structure is above the light emitting layer and under a top surface of the top spacer, and a bottom surface

According to another aspect of embodiments of the present disclosure, a micro-LED chip includes multiple micro-LEDs. At least one micro-LED of the multiple micro-LEDs includes: a first type conductive layer; a second type conductive layer stacked on the first type conductive layer; and a light emitting layer formed between the first type conductive layer and the second type conductive layer. The light emitting layer is continuously formed on the whole micro-LED chip, the multiple micro-LEDs sharing the light emitting layer. The micro-LED chip further includes: a top spacer formed on a top surface of the light emitting layer; a bottom spacer formed on a bottom surface of the light emitting layer, wherein an edge of the top spacer is aligned with an edge of the light emitting layer, and an edge of the bottom spacer is aligned with the edge of the light emitting layer; and an isolation structure formed between adjacent micro-LEDs, wherein at least a portion of the isolation structure is formed in the light emitting layer, a bottom surface of the isolation structure is under the light emitting

According to another aspect of embodiments of the present disclosure, a micro-LED chip includes multiple micro-LEDs. At least one micro-LED of the multiple micro-LEDs includes: a first type conductive layer; a second type con-

According to another aspect of embodiments of the present disclosure, a micro-LED chip includes multiple micro-LEDs. At least one micro-LED of the multiple micro-LEDs includes: a first type conductive layer; a second type conductive layer stacked on the first type conductive layer; and a light emitting layer formed between the first type conductive layer and the second type conductive layer. The light emitting layer is continuously formed on the whole micro-

LED chip, the multiple micro-LEDs sharing the light emitting layer. A bottom edge of the second type conductive layer is aligned with a top edge of the first type conductive layer.

According to another aspect of embodiments of the present disclosure, a micro-LED chip includes multiple micro-LEDs. At least one micro-LED of the multiple micro-LEDs includes: a first type conductive layer; a second type conductive layer stacked on the first type conductive layer; and a light emitting layer formed between the first type conductive layer and the second type conductive layer. The light emitting layer is continuously formed on the whole micro-LED chip, the multiple micro-LEDs sharing the light emitting layer. A profile of the second type conductive layer perpendicularly projected on a top surface of the first type conductive layer is surrounded by an edge of the first type conductive layer.

According to another aspect of embodiments of the present disclosure, a micro-LED chip includes multiple micro-LEDs. At least one micro-LED of the multiple micro-LEDs includes: a first type conductive layer; a second type conductive layer stacked on the first type conductive layer; and a light emitting layer formed between the first type conductive layer and the second type conductive layer. The light emitting layer is continuously formed on the whole micro-LED chip, the multiple micro-LEDs sharing the light emitting layer. A profile of the first type conductive layer perpendicularly projected on a bottom surface of the second type conductive layer is surrounded by an edge of the second type conductive layer.

According to another aspect of embodiments of the present disclosure, a micro-LED chip includes multiple micro-LEDs. At least one micro-LED of the multiple micro-LEDs includes: a first type conductive layer; a second type conductive layer stacked on the first type conductive layer; and a light emitting layer formed between the first type conductive layer and the second type conductive layer, at least one part of the light emitting layer being formed between adjacent micro-LEDs. the micro-LED chip further comprises a metal layer formed on the light emitting layer between the adjacent micro-LEDs.

According to another aspect of embodiments of the present disclosure, a micro-LED chip includes multiple micro-LEDs. At least one micro-LED includes: a first type conductive layer; a second type conductive layer stacked on the first type conductive layer; and a light emitting layer formed between the first type conductive layer and the second type conductive layer. The light emitting layer extends along a horizontal level from a top edge of the first type conductive layer and from a bottom edge of the second type conductive layer, such that an edge of the light emitting layer does not contact the top edge of the first type conductive layer and the bottom edge of the second type conductive layer, and the bottom edge of the second type conductive layer is aligned with the top edge of the first type conductive layer. The micro-LED chip further includes a metal layer formed on the light emitting layer between adjacent micro-LEDs.

According to another aspect of embodiments of the present disclosure, a micro-LED chip includes multiple micro-LEDs. At least one micro-LED of the multiple micro-LEDs includes: a first type conductive layer; a second type conductive layer stacked on the first type conductive layer; and a light emitting layer formed between the first type conductive layer and the second type conductive layer. The light emitting layer extends along a horizontal level from a top edge of the first type conductive layer and a bottom edge of the second type conductive layer, such that an edge of the

light emitting layer does not contact the top edge of the first type conductive layer and the bottom edge of the second type conductive layer, and a profile of the second type conductive layer perpendicularly projected on a top surface of the first type conductive layer is surrounded by an edge of the first type conductive layer. The micro-LED chip further includes a metal layer formed on a portion of the light emitting layer that extends from the top edge of the first type conductive layer.

According to another aspect of embodiments of the present disclosure, a micro-LED chip includes multiple micro-LEDs. At least one micro-LED of the multiple micro-LEDs includes: a first type conductive layer; a second type conductive layer stacked on the first type conductive layer; and a light emitting layer formed between the first type conductive layer and the second type conductive layer. The light emitting layer extends along a horizontal level from a top edge of the first type conductive layer and a bottom edge of the second type conductive layer, such that an edge of the light emitting layer does not touch the top edge of the first type conductive layer and the bottom edge of the second type conductive layer. A profile of the first type conductive layer perpendicularly projected on a bottom surface of the second type conductive layer is surrounded by the bottom edge of the second type conductive layer. The micro-LED chip further includes a metal layer formed on a portion of the light emitting layer that extends from the second type conductive layer.

BRIEF DESCRIPTION OF DRAWING(S)

FIG. 1A is a cross-sectional view of a micro-LED structure, according to a first embodiment of the present disclosure.

FIG. 1B is a cross-sectional view of a micro-LED structure, according to a first variation of the first embodiment of the present disclosure.

FIG. 1C is a cross-sectional view of a micro-LED structure, according to a second variation of the first embodiment of the present disclosure.

FIG. 1D is a cross-sectional view of a micro-LED structure, according to a third variation of the first embodiment of the present disclosure.

FIG. 2A is a cross-sectional view of a micro-LED structure, according to a second embodiment of the present disclosure.

FIG. 2B is a cross-sectional view of a micro-LED structure, according to a first variation of the second embodiment of the present disclosure.

FIG. 2C is a cross-sectional view of a micro-LED structure, according to a second variation of the second embodiment of the present disclosure.

FIG. 2D is a cross-sectional view of a micro-LED structure, according to a third variation of the second embodiment of the present disclosure.

FIG. 3A is a cross-sectional view of a micro-LED structure, according to a third embodiment of the present disclosure.

FIG. 3B is a cross-sectional view of a micro-LED structure, according to a first variation of the third embodiment of the present disclosure.

FIG. 3C is a cross-sectional view of a micro-LED structure, according to a second variation of the third embodiment of the present disclosure.

FIG. 3D is a cross-sectional view of a micro-LED structure, according to a third variation of the third embodiment of the present disclosure.

13

FIG. 14C is a cross-sectional view of a micro-LED chip, according to another variation of the fourteenth embodiment of the present disclosure.

FIG. 14D is a cross-sectional view of a micro-LED chip, according to another variation of the fourteenth embodiment of the present disclosure.

FIG. 14E is a cross-sectional view of a micro-LED chip, according to another variation of the fourteenth embodiment of the present disclosure.

FIG. 15A is a cross-sectional view of a micro-LED chip, according to a fifteenth embodiment of the present disclosure.

FIG. 15B is a cross-sectional view of a micro-LED chip, according to a variation of the fifteenth embodiment of the present disclosure.

FIG. 15C is a cross-sectional view of a micro-LED chip, according to another variation of the fifteenth embodiment of the present disclosure.

FIG. 15D is a cross-sectional view of a micro-LED chip, according to another variation of the fifteenth embodiment of the present disclosure.

FIG. 15E is a cross-sectional view of a micro-LED chip, according to another variation of the fifteenth embodiment of the present disclosure.

FIG. 16A is a cross-sectional view of a micro-LED chip, according to a sixteenth embodiment of the present disclosure.

FIG. 16B is a cross-sectional view of a micro-LED chip, according to a variation of the sixteenth embodiment of the present disclosure.

FIG. 17A is a cross-sectional view of a micro-LED chip, according to a seventeenth embodiment of the present disclosure.

FIG. 17B is a cross-sectional view of a micro-LED chip, according to a variation of the seventeenth embodiment of the present disclosure.

FIG. 18 is a cross-sectional view of a micro-LED structure, according to a comparative example.

DETAILED DESCRIPTION

The text below provides a detailed description of the present disclosure in conjunction with specific embodiments illustrated in the attached drawings. However, these embodiments do not limit the present disclosure. The scope of protection for the present disclosure covers changes made to the structure, method, or function by persons having ordinary skill in the art on the basis of these embodiments.

To facilitate the presentation of the drawings in the present disclosure, the sizes of certain structures or portions may be enlarged relative to other structures or portions. Therefore, the drawings in the present application are only for the purpose of illustrating the basic structure of the subject matter of the present application. The same numbers in different drawings represent the same or similar elements unless otherwise represented.

Additionally, terms in the text indicating relative spatial position, such as “front,” “back,” “upper,” “lower,” “above,” “below,” and so forth, are used for explanatory purposes in describing the relationship between a unit or feature depicted in a drawing and another unit or feature therein. Terms indicating relative spatial position may refer to positions other than those depicted in the drawings when a device is being used or operated. For example, if a device shown in a drawing is flipped over, a unit which is described as being positioned “below” or “under” another unit or feature will be located “above” the other unit or feature.

14

Therefore, the illustrative term “below” may include positions both above and below. A device may be oriented in other ways (rotated 90 degrees or facing another direction), and descriptive terms that appear in the text and are related to space should be interpreted accordingly. When a component or layer is said to be “above” another member or layer or “connected to” another member or layer, it may be directly above the other member or layer or directly connected to the other member or layer, or there may be an intermediate component or layer.

First Embodiment

FIG. 1A is a cross-sectional view of a micro-light emitting diode (micro-LED) structure 1000, according to a first embodiment of the present disclosure. As illustrated in FIG. 1A, the micro-LED structure 1000 includes a first type conductive layer 101, a second type conductive layer 102 stacked on the first type conductive layer 101, and a light emitting layer 103 formed between the first type conductive layer 101 and the second type conductive layer 102. The light emitting layer 103 extends along a horizontal level away from a top edge 101a of the first type conductive layer 101 and a bottom edge 102a of the second type conductive layer 102, such that an edge 103a of the light emitting layer 103 and does not contact the top edge 101a of the first type conductive layer 101 and the bottom edge 102a of the second type conductive layer 102. The bottom edge 102a of the second type conductive layer 102 is aligned with the top edge 101a of the first type conductive layer 101.

The first type conductive layer 101 and the second type conductive layer 102 may be any type of conductive layers. In one embodiment, the first type conductive layer 101 may be an n-type conductive semiconductor containing one or more n-type dopants, and the second type conductive layer 102 may be a p-type conductive semiconductor layer containing one or more p-type dopants. In another embodiment, the first type conductive layer 101 may be a p-type conductive semiconductor, and the second type conductive layer 102 may be an n-type conductive semiconductor layer. As illustrated in FIG. 1A, a top area of the first type conductive layer 101 is larger than a bottom area of the first type conductive layer 101. A top area of the second type conductive layer 102 is smaller than a bottom area of the second type conductive layer 102.

The light emitting layer 103 may have a quantum well structure in which quantum well layers and barrier layers are alternately stacked. In one embodiment, the light emitting layer 103 may include one pair of quantum well layers and a barrier layer interposed between the quantum well layers. In another embodiment, the light emitting layer 103 may include multiple pairs of quantum well layers and a barrier layer interposed between adjacent quantum well layers. The quantum well layers are made of, for example, GaAs, AlGaAs, InGaAs, GaAsP, AlGaInP, GaInAsP, GaInP, AlInP, GaP, InP, or the like. The barrier layers are formed of, for example, GaAs, AlGaAs, InGaAs, GaAsP, AlGaInP, GaInAsP, GaInP, AlInP, GaP, InP, or the like.

As illustrated in FIG. 1A, the micro-LED structure 1000 also includes a top spacer 107 formed on a top surface 103b of the light emitting layer 103, and a bottom spacer 108 formed on a bottom surface 103c of the light emitting layer 103. The top spacer 107 and the bottom spacer 108 may be made of GaAs, AlGaAs, InGaAs, GaAsP, AlGaInP, GaInAsP, GaInP, AlInP, GaP, InP or the like. The top spacer 107 and the bottom spacer 108 are configured to control the carrier injection efficiency to improve the performance and

15

reliability of the micro-LED. An edge **107a** of the top spacer **107** and an edge **108a** of the bottom spacer **108** are aligned with the edge **103a** of the light emitting layer **103**. A thickness of the top spacer **107** is larger than a thickness of the light emitting layer **103**. A thickness of the bottom spacer **108** is larger than the thickness of the light emitting layer **103**.

In the embodiment illustrated in FIG. 1A, the micro-LED structure **1000** further includes a reflective structure **104** surrounding the first type conductive layer **101**. The reflective structure **104** is attached on a sidewall surface **101b** of the first type conductive layer **101**. The reflective structure **104** on the sidewall of the first type conductive layer **101** is inclined relative to a surface **110a** of a substrate **110**. An inclined angle of the reflective structure **104** is approximately 30° to approximately 75° relative to the surface **110a** of the substrate **110**. The reflective structure **104** on the sidewall surface **101b** of the first type conductive layer **101** is made of an ODR (omnidirectional reflector) structure or a DBR (distributed bragg reflection) structure. The reflective structure **104** is configured to focus light on the second type conductive layer **102**.

In the embodiment illustrated in FIG. 1A, the micro-LED structure **1000** further includes a bottom connection structure **105** formed under the first type conductive layer **101**, and electrically connected with the first type conductive layer **101**. The bottom connection structure **105** may be formed of electrically conductive material, such as, for example, metal. The bottom connection structure **105** may be reflective.

As illustrated in FIG. 1A, the micro-LED structure **1000** further includes the substrate **110** under the first type conductive layer **101**, and is electrically connected with the bottom connection structure **105** by a connecting pad **106** in the substrate **110**. In some embodiments, the substrate **110** may be made of one of more of the materials from the III-V groups, such as, for example, GaN. In some other embodiments, the substrate **110** may include an IC circuit. The connecting pad **106** may be made of conductive materials, such as, for example, Cu.

As illustrated in FIG. 1A, the micro-LED structure **1000** further includes an isolation layer **109** surrounding the first type conductive layer **101** and under the light emitting layer **103**. The isolation layer **109** may be made of a light absorption material which includes, for example, impurity doped SiO₂ or Si₃N₄.

As illustrated in FIG. 1A, the micro-LED structure **1000** further includes a microlens **111** formed on the second type conductive layer **102** and on a top surface of **107b** the top spacer **107**. The microlens **111** is configured to converge light emitted by the light emitting layer **103**.

First Variation of First Embodiment

FIG. 1B is a cross-sectional view of a micro-LED structure **1001**, according to a first variation of the first embodiment of the present disclosure. The embodiment illustrated in FIG. 1B differs from the embodiment illustrated in FIG. 1A in that the sidewall surface **101b** of the first type conductive layer **101** is curved, and a reflective structure **1041** is formed on the sidewall surface **101b** of the first type conductive layer **101** has a curved surface **104a**. Except for the reflective structure **1041**, the components of the micro-LED structure **1001** illustrated in FIG. 1B are the same as the

16

components of the micro-LED structure **1000** illustrated in FIG. 1A, and therefore detailed descriptions of these components are not repeated.

Second Variation of First Embodiment

FIG. 1C is a cross-sectional view of a micro-LED structure **1002**, according to a second variation of the first embodiment of the present disclosure. The embodiment illustrated in FIG. 1C differs from the embodiment illustrated in FIG. 1A in that a reflective structure **1042** is attached on a bottom surface **101c** of the first type conductive layer **101**. The reflective structure **1042** is electrically conductive. The bottom connection structure **105** is formed at the bottom of the reflective structure **1042**, and is electrically connected with the reflective structure **1042**. The reflective structure **1042** on the bottom surface **101c** of the first type conductive layer **101** may be made of metal. Except for the reflective structure **1042**, the components of the micro-LED structure **1002** illustrated in FIG. 1C are the same as the components of the micro-LED structure **1000** illustrated in FIG. 1A, and therefore detailed descriptions of these components are not repeated.

Third Variation of First Embodiment

FIG. 1D is a cross-sectional view of a micro-LED structure **1003**, according to a third variation of the first embodiment of the present disclosure. The embodiment illustrated in FIG. 1D differs from the embodiment illustrated in FIG. 1A in that a reflective structure **1043** is attached on both of the sidewall surface **101b** and the bottom surface **101c** of the first type conductive layer **101**. The reflective structure **1043** is electrically conductive. The bottom connection structure **105** is formed at the bottom of the reflective structure **1043**, and is electrically connected with the reflective structure **1043**. Except for the reflective structure **1043**, the components of the micro-LED structure **1003** illustrated in FIG. 1D are the same as the components of the micro-LED structure **1000** illustrated in FIG. 1A, and therefore detailed descriptions of these components are not repeated.

Second Embodiment

FIG. 2A is a cross-sectional view of a micro-LED structure **2000**, according to a second embodiment of the present disclosure. The second embodiment illustrated in FIG. 2A differs from the first embodiment illustrated in FIG. 1A in that the bottom edge **102a** of the second type conductive layer **102** is not aligned with the top edge **101a** of the first type conductive layer **101**. Instead, a profile of the second type conductive layer **102** perpendicularly projected on the top surface **101d** of the first type conductive layer **101** is surrounded by the top edge **101a** of the first type conductive layer **101**. The components of the micro-LED structure **2000** of the second embodiment illustrated in FIG. 2A are the same as the components of the micro-LED structure **1000** of the first embodiment illustrated in FIG. 1A, and therefore detailed descriptions of these components are not repeated.

First Variation of Second Embodiment

FIG. 2B is a cross-sectional view of a micro-LED structure **2001**, according to a first variation of the second embodiment of the present disclosure. The embodiment illustrated in FIG. 2B differs from the embodiment illustrated in FIG. 2A in that the sidewall surface **101b** of the first

17

type conductive layer **101** is curved, and the reflective structure **1041** is formed on the sidewall surface **101b** of the first type conductive layer **101** has the curved surface **104a**. Except for the reflective structure **1041**, the components of the micro-LED structure **2001** illustrated in FIG. 2B are the same as the components of the micro-LED structure **2000** illustrated in FIG. 2A, and therefore detailed descriptions of these components are not repeated.

Second Variation of Second Embodiment

FIG. 2C is a cross-sectional view of a micro-LED structure **2002**, according to a second variation of the second embodiment of the present disclosure. The embodiment illustrated in FIG. 2C differs from the embodiment illustrated in FIG. 2A in that the reflective structure **1042** is attached on the bottom surface **101c** of the first type conductive layer **101**. The reflective structure **1042** is electrically conductive. The bottom connection structure **105** is formed at the bottom of the reflective structure **1042**, and is electrically connected with the bottom of the reflective structure **1042**. The reflective structure **1042** on the bottom surface **101c** of the first type conductive layer **101** may be made of metal. Except for the reflective structure **1042**, the components of the micro-LED structure **2002** illustrated in FIG. 2C are the same as the components of the micro-LED structure **2000** illustrated in FIG. 2A, and therefore detailed descriptions of these components are not repeated.

Third Variation of Second Embodiment

FIG. 2D is a cross-sectional view of a micro-LED structure **2003**, according to a third variation of the second embodiment of the present disclosure. The embodiment illustrated in FIG. 2D differs from the embodiment illustrated in FIG. 2A in that the reflective structure **1043** is attached on both of the sidewall surface **101b** and the bottom surface **101c** of the first type conductive layer **101**. The reflective structure **1043** is electrically conductive. The bottom connection structure **105** is formed at the bottom of the reflective structure **1043**, and is electrically connected with the reflective structure **1043**. Except for the reflective structure **1043**, the components of the micro-LED structure **2003** illustrated in FIG. 2D are the same as the components of the micro-LED structure **2000** illustrated in FIG. 2A, and therefore detailed descriptions of these components are not repeated.

Third Embodiment

FIG. 3A is a cross-sectional view of a micro-LED structure **3000**, according to a third embodiment of the present disclosure. The third embodiment illustrated in FIG. 3A differs from the first embodiment illustrated in FIG. 1A in that the bottom edge **102a** of the second type conductive layer **102** is not aligned with the top edge **101a** of the first type conductive layer **101**. Instead, a profile of the first type conductive layer **101** perpendicularly projected on a bottom surface **102b** of the second type conductive layer **102** is surrounded by the bottom edge **102a** of the second type conductive layer **102**. The components of the micro-LED structure **3000** of the third embodiment illustrated in FIG. 3A are the same as the components of the micro-LED structure **1000** of the first embodiment illustrated in FIG. 1A, and therefore detailed descriptions of these components are not repeated.

18

First Variation of Third Embodiment

FIG. 3B is a cross-sectional view of a micro-LED structure **3001**, according to a first variation of the third embodiment of the present disclosure. The embodiment illustrated in FIG. 3B differs from the embodiment illustrated in FIG. 3A in that the sidewall surface **101b** of the first type conductive layer **101** is curved, and the reflective structure **1041** is formed on the sidewall surface **101b** of the first type conductive layer **101** has the curved surface **104a**. Except for the reflective structure **104**, the components of the micro-LED structure **3001** illustrated in FIG. 3B are the same as the components of the micro-LED structure **3000** illustrated in FIG. 3A, and therefore detailed descriptions of these components are not repeated.

Second Variation of Third Embodiment

FIG. 3C is a cross-sectional view of a micro-LED structure **3002**, according to a second variation of the third embodiment of the present disclosure. The embodiment illustrated in FIG. 3C differs from the embodiment illustrated in FIG. 3A in that the reflective structure **1042** is attached on the bottom surface **101c** of the first type conductive layer **101**. The reflective structure **1042** is electrically conductive. The bottom connection structure **105** is formed at the bottom of the reflective structure **1042**, and is electrically connected with the bottom of the reflective structure **1042**. The reflective structure **1042** on the bottom surface **101c** of the first type conductive layer **101** may be made of metal. Except for the reflective structure **1042**, the components of the micro-LED structure **3002** illustrated in FIG. 3C are the same as the components of the micro-LED structure **3000** illustrated in FIG. 3A, and therefore detailed descriptions of these components are not repeated.

Third Variation of Third Embodiment

FIG. 3D is a cross-sectional view of a micro-LED structure **3003**, according to a third variation of the third embodiment of the present disclosure. The embodiment illustrated in FIG. 3D differs from the embodiment illustrated in FIG. 3A in that the reflective structure **1043** is attached on both of the sidewall surface **101b** and the bottom surface **101c** of the first type conductive layer **101**. The reflective structure **1043** is electrically conductive. The bottom connection structure **105** is formed at the bottom of the reflective structure **1043**, and is electrically connected with the reflective structure **1043**. Except for the reflective structure **1042**, the components of the micro-LED structure **3003** illustrated in FIG. 3D are the same as the components of the micro-LED structure **3000** illustrated in FIG. 3A, and therefore detailed descriptions of these components are not repeated.

Fourth Embodiment

FIG. 4A is a cross-sectional view of a micro-LED structure **4000**, according to a fourth embodiment of the present disclosure. The fourth embodiment illustrated in FIG. 4A differs from the third embodiment illustrated in FIG. 3A in that the light emitting layer **103** extends along a horizontal level away from the top edge **101a** of the first type conductive layer **101**, and the edge **103a** of the light emitting layer **103** is aligned with the bottom edge **102a** of the second type conductive layer **102**. In addition, the micro-LED structure **4000** further includes a top isolation layer **114** surrounding the light emitting layer **103**. The top isolation layer **114** may

be made of one or more electrical insulating dielectric materials, such as, for example, SiO_2 , Si_3N_4 , Al_2O_3 , TiO_2 , HfO_2 , AlN or the like. The top isolation layer **114** is configured to isolate the micro-LED structure **4000** from an adjacent micro-LED structure (not shown). The microlens **111** is formed on the second type conductive layer **102** and on a top surface **114a** of the isolation layer **114**. The other components of the micro-LED structure **4000** of the fourth embodiment illustrated in FIG. 4A are the same as the components of the micro-LED structure **3000** of the first embodiment illustrated in FIG. 3A, and therefore detailed descriptions of these components are not repeated.

First Variation of Fourth Embodiment

FIG. 4B is a cross-sectional view of a micro-LED structure **4001**, according to a first variation of the fourth embodiment of the present disclosure. The embodiment illustrated in FIG. 4B differs from the embodiment illustrated in FIG. 4A in that the sidewall surface **101b** of the first type conductive layer **101** is curved, and the reflective structure **1041** is formed on the sidewall surface **101b** of the first type conductive layer **101** and has the curved surface **104a**. Except for the reflective structure **1041**, the components of the micro-LED structure **4001** illustrated in FIG. 4B are the same as the components of the micro-LED structure **4000** illustrated in FIG. 4A, and therefore detailed descriptions of these components are not repeated.

Second Variation of Fourth Embodiment

FIG. 4C is a cross-sectional view of a micro-LED structure **4002**, according to a second variation of the fourth embodiment of the present disclosure. The embodiment illustrated in FIG. 4C differs from the embodiment illustrated in FIG. 4A in that the reflective structure **1042** is attached on the bottom surface **101c** of the first type conductive layer **101**. The reflective structure **1042** is electrically conductive. The bottom connection structure **105** is formed at the bottom of the reflective structure **1042**, and is electrically connected with the bottom of the reflective structure **1042**. The reflective structure **1042** on the bottom surface **101c** of the first type conductive layer **101** may be made of metal. Except for the reflective structure **1042**, the components of the micro-LED structure **4002** illustrated in FIG. 4C are the same as the components of the micro-LED structure **4000** illustrated in FIG. 4A, and therefore detailed descriptions of these components are not repeated.

Third Variation of Fourth Embodiment

FIG. 4D is a cross-sectional view of a micro-LED structure **4003**, according to a third variation of the fourth embodiment of the present disclosure. The embodiment illustrated in FIG. 4D differs from the embodiment illustrated in FIG. 4A in that the reflective structure **1043** is attached on both of the sidewall surface **101b** and the bottom surface **101c** of the first type conductive layer **101**. The reflective structure **1043** is electrically conductive. The bottom connection structure **105** is formed at the bottom of the reflective structure **1043**, and is electrically connected with the reflective structure **1043**. Except for the reflective structure **1042**, the components of the micro-LED structure **4003** illustrated in FIG. 4D are the same as the components of the micro-LED structure **4000** illustrated in FIG. 4A, and therefore detailed descriptions of these components are not repeated.

Fifth Embodiment

FIG. 5A is a cross-sectional view of a micro-LED structure **5000**, according to a fifth embodiment of the present disclosure. The fifth embodiment illustrated in FIG. 5A differs from the second embodiment illustrated in FIG. 2A in that the light emitting layer **103** extends along a horizontal level from the bottom edge **102a** of the second type conductive layer **102**, and the edge **103a** of the light emitting layer **103** is aligned with the top edge **101a** of the first type conductive layer **101**. In addition, the micro-LED structure **4000** further includes the top isolation layer **114** surrounding the light emitting layer **103**. The microlens **111** is formed on the second type conductive layer **102** and on the top surface **114a** of the isolation layer **114**. The other components of the micro-LED structure **5000** of the fourth embodiment illustrated in FIG. 5A are the same as the components of the micro-LED structure **2000** of the second embodiment illustrated in FIG. 2A, and therefore detailed descriptions of these components are not repeated.

First Variation of Fifth Embodiment

FIG. 5B is a cross-sectional view of a micro-LED structure **5001**, according to a first variation of the fifth embodiment of the present disclosure. The embodiment illustrated in FIG. 5B differs from the embodiment illustrated in FIG. 5A in that the sidewall surface **101b** of the first type conductive layer **101** is curved, and the reflective structure **1041** formed on the sidewall surface **101b** of the first type conductive layer **101** has the curved surface **104a**. Except for the reflective structure **1041**, the components of the micro-LED structure **5001** illustrated in FIG. 5B are the same as the components of the micro-LED structure **5000** illustrated in FIG. 5A, and therefore detailed descriptions of these components are not repeated.

Second Variation of Fifth Embodiment

FIG. 5C is a cross-sectional view of a micro-LED structure **5002**, according to a second variation of the fifth embodiment of the present disclosure. The embodiment illustrated in FIG. 5C differs from the embodiment illustrated in FIG. 5A in that the reflective structure **1042** is attached on the bottom surface **101c** of the first type conductive layer **101**. The reflective structure **1042** is electrically conductive. The bottom connection structure **105** is formed at the bottom of the reflective structure **1042**, and is electrically connected with the bottom of the reflective structure **1042**. The reflective structure **1042** on the bottom surface **101c** of the first type conductive layer **101** is made of metal. Except for the reflective structure **1042**, the components of the micro-LED structure **5002** illustrated in FIG. 5C are the same as the components of the micro-LED structure **5000** illustrated in FIG. 5A, and therefore detailed descriptions of these components are not repeated.

Third Variation of Fifth Embodiment

FIG. 5D is a cross-sectional view of a micro-LED structure **5003**, according to a third variation of the fifth embodiment of the present disclosure. The embodiment illustrated in FIG. 5D differs from the embodiment illustrated in FIG. 5A in that the reflective structure **1043** is attached both on the sidewall surface **101b** and the bottom surface **101c** of the first type conductive layer **101**. The reflective structure **1043** is electrically conductive. The bottom connection structure

21

105 is formed at the bottom of the reflective structure **1043**, and is electrically connected with the reflective structure **1043**. Except for the reflective structure **1042**, the components of the micro-LED structure **5003** illustrated in FIG. 5D are the same as the components of the micro-LED structure **5000** illustrated in FIG. 5A, and therefore detailed descriptions of these components are not repeated.

Sixth Embodiment

FIG. 6A is a cross-sectional view of a micro-LED chip **6000**, according to a sixth embodiment of the present disclosure. The micro-LED chip **6000** may include multiple micro-LEDs. At least one of the micro-LEDs included in the micro-LED chip **6000** may have any one of the micro-LED structures **1000**, **1001**, . . . **3003** described above.

In the embodiment illustrated in FIG. 6A, the micro-LED chip **6000** includes two micro-LEDs **610** and **620**. Each one of the micro-LEDs **610** and **620** includes the micro-LED structure **1000** described in the first embodiment illustrated in FIG. 1A. Hereinafter in the description and in FIG. 6A, the micro-LEDs **610** and **620** in FIG. 6A are also referred to as micro-LEDs **610(1000)** and **620(1000)**.

As illustrated in FIG. 6A, each one of the micro-LEDs **610(1000)** and **620(1000)** includes the first type conductive layer **101**, the second type conductive layer **102** stacked on the first type conductive layer **101**, and the light emitting layer **103** formed between the first type conductive layer **101** and the second type conductive layer **102**. The light emitting layer **103** is continuously formed on the whole micro-LED chip **6000**. The first and second micro-LEDs **610(1000)** and **620(1000)** share the light emitting layer **103**. The light emitting layer **103** extends along a horizontal level away from the top edge **101a** of the first type conductive layer **101** and the bottom edge **102a** of the second type conductive layer **102**, such that the edge **103a** of the light emitting layer **103** does not contact the top edge **101a** of the first type conductive layer **101** and the bottom edge **102a** of the second type conductive layer **102**. The bottom edge **102a** of the second type conductive layer **102** is aligned with the top edge **101a** of the first type conductive layer **101**.

As illustrated in FIG. 6A, each one of the micro-LEDs **610(1000)** and **620(1000)** also includes the top spacer **107** formed on the top surface **103b** of the light emitting layer **103**, and the bottom spacer **108** formed on the bottom surface **103c** of the light emitting layer **103**. Both of the top spacer **107** and the bottom spacer **108** are continuously formed on the whole micro-LED chip **6000**, and are shared by the first and second micro-LEDs **610(1000)** and **620(1000)**. The edge **107a** of the top spacer **107** is aligned with the edge **103a** of the light emitting layer **103**. The edge **108a** of the bottom spacer **108** is aligned with the edge **103a** of the light emitting layer **103**.

As illustrated in FIG. 6A, each one of the micro-LEDs **610(1000)** and **620(1000)** also includes the reflective structure **104** surrounding the first type conductive layer **101**, the bottom connection structure **105** formed under the first type conductive layer **101**, the connecting pad **106** formed in the substrate **110**, the isolation layer **109** surrounding the first type conductive layer **101** and under the light emitting layer **103**, the substrate **110** under the first type conductive layer **101** and electrically connected with the bottom connection structure **105** by the connecting pad **106**, and the microlens **111** formed on the second type conductive layer **102** and on the top surface of **107b** the top spacer **107**.

Except for the light emitting layer **103**, the top spacer **107**, and the bottom spacer **108** that are continuously formed on

22

the whole micro-LED chip **6000** and are shared by the first and second micro-LEDs **610(1000)** and **620(1000)**, the components of the first and second micro-LEDs **610(1000)** and **620(1000)** of the sixth embodiment illustrated in FIG. 6A are the same as the components of the micro-LED structure **1000** of the first embodiment illustrated in FIG. 1A. Therefore, detailed descriptions of these components are not repeated.

First Variation of Sixth Embodiment

FIG. 6B is a cross-sectional view of a micro-LED chip **6001**, according to a first variation of the sixth embodiment of the present disclosure. The first variation of the sixth embodiment illustrated in FIG. 6B differs from the sixth embodiment illustrated in FIG. 6A in that each one of the micro-LEDs **610** and **620** in the micro-LED chip **6001** includes the micro-LED structure **1001** described in the first variation of the first embodiment illustrated in FIG. 1B. Therefore, the micro-LEDs **610** and **620** in FIG. 6B are also referred to as micro-LEDs **610(1001)** and **620(1001)**.

More specifically, as illustrated in FIG. 6B, the sidewall surface **101b** of the first type conductive layer **101** is curved, and the reflective structure **1041** formed on the sidewall surface **101b** of the first type conductive layer **101** has the curved surface **104a**. Except for the reflective structure **1041**, the components of the micro-LED chip **6001** illustrated in FIG. 6B are the same as the components of the micro-LED chip **6000** illustrated in FIG. 6A, and therefore detailed descriptions of these components are not repeated.

Second Variation of Sixth Embodiment

FIG. 6C is a cross-sectional view of a micro-LED chip **6002**, according to a second variation of the sixth embodiment of the present disclosure. The second variation of the sixth embodiment illustrated in FIG. 6C differs from the sixth embodiment illustrated in FIG. 6A in that each one of the micro-LEDs **610** and **620** in the micro-LED chip **6002** includes the micro-LED structure **1002** described in the second variation of the first embodiment illustrated in FIG. 1C. Therefore, the micro-LEDs **610** and **620** in FIG. 6C are referred to as micro-LEDs **610(1002)** and **620(1002)**.

More specifically, as illustrated in FIG. 6C, the reflective structure **1042** is attached on the bottom surface **101c** of the first type conductive layer **101**. The reflective structure **1042** is electrically conductive. The bottom connection structure **105** is formed at the bottom of the reflective structure **1042**, and is electrically connected with the bottom of the reflective structure **1042**. The reflective structure **1042** on the bottom surface **101c** of the first type conductive layer **01** may be made of metal. Except for the reflective structure **1042**, the components of the micro-LED chip **6002** illustrated in FIG. 6C are the same as the components of the micro-LED chip **6000** illustrated in FIG. 6A, and therefore detailed descriptions of these components are not repeated.

Third Variation of Sixth Embodiment

FIG. 6D is a cross-sectional view of a micro-LED chip **6003**, according to a third variation of the sixth embodiment of the present disclosure. The embodiment illustrated in FIG. 6D differs from the embodiment illustrated in FIG. 6A in that each one of the micro-LEDs **610** and **620** in the micro-LED chip **6003** includes the micro-LED structure **1003** described in the third variation of the first embodiment

illustrated in FIG. 1D. Therefore, the micro-LEDs **610** and **620** in FIG. 6C are referred to as micro-LEDs **610(1003)** and **620(1003)**.

More specifically, as illustrated in FIG. 6D, the reflective structure **1043** is attached both on the sidewall surface **101b** and the bottom surface **101c** of the first type conductive layer **101**. The reflective structure **1043** is electrically conductive. The bottom connection structure **105** is formed at the bottom of the reflective structure **1043**, and is electrically connected with the reflective structure **1043**. Except for the reflective structure **1043**, the components of the micro-LED chip **6003** illustrated in FIG. 6D are the same as the components of the micro-LED chip **6000** illustrated in FIG. 6A, and therefore detailed descriptions of these components are not repeated.

Fourth Variation of Sixth Embodiment

FIG. 6E is a cross-sectional view of a micro-LED chip **6004**, according to a fourth variation of the sixth embodiment of the present disclosure. The embodiment illustrated in FIG. 6E differs from the embodiment illustrated in FIG. 6A in that each one of the micro-LEDs **610** and **620** in the micro-LED chip **6004** includes the micro-LED structure **2000** described in the second embodiment illustrated in FIG. 2A. Therefore, the micro-LEDs **610** and **620** in FIG. 6E are referred to as micro-LEDs **610(2000)** and **620(2000)**.

More specifically, as illustrated in FIG. 6E, the bottom edge **102a** of the second type conductive layer **102** is not aligned with the top edge **101a** of the first type conductive layer **101**. Instead, a profile of the second type conductive layer **102** perpendicularly projected on the top surface **101d** of the first type conductive layer **101** is surrounded by the top edge **101a** of the first type conductive layer **101**. The components of the micro-LED chip **6004** of the embodiment illustrated in FIG. 6E are the same as the components of the micro-LED chip **6000** of the embodiment illustrated in FIG. 6A, and therefore detailed descriptions of these components are not repeated.

In the embodiment illustrated in FIG. 6E, each of the micro-LEDs **610** and **620** in the micro-LED chip **6004** includes the micro-LED structure **2000** described in the second embodiment illustrated in FIG. 2A. Alternatively, in other embodiments, each of the micro-LEDs **610** and **620** in the micro-LED chip **6004** may include the micro-LED structure **2001**, **2002**, or **2003** described in the first, second, or third variation of the second embodiment illustrated in FIG. 2B, 2C, or 2D, respectively. For example, FIG. 6G is a cross-sectional view of a micro-LED chip **6006** which includes micro-LEDs **610(2001)** and **620(2001)** each having the micro-LED structure **2001** described in the first variation of the second embodiment illustrated in FIG. 2B; FIG. 6H is a cross-sectional view of a micro-LED chip **6007** which includes micro-LEDs **610(2002)** and **620(2002)** each having the micro-LED structure **2002** described in the second variation of the second embodiment illustrated in FIG. 2C; FIG. 6I is a cross-sectional view of a micro-LED chip **6008** which includes micro-LEDs **610(2003)** and **620(2003)** each having the micro-LED structure **2003** described in the third variation of the second embodiment illustrated in FIG. 2D.

Fifth Variation of Sixth Embodiment

FIG. 6F is a cross-sectional view of a micro-LED chip **6005**, according to a fifth variation of the sixth embodiment of the present disclosure. The embodiment illustrated in FIG. 6F differs from the embodiment illustrated in FIG. 6A

in that each one of the micro-LEDs **610** and **620** in the micro-LED chip **6005** includes the micro-LED structure **3000** described in the second embodiment illustrated in FIG. 3A. Therefore, the micro-LEDs **610** and **620** in FIG. 6F are referred to as micro-LEDs **610(3000)** and **620(3000)**.

More specifically, as illustrated in FIG. 6F, the bottom edge **102a** of the second type conductive layer **102** is not aligned with the top edge **101a** of the first type conductive layer **101**. Instead, a profile of the first type conductive layer **101** perpendicularly projected on the bottom surface **102b** of the second type conductive layer **102** is surrounded by the bottom edge **102a** of the second type conductive layer **102**. The components of the micro-LED chip **6005** of the embodiment illustrated in FIG. 6F are the same as the components of micro-LED chip **6000** of the embodiment illustrated in FIG. 6A, and therefore detailed descriptions of these components are not repeated.

In the embodiment illustrated in FIG. 6F, each of the micro-LEDs **610** and **620** in the micro-LED chip **6005** includes the micro-LED structure **3000** described in the third embodiment illustrated in FIG. 3A. Alternatively, in other embodiments, each of the micro-LEDs **610** and **620** in the micro-LED chip **6005** may include the micro-LED structure **3001**, **3002**, or **3003** described in the first, second, or third variation of the third embodiment illustrated in FIG. 3B, 3C, or 3D, respectively. For example, FIG. 6J is a cross-sectional view of a micro-LED chip **6009** which includes micro-LEDs **610(3001)** and **620(3001)** each having the micro-LED structure **3001** described in the first variation of the third embodiment illustrated in FIG. 3B; FIG. 6K is a cross-sectional view of a micro-LED chip **6010** which includes micro-LEDs **610(3002)** and **620(3002)** each having the micro-LED structure **3002** described in the second variation of the third embodiment illustrated in FIG. 3C; and FIG. 6L is a cross-sectional view of a micro-LED chip **6011** which includes micro-LEDs **610(3003)** and **620(3003)** each having the micro-LED structure **3003** described in the third variation of the second embodiment illustrated in FIG. 3D.

Seventh Embodiment

FIG. 7 is a cross-sectional view of a micro-LED chip **7000**, according to a seventh embodiment of the present disclosure. The micro-LED chip **7000** may include multiple micro-LEDs. At least one of the micro-LEDs included in the micro-LED chip **7000** may have any one of the micro-LED structures **1000**, **1001**, . . . **3003** described above.

The micro-LED chip **7000** illustrated in FIG. 7 differs from the micro-LED chip **6000** illustrated in FIG. 6A in that the micro-LED chip **7000** further includes an isolation structure **112** formed between adjacent micro-LEDs **610(1000)** and **620(1000)**. The isolation structure **112** may be made of a light absorption material, which may be a dielectric material including, for example, impurity doped SiO₂ or Si₃N₄. In other embodiments, the isolation structure **112** may be made of a reflective material such as, for example, metal. The isolation structure **112** is configured to electrically isolate the adjacent micro-LEDs **610(1000)** and **620(1000)** from each other.

At least a portion of the isolation structure **112** is formed in the light emitting layer **103**. In the embodiment illustrated in FIG. 7, a top surface **112a** of the isolation structure **112** is aligned with the top surface **103b** of the light emitting layer **103**, and a bottom surface **112b** of the isolation structure **112** is aligned with the bottom surface **103c** of the light emitting layer **103**. The isolation structure **112** may be formed surrounding at least one of the micro-LEDs **610**

25

(1000) and 620(1000). Additionally or alternatively, at least a portion of the isolation structure 112 may be formed in the top spacer 107 or the bottom spacer 108.

Except for the isolation structure, the components of the micro-LED chip 7000 of the embodiment illustrated in FIG. 7 are the same as the components of the micro-LED chip 6000 of the embodiment illustrated in FIG. 6A, and therefore detailed descriptions of these components are not repeated.

Eighth Embodiment

FIG. 8A is a cross-sectional view of a micro-LED chip 8000, according to an eighth embodiment of the present disclosure. The micro-LED chip 8000 illustrated in FIG. 8A differs from the micro-LED chip 7000 illustrated in FIG. 7 in that the bottom surface 112b of the isolation structure 112 is below the light emitting layer 103. More specifically, as illustrated in FIG. 8A, the bottom surface 112b of the isolation structure 112 is below the bottom surface 103c of the light emitting layer 103 and above the bottom surface 108b of the bottom spacer 108. Except for the isolation structure 112, the components of the micro-LED chip 8000 illustrated in FIG. 8A are the same as the components of the micro-LED chip 7000 illustrated in FIG. 7, and therefore detailed descriptions of these components are not repeated.

First Variation of Eighth Embodiment

FIG. 8B is a cross-sectional view of a micro-LED chip 8001, according to a first variation of the eighth embodiment of the present disclosure. The micro-LED chip 8001 illustrated in FIG. 8B differs from the micro-LED chip 8000 illustrated in FIG. 8A in that the bottom surface 112b of the isolation structure 112 is aligned with the bottom surface 108b of the bottom spacer 108. Except for the isolation structure 112, the components of the micro-LED chip 8001 illustrated in FIG. 8B are the same as the components of the micro-LED chip 8000 illustrated in FIG. 8A, and therefore detailed descriptions of these components are not repeated.

Second Variation of Eighth Embodiment

FIG. 8C is a cross-sectional view of a micro-LED chip 8002, according to a second variation of the eighth embodiment of the present disclosure. The micro-LED chip 8002 illustrated in FIG. 8C differs from the micro-LED chip 8000 illustrated in FIG. 8A in that the bottom surface 112b of the isolation structure 112 is below the bottom surface 108b of the bottom spacer 108 and is disposed in the isolation layer 109. Except for the isolation structure 112, the components of the micro-LED chip 8002 illustrated in FIG. 8C are the same as the components of the micro-LED chip 8000 illustrated in FIG. 8A, and therefore detailed descriptions of these components are not repeated.

In the embodiments illustrated in FIGS. 8A-8C, an area of the top surface 112a of the isolation structure 112 is equal to an area of the bottom surface 112b of the isolation structure 112. Alternatively, in other embodiments, the area of the top surface 112a of the isolation structure 112 may be larger or smaller than the area of the bottom surface 112b of the isolation structure 112. Still alternatively, in some embodiments, a cross-sectional area of the isolation structure 112 at an interface between the light emitting layer 103 and the bottom spacer 108, or at the bottom surface 108b of the

26

bottom spacer 108, may be larger than the area of the bottom surface 112b of the isolation structure 112.

Ninth Embodiment

FIG. 9A is a cross-sectional view of a micro-LED chip 9000, according to a ninth embodiment of the present disclosure. The micro-LED chip 9000 illustrated in FIG. 9A differs from the micro-LED chip 7000 illustrated in FIG. 7 in that the top surface 112a of the isolation structure 112 is above the light emitting layer 103. More specifically, as illustrated in FIG. 9A, the top surface 112a of the isolation structure 112 is above the top surface 103b of the light emitting layer 103 and below the top surface 107b of the top spacer 107. Except for the isolation structure 112, the components of the micro-LED chip 9000 illustrated in FIG. 9A are the same as the components of the micro-LED chip 7000 illustrated in FIG. 7, and therefore detailed descriptions of these components are not repeated.

First Variation of Ninth Embodiment

FIG. 9B is a cross-sectional view of a micro-LED chip 9001, according to a first variation of the ninth embodiment of the present disclosure. The micro-LED chip 9001 illustrated in FIG. 9B differs from the micro-LED chip 9000 illustrated in FIG. 9A in that the top surface 112a of the isolation structure 112 is aligned with the top surface 107b of the top spacer 107. Except for the isolation structure 112, the components of the micro-LED chip 9001 illustrated in FIG. 9B are the same as the components of the micro-LED chip 9000 illustrated in FIG. 9A, and therefore detailed descriptions of these components are not repeated.

Second Variation of Ninth Embodiment

FIG. 9C is a cross-sectional view of a micro-LED chip 9002, according to a second variation of the ninth embodiment of the present disclosure. The micro-LED chip 9002 illustrated in FIG. 9C differs from the micro-LED chip 9000 illustrated in FIG. 9A in that the top surface 112a of the isolation structure 112 is above the top surface 107b of the top spacer 107 and between adjacent second type conductive layers 102. Except for the isolation structure 112, the components of the micro-LED chip 9002 illustrated in FIG. 9C are the same as the components of the micro-LED chip 9000 illustrated in FIG. 9A, and therefore detailed descriptions of these components are not repeated.

In the embodiment illustrated in FIGS. 9A-9C, the area of the top surface 112a of the isolation structure 112 is equal to the area of the bottom surface 112b of the isolation structure 112. Alternatively, in other embodiments, the area of the top surface 112a of the isolation structure 112 may be larger or smaller than the area of the bottom surface 112b of the isolation structure 112. Still alternatively, in some embodiments, a cross-sectional area of the isolation structure 112 at the top surface 103b of the light emitting layer 103 or at the top surface 107b of the top spacer 107 may be larger than the area of the bottom surface 112b of the isolation structure 112.

Tenth Embodiment

FIG. 10A is a cross-sectional view of a micro-LED chip 10000, according to a tenth embodiment of the present disclosure. The micro-LED chip 10000 illustrated in FIG. 10A differs from the micro-LED chip 7000 illustrated in

27

FIG. 7 in that the top surface **112a** of the isolation structure **112** is above the light emitting layer **103**, and the bottom surface **112b** of the isolation structure **112** is below the light emitting layer **103**. More specifically, as illustrated in FIG. 10A, the top surface **112a** of the isolation structure **112** is above the top surface **103b** of the light emitting layer **103** and below the top surface **107b** of the top spacer **107**, and the bottom surface **112b** of the isolation structure **112** is below the bottom surface **103c** of the light emitting layer **103** and above the bottom surface **108b** of the bottom spacer **108**. Except for the isolation structure **112**, the components of the micro-LED chip **10000** illustrated in FIG. 10A are the same as the components of the micro-LED chip **7000** illustrated in FIG. 7, and therefore detailed descriptions of these components are not repeated.

First Variation of Tenth Embodiment

FIG. 10B is a cross-sectional view of a micro-LED chip **10001**, according to a first variation of the tenth embodiment of the present disclosure. The micro-LED chip **10001** illustrated in FIG. 10B differs from the micro-LED chip **10000** illustrated in FIG. 10A in that the bottom surface **112b** of the isolation structure **112** is aligned with the bottom surface **108b** of the bottom spacer **108**. Except for the isolation structure **112**, the components of the micro-LED chip **10001** illustrated in FIG. 10B are the same as the components of the micro-LED chip **10000** illustrated in FIG. 10A, and therefore detailed descriptions of these components are not repeated.

Second Variation of Tenth Embodiment

FIG. 10C is a cross-sectional view of a micro-LED chip **10002**, according to a second variation of the tenth embodiment of the present disclosure. The micro-LED chip **10002** illustrated in FIG. 10C differs from the micro-LED chip **10000** illustrated in FIG. 10A in that the bottom surface **112b** of the isolation structure **112** is below the bottom surface **108b** of the bottom spacer **108** and is disposed in the isolation layer **109**. Except for the isolation structure **112**, the components of the micro-LED chip **10002** illustrated in FIG. 10C are the same as the components of the micro-LED chip **10000** illustrated in FIG. 10A, and therefore detailed descriptions of these components are not repeated.

In the embodiment illustrated in FIGS. 10A-10C, the area of the top surface **112a** of the isolation structure **112** is equal to the area of the bottom surface **112b** of the isolation structure **112**. Alternatively, in other embodiments, the area of the top surface **112a** of the isolation structure **112** may be larger or smaller than the area of the bottom surface **112b** of the isolation structure **112**. Still alternatively, in some embodiments, a cross-sectional area of the isolation structure **112** at the bottom surface **103c** of the light emitting layer **103** or at the bottom surface **108b** of the bottom spacer **108** may be larger than the area of the bottom surface **112b** of the isolation structure **112**.

Eleventh Embodiment

FIG. 11A is a cross-sectional view of a micro-LED chip **11000**, according to an eleventh embodiment of the present disclosure. The micro-LED chip **11000** illustrated in FIG. 11A differs from the micro-LED chip **7000** illustrated in FIG. 7 in that the top surface **112a** of the isolation structure **112** is aligned with the top surface **107b** of the top spacer **107**, and the bottom surface **112b** of the isolation structure

28

112 is below the light emitting layer **103**. More specifically, in the embodiment illustrated in FIG. 11A, the bottom surface **112b** of the isolation structure **112** is below the bottom surface **103c** of the light emitting layer **103** and above the bottom surface **108b** of the bottom spacer **108**. Except for the isolation structure **112**, the components of the micro-LED chip **11000** illustrated in FIG. 11A are the same as the components of the micro-LED chip **7000** illustrated in FIG. 7, and therefore detailed descriptions of these components are not repeated.

First Variation of Eleventh Embodiment

FIG. 11B is a cross-sectional view of a micro-LED chip **11001**, according to a first variation of the eleventh embodiment of the present disclosure. The micro-LED chip **11001** illustrated in FIG. 11B differs from the micro-LED chip **11000** illustrated in FIG. 11A in that the bottom surface **112b** of the isolation structure **112** is aligned with the bottom surface **108b** of the bottom spacer **108**. Except for the isolation structure **112**, the components of the micro-LED chip **11001** illustrated in FIG. 11A are the same as the components of the micro-LED chip **11000** illustrated in FIG. 11A, and therefore detailed descriptions of these components are not repeated.

Second Variation of Eleventh Embodiment

FIG. 11C is a cross-sectional view of a micro-LED chip **11002**, according to a second variation of the eleventh embodiment of the present disclosure. The micro-LED chip **11002** illustrated in FIG. 11C differs from the micro-LED chip **11000** illustrated in FIG. 11A in that the bottom surface **112b** of the isolation structure **112** is below the bottom surface **108b** of the bottom spacer **108** and is disposed in the isolation layer **109**. Except for the isolation structure **112**, the components of the micro-LED chip **11002** illustrated in FIG. 11C are the same as the components of the micro-LED chip **11000** illustrated in FIG. 11A, and therefore detailed descriptions of these components are not repeated.

In the embodiment illustrated in FIGS. 11A-11C, the area of the top surface **112a** of the isolation structure **112** is equal to the area of the bottom surface **112b** of the isolation structure **112**. Alternatively, in other embodiments, the area of the top surface **112a** of the isolation structure **112** may be larger or smaller than the area of the bottom surface **112b** of the isolation structure **112**. Still alternatively, in some embodiments, a cross-sectional area of the isolation structure **112** at the bottom surface **103c** of the light emitting layer **103** or the bottom surface **108b** of the bottom spacer **108** may be larger than the area of the bottom surface **112b** of the isolation structure **112**.

Third Variation of Eleventh Embodiment

FIG. 11D is a cross-sectional view of a micro-LED chip **11003**, according to a third variation of the eleventh embodiment of the present disclosure. The micro-LED chip **11003** illustrated in FIG. 11D differs from the micro-LED chip **11001** illustrated in FIG. 11B in that the area of the top surface **112a** of the isolation structure **112** is larger than an area of the bottom surface **112b** of the isolation structure **112**. Except for the isolation structure **112**, the components of the micro-LED chip **11003** illustrated in FIG. 11D are the same as the components of the micro-LED chip **11001**

29

illustrated in FIG. 11B, and therefore detailed descriptions of these components are not repeated.

Twelfth Embodiment

FIG. 12A is a cross-sectional view of a micro-LED chip 12000, according to a twelfth embodiment of the present disclosure. The micro-LED chip 12000 illustrated in FIG. 12A differs from the micro-LED chip 7000 illustrated in FIG. 7 in that the top surface 112a of the isolation structure 112 is above the top surface 107b of the top spacer 107 and between adjacent second type conductive layers 102, and the bottom surface 112b of the isolation structure 112 is below the light emitting layer 103. More specifically, in the embodiment illustrated in FIG. 12A, the bottom surface 112b of the isolation structure 112 is below the bottom surface 103c of the light emitting layer 103 and above the bottom surface 108b of the bottom spacer 108. Except for the isolation structure 112, the components of the micro-LED chip 12000 illustrated in FIG. 12A are the same as the components of the micro-LED chip 7000 illustrated in FIG. 7, and therefore detailed descriptions of these components are not repeated.

First Variation of Twelfth Embodiment

FIG. 12B is a cross-sectional view of a micro-LED chip 12001, according to a first variation of the twelfth embodiment of the present disclosure. The micro-LED chip 12001 illustrated in FIG. 12B differs from the micro-LED chip 12000 illustrated in FIG. 12A in that the bottom surface 112b of the isolation structure 112 is aligned with the bottom surface 108b of the bottom spacer 108. Except for the isolation structure 112, the components of the micro-LED chip 12001 illustrated in FIG. 12B are the same as the components of the micro-LED chip 12000 illustrated in FIG. 12A, and therefore detailed descriptions of these components are not repeated.

Second Variation of Twelfth Embodiment

FIG. 12C is a cross-sectional view of a micro-LED chip 12002, according to a second variation of the twelfth embodiment of the present disclosure. The micro-LED chip 12002 illustrated in FIG. 12C differs from the micro-LED chip 12000 illustrated in FIG. 12A in that the bottom surface 112b of the isolation structure 112 is below the bottom surface 108b of the bottom spacer 108 and disposed in the isolation layer 109. Except for the isolation structure 112, the components of the micro-LED chip 12002 illustrated in FIG. 12C are the same as the components of the micro-LED chip 12000 illustrated in FIG. 12A, and therefore detailed descriptions of these components are not repeated.

Third Variation of Twelfth Embodiment

FIG. 12D is a cross-sectional view of a micro-LED chip 12003, according to a third variation of the twelfth embodiment of the present disclosure. The micro-LED chip 12003 illustrated in FIG. 12D differs from the micro-LED chip 12002 illustrated in FIG. 12C in that the area of the top surface 112a of the isolation structure 112 is smaller than the area of the bottom surface 112b of the isolation structure 112, and a cross-sectional area 112c of the isolation structure 112 at the bottom surface 103c of the light emitting layer 103 is larger than the area of the bottom surface 112b of the isolation structure 112. Except for the isolation structure

30

112, the components of the micro-LED chip 12003 illustrated in FIG. 12D are the same as the components of the micro-LED chip 12002 illustrated in FIG. 12C, and therefore detailed descriptions of these components are not repeated.

Thirteenth Embodiment

FIG. 13A is a cross-sectional view of a micro-LED chip 13000, according to a thirteenth embodiment of the present disclosure. The micro-LED chip 13000 may include multiple micro-LEDs. At least one of the micro-LEDs included in the micro-LED chip 13000 may have any one of the micro-LED structures 1000, 1001, . . . 3003 described above.

The micro-LED chip 13000 illustrated in FIG. 13A differs from the micro-LED chip 6000 illustrated in FIG. 6A in that the micro-LED chip 13000 further includes a metal layer 113 formed over the light emitting layer 103 between adjacent micro-LEDs 610(1000) and 620(1000). The metal layer 113 is formed over the top surface 103b of the light emitting layer 103 and does not contact the first type conductive layer 101 or the second type conductive layer 102. More specifically, in the embodiment illustrated in FIG. 13A, the top spacer 107 is formed on top of the light emitting layer 103, and the metal layer 113 is formed on the top surface 107b of the top spacer 107.

A lateral dimensional value d1 of the metal layer 113 is not more than a distance d2 between the edge 103a of the light emitting layer 103 and the top edge 101a of the first type conductive layer 101, or the lateral dimensional value d1 of the metal layer 113 is not more than a distance d3 between the edge 103a of the light emitting layer 103 and the bottom edge 102a of the second type conductive layer 102. The lateral dimensional value d1 of the metal layer 113 may be from approximately 2 nm to approximately 10 μm.

In some embodiments, a center point of the metal layer 113 may be aligned with a center point between the adjacent micro-LEDs 610(1000) and 620(1000). Alternatively, in some embodiments, the center point of the metal layer 113 is closer to one of the adjacent micro-LEDs 610(1000) and 620(1000) than the other one of the micro-LEDs 610(1000) and 620(1000).

The metal layer 113 may include a high work function metal material having a work function that matches the work function of a material of the light emitting layer 103. The high work function metal material may include at least one of gold, platinum, palladium, beryllium, cobalt, nickel, or tungsten.

Variation of Thirteenth Embodiment

FIG. 13B is a cross-sectional view of a micro-LED chip 13001, according to a variation of the thirteenth embodiment of the present disclosure. The micro-LED chip 13001 illustrated in FIG. 13B differs from the micro-LED chip 13000 illustrated in FIG. 13A in that the micro-LED chip 13001 includes multiple metal layers 113 formed on the top surface 107b of the top spacer 107 and between adjacent micro-LEDs 610(1000) and 620(1000). The metal layers 113 are arranged in parallel along the top surface 103b of the light emitting layer 103.

In the embodiment illustrated in FIG. 13B, there are three (3) metal layers 113. In other embodiments, the number of the metal layers 113 may be more than three.

In the embodiments illustrated in FIGS. 13A and 13B, each of the micro-LEDs 610 and 620 includes the micro-

31

LED structure **1000** described in the first embodiment illustrated in FIG. 1A. Alternatively, in other embodiments, each of the micro-LEDs **610** and **620** may include any one of the micro-LED structures **1001**, **1002**, and **1003** described in the first, second, and third variations of the first embodiment illustrated in FIGS. 1B, 1C, and 1D, respectively. For example, FIG. 13C is a cross-sectional view of a micro-LED chip **13002** which includes micro-LEDs **610(1001)** and **620(1001)** each having the micro-LED structure **1001** described in the first variation of the first embodiment illustrated in FIG. 1B; FIG. 13D is a cross-sectional view of a micro-LED chip **13003** which includes micro-LEDs **610(1002)** and **620(1002)** each having the micro-LED structure **1002** described in the second variation of the first embodiment illustrated in FIG. 1C; and FIG. 13E is a cross-sectional view of a micro-LED chip **13004** which includes micro-LEDs **610(1003)** and **620(1003)** each having the micro-LED structure **1003** described in the third variation of the first embodiment illustrated in FIG. 1D.

Fourteenth Embodiment

FIG. 14A is a cross-sectional view of a micro-LED chip **14000**, according to a fourteenth embodiment of the present disclosure. The embodiment illustrated in FIG. 14A differs from the embodiment illustrated in FIG. 13A in that micro-LED chip **14000** includes two micro-LEDs **610(2000)** and **620(2000)**, and each one of the micro-LEDs **610(2000)** and **620(2000)** includes the micro-LED structure **2000** described in the second embodiment illustrated in FIG. 2A. The other components of the micro-LED chip **14000** of the embodiment illustrated in FIG. 14A are the same as the components of the micro-LED chip **13000** of the embodiment illustrated in FIG. 13A, and therefore detailed descriptions of these components are not repeated.

Variation of Fourteenth Embodiment

FIG. 14B is a cross-sectional view of a micro-LED chip **14001**, according to a variation of the fourteenth embodiment of the present disclosure. The micro-LED chip **14001** illustrated in FIG. 14B differs from the micro-LED chip **14000** illustrated in FIG. 14A in that the micro-LED chip **14001** includes multiple metal layers **113** formed on the top surface **107b** of the top spacer **107** and between adjacent micro-LEDs **610(2000)** and **620(2000)**. The other components of the micro-LED chip **14001** of the embodiment illustrated in FIG. 14B are the same as the components of the micro-LED chip **13000** of the embodiment illustrated in FIG. 13A, and therefore detailed descriptions of these components are not repeated.

In the embodiments illustrated in FIGS. 14A and 14B, each of the micro-LEDs **610** and **620** includes the micro-LED structure **2000** described in the second embodiment illustrated in FIG. 2A. Alternatively, in other embodiments, each of the micro-LEDs **610** and **620** may include any one of the micro-LED structures **2001**, **2002**, and **2003** described in the first, second, and third variations of the second embodiment illustrated in FIGS. 2B, 2C, and 2D, respectively. For example, FIG. 14C is a cross-sectional view of a micro-LED chip **14002** which includes micro-LEDs **610(2001)** and **620(2001)** each having the micro-LED structure **2001** described in the first variation of the second embodiment illustrated in FIG. 2B; FIG. 14D is a cross-sectional view of a micro-LED chip **14003** which includes micro-LEDs **610(2002)** and **620(2002)** each having the micro-LED structure **2002** described in the second variation of the

32

second embodiment illustrated in FIG. 2C; and FIG. 14E is a cross-sectional view of a micro-LED chip **14004** which includes micro-LEDs **610(2003)** and **620(2003)** each having the micro-LED structure **2003** described in the third variation of the second embodiment illustrated in FIG. 2D.

Fifteenth Embodiment

FIG. 15A is a cross-sectional view of a micro-LED chip **15000**, according to a fifteenth embodiment of the present disclosure. The embodiment illustrated in FIG. 15A differs from the embodiment illustrated in FIG. 13A in that micro-LED chip **15000** includes two micro-LEDs **610(3000)** and **620(3000)**, and each one of the micro-LEDs **610(3000)** and **620(3000)** includes the micro-LED structure **3000** described in the second embodiment illustrated in FIG. 3A. The other components of the micro-LED chip **15000** of the embodiment illustrated in FIG. 15A are the same as the components of the micro-LED chip **13000** of the embodiment illustrated in FIG. 13A, and therefore detailed descriptions of these components are not repeated.

Variation of Fifteenth Embodiment

FIG. 15B is a cross-sectional view of a micro-LED chip **15001**, according to a variation of the fifteenth embodiment of the present disclosure. The micro-LED chip **15001** illustrated in FIG. 15B differs from the micro-LED chip **15000** illustrated in FIG. 15A in that the micro-LED chip **15001** includes multiple metal layers **113** formed on the top surface **107b** of the top spacer **107** and between adjacent micro-LEDs **610(3000)** and **620(3000)**. The components of the micro-LED chip **15001** of the embodiment illustrated in FIG. 15B are the same as the components of the micro-LED chip **13000** of the embodiment illustrated in FIG. 13A, and therefore detailed descriptions of these components are not repeated.

In the embodiments illustrated in FIGS. 15A and 15B, each of the micro-LEDs **610** and **620** includes the micro-LED structure **3000** described in the third embodiment illustrated in FIG. 3A. Alternatively, in other embodiments, each of the micro-LEDs **610** and **620** may include any one of the micro-LED structures **3001**, **3002**, and **3003** described in the first, second, and third variations of the third embodiment illustrated in FIGS. 3B, 3C, and 3D, respectively. For example, FIG. 15C is a cross-sectional view of a micro-LED chip **15002** which includes micro-LEDs **610(3001)** and **620(3001)** each having the micro-LED structures **3001** described in the first variation of the third embodiment illustrated in FIG. 3B; FIG. 15D is a cross-sectional view of a micro-LED chip **15003** which includes micro-LEDs **610(3002)** and **620(3002)** each having the micro-LED structures **3002** described in the second variation of the third embodiment illustrated in FIG. 3C; and FIG. 15E is a cross-sectional view of a micro-LED chip **15004** which includes micro-LEDs **610(3003)** and **620(3003)** each having the micro-LED structures **3003** described in the third variation of the third embodiment illustrated in FIG. 3D.

Sixteenth Embodiment

FIG. 16A is a cross-sectional view of a micro-LED chip **16000**, according to a sixteenth embodiment of the present disclosure. The micro-LED chip **16000** may include multiple micro-LEDs. At least one of the micro-LEDs included

33

in the micro-LED chip **16000** may have any one of the micro-LED structures **4000**, **4001**, . . . **5003** described above.

In the embodiment illustrated in FIG. **16A**, the micro-LED chip **16000** includes two micro-LEDs **610(4000)** and **620(4000)**. Each one of the micro-LEDs **610(4000)** and **620(4000)** includes the micro-LED structure **4000** described in the fourth embodiment illustrated in FIG. **4A**.

As illustrated in FIG. **16A**, each one of the micro-LEDs **610(4000)** and **620(4000)** includes the first type conductive layer **101**, the second type conductive layer **102** stacked on the first type conductive layer **101**, and the light emitting layer **103** formed between the first type conductive layer **101** and the second type conductive layer **102**. The profile of the first type conductive layer **101** perpendicularly projected on the bottom surface **102b** of the second type conductive layer **102** is surrounded by the bottom edge **102a** of the second type conductive layer **102**. The light emitting layer **103** extends along a horizontal level away from the top edge **101a** of the first type conductive layer **101** and the edge **103a** of the light emitting layer **103** is aligned with the bottom edge **102a** of the second type conductive layer **102**.

Each one of the micro-LEDs **610(4000)** and **620(4000)** further includes the top spacer **107** formed on the light emitting layer **103** and the bottom spacer formed under the light emitting layer **103**. Both of the edge **107a** of the top spacer **107** and the edge **108a** of the bottom spacer **108** are aligned with the edge **103a** of the light emitting layer **103**, which is aligned with the bottom edge **102a** of the second type conductive layer **102**.

The micro-LED chip **16000** further includes the top isolation layer **114** surrounding the light emitting layer **103**, and the microlens **111** formed on the second type conductive layer **102** and on a top surface **114a** of the isolation layer **114**.

Except that the edge **103a** of the light emitting layer **103**, the edge **107a** of the top spacer **107**, and the edge **108a** of the bottom spacer **108** are aligned with the bottom edge **102a** of the second type conductive layer **102**, the components of the micro-LED chip **16000** illustrated in FIG. **16A** are the same as the components of micro-LED chip **6005** illustrated in FIG. **6F**, and therefore detailed descriptions of these components are not repeated.

Variation of Sixteenth Embodiment

FIG. **16B** is a cross-sectional view of a micro-LED chip **16001**, according to a variation of the sixteenth embodiment of the present disclosure. The embodiment illustrated in FIG. **16B** differs from the embodiment illustrated in FIG. **16A** in that the micro-LED chip **16001** further includes the metal layer **113** formed on the top surface **114a** of the isolation layer **114** and between adjacent micro-LEDs **610(4000)** and **620(4000)**. In the embodiment illustrated in FIG. **16B**, there is only one metal layer **113** between micro-LEDs **610(4000)** and **620(4000)**. Alternatively, in other embodiments, there are multiple metal layers **113** between micro-LEDs **610(4000)** and **620(4000)**, and the number of the multiple metal layers **113** between micro-LEDs **610(4000)** and **620(4000)** may be more than two. Except for the metal layer **113**, the components of the micro-LED chip **16001** illustrated in FIG. **16B** are the same as the components of the micro-LED chip **16000** illustrated in FIG. **16A**, and therefore detailed descriptions of these components are not repeated.

In the embodiments illustrated in FIGS. **16A** and **16B**, each of the micro-LEDs **610** and **620** includes the micro-

34

LED structure **4000** described in the fourth embodiment illustrated in FIG. **4A**. Alternatively, in other embodiments, each of the micro-LEDs **610** and **620** may include any one of the micro-LED structures **4001**, **4002**, and **4003** described in the first, second, and third variations of the fourth embodiment illustrated in FIGS. **4B**, **4C**, and **4D**, respectively.

Seventeenth Embodiment

FIG. **17A** is a cross-sectional view of a micro-LED chip **17000**, according to a seventeenth embodiment of the present disclosure. The micro-LED chip **17000** includes two micro-LEDs **610(5000)** and **620(5000)**. Each one of the micro-LEDs **610(5000)** and **620(5000)** includes the micro-LED structure **5000** described in the fifth embodiment illustrated in FIG. **5A**.

The micro-LED chip **17000** differs from the micro-LED chip **16000** in that a profile of the second type conductive layer **102** perpendicularly projected on the top surface **101d** of the first type conductive layer **101** is surrounded by the top edge **101a** of the first type conductive layer **101**. In addition, the edge **103a** of the light emitting layer **103**, the edge **107a** of the top spacer **107**, and the edge **108a** of the bottom spacer **108** are aligned with the top edge **101a** of the first type conductive layer **101**.

In addition, the micro-LED chip **17000** further includes the top isolation layer **114** surrounding the light emitting layer **103**. The microlens **111** is formed on the second type conductive layer **102** and on the top surface **114a** of the isolation layer **114**. The other components of the micro-LED chip **17000** of the fourth embodiment illustrated in FIG. **17A** are the same as the components of the micro-LED chip **16000** of the second embodiment illustrated in FIG. **16A**, and therefore detailed descriptions of these components are not repeated.

Variation of Seventeenth Embodiment

FIG. **17B** is a cross-sectional view of a micro-LED chip **17001**, according to a variation of the seventeenth embodiment of the present disclosure. The embodiment illustrated in FIG. **17B** differs from the embodiment illustrated in FIG. **17A** in that the micro-LED chip **17001** further includes the metal layer **113** formed on the top surface **114a** of the isolation layer **114** and between adjacent micro-LEDs **610(5000)** and **620(5000)**. In the embodiment illustrated in FIG. **17B**, there is only one metal layer **113** between micro-LEDs **610(5000)** and **620(5000)**. Alternatively, in other embodiments, there are multiple metal layers **113** between micro-LEDs **610(5000)** and **620(5000)**, and the number of the multiple metal layers **113** between micro-LEDs **610(5000)** and **620(5000)** may be more than two. Except for the metal layer **113**, the components of the micro-LED chip **17001** illustrated in FIG. **17B** are the same as the components of the micro-LED chip **17000** illustrated in FIG. **17A**, and therefore detailed descriptions of these components are not repeated.

In the embodiments illustrated in FIGS. **17A** and **17B**, each of the micro-LEDs **610** and **620** includes the micro-LED structure **5000** described in the fifth embodiment illustrated in FIG. **5A**. Alternatively, in other embodiments, each of the micro-LEDs **610** and **620** may include any one of the micro-LED structures **5001**, **5002**, and **5003** described in the first, second, and third variations of the fifth embodiment illustrated in FIGS. **5B**, **5C**, and **5D**, respectively.

Comparative Example

FIG. **18** is a cross-sectional view of a micro-LED structure **1**, according to a comparative example. The micro-LED

35

structure 1 in the comparative example illustrated in FIG. 18 differs from the micro-LED structure 1000 of the first embodiment illustrated in FIG. 1A in that the light emitting layer 103 does not extend along the horizontal level away from the top edge 101a of the first type conductive layer 101 and the bottom edge 102a of the second type conductive layer 102. Instead, in the comparative example, the edge 103a of the light emitting layer 103 is aligned with both of the top edge 101a of the first type conductive layer 101 and the bottom edge 102a of the second type conductive layer 102.

As explained previously, the light emitting layer 103 may include multiple pairs of quantum well layers. The number of multi-quantum well (MQW) pairs in the light emitting layer 103 is related to exposed sidewall area of the light emitting layer 103 which is generated by, for example, inductively coupled plasma etching. The larger the exposed sidewall area, the more MQW pairs, resulting in larger surface recombination carrier loss.

In the comparative example, the sidewall of the light emitting layer 103 is aligned with the edges 101a and 102a of the first type conductive layer 101 and the second type conductive layer 102. As a result, large surface recombination carrier loss may occur in the micro-LED between the first type conductive layer 101 and the second type conductive layer 102, negatively impacting the light emission efficiency of the micro-LED.

In contrast, in the first embodiment as well as other embodiments of the present disclosure, the light emitting layer 103 extends away from the top edge 101a of the first type conductive layer 101 and the bottom edge 102a of the second type conductive layer 102, such that the edge 103a of the light emitting layer 103 does not contact the top edge 101a of the first type conductive layer 101 and the bottom edge 102a of the second type conductive layer 102. As a result, surface recombination carrier loss may not occur in the micro-LED between the first type conductive layer 101 and the second type conductive layer 102. Consequently, light emission efficiency of the micro-LED will be improved.

While illustrative embodiments have been described herein, the scope of the present disclosure covers any and all embodiments having equivalent elements, modifications, omissions, combinations (e.g., of aspects across various embodiments), adaptations and/or alterations as would be appreciated by those skilled in the art based on the present disclosure. For example, features included in different embodiments shown in different figures may be combined. The limitations in the claims are to be interpreted broadly based on the language employed in the claims and not limited to examples described in the present specification or during the prosecution of the application. The examples are to be construed as non-exclusive. It is intended, therefore, that the specification and examples be considered as illustrative only, with a true scope and spirit being indicated by the following claims and their full scope of equivalents.

What is claimed is:

1. A micro-LED chip including multiple micro-LEDs, wherein,

at least one micro-LED comprises:

a first type conductive layer;

a second type conductive layer stacked on the first type conductive layer; and

a light emitting layer formed between the first type conductive layer and the second type conductive layer, wherein the light emitting layer extends along a horizontal level from a top edge of the first type

36

conductive layer and from a bottom edge of the second type conductive layer, such that an edge of the light emitting layer does not contact the top edge of the first type conductive layer and the bottom edge of the second type conductive layer, and the bottom edge of the second type conductive layer is aligned with the top edge of the first type conductive layer, the micro-LED chip further comprises a metal layer formed on the light emitting layer between adjacent micro-LEDs, and

a lateral dimensional value of the metal layer is not more than a distance between an edge of the light emitting layer and the top edge of the first type conductive layer.

2. The micro-LED chip according to claim 1, wherein the metal layer is formed on a surface of the light emitting layer and does not touch the first type conductive layer or the second type conductive layer.

3. The micro-LED chip according to claim 1, wherein the lateral dimensional value of the metal layer is from 2 nm to 10 μ m.

4. The micro-LED chip according to claim 1, wherein a center point of the metal layer is aligned with a center point between the adjacent micro-LEDs.

5. The micro-LED chip according to claim 1, wherein a center point of the metal layer is near one of the adjacent micro-LEDs.

6. The micro-LED chip according to claim 1, further comprising a plurality of metal layers, and the plurality of metal layers are arranged in parallel along a surface of the light emitting layer.

7. The micro-LED chip according to claim 1, wherein a material of the metal layer includes a high work functional metal material which matches a material of the light emitting layer.

8. The micro-LED chip according to claim 7, wherein the high work functional metal material includes at least one of gold, platinum, palladium, beryllium, cobalt, nickel, or tungsten.

9. The micro-LED chip according to claim 1, wherein the light emitting layer is continuously formed on the chip.

10. The micro-LED chip according to claim 1, wherein the multiple micro-LEDs share the light emitting layer.

11. The micro-LED chip according to claim 1, further comprising:

a top spacer formed on a top surface of the light emitting layer,

a bottom spacer formed on a bottom surface of the light emitting layer,

wherein an edge of the top spacer is aligned with an edge of the light emitting layer, and

an edge of the bottom spacer is aligned with the edge of the light emitting layer.

12. The micro-LED chip according to claim 11, wherein a thickness of the top spacer is larger than a thickness of the light emitting layer, and a thickness of the bottom spacer is larger than the thickness of the light emitting layer.

13. The micro-LED chip according to claim 11, further comprising, in the at least one micro-LED, a microlens formed on the second type conductive layer and on a top surface of the top spacer.

14. The micro-LED chip according to claim 1, wherein a top area of the first type conductive layer is larger than a bottom area of the first type conductive layer, and a top area of the second type conductive layer is smaller than a bottom area of the second type conductive layer.

37

15. The micro-LED chip according to claim 1, wherein the light emitting layer includes only one pair of quantum well layers, or multiple pairs of quantum well layers.

16. The micro-LED chip according to claim 1, wherein the at least one micro-LED further comprises a reflective structure formed surrounding the first type conductive layer.

17. The micro-LED chip according to claim 16, wherein, in the at least one micro-LED, the reflective structure is attached on a sidewall surface of the first type conductive layer, and

the at least one micro-LED further comprises a bottom connection structure formed under the first type conductive layer and electrically connected with the first type conductive layer.

18. The micro-LED chip according to claim 17, further comprising a substrate under the first type conductive layer, and electrically connected with the bottom connection structure by a connecting pad in the substrate.

19. The micro-LED chip according to claim 17, wherein the bottom connection structure is made of a reflective and electrically conductive material.

20. A micro-LED chip including multiple micro-LEDs, wherein,

at least one micro-LED comprises:

a first type conductive layer;

a second type conductive layer stacked on the first type conductive layer; and

a light emitting layer formed between the first type conductive layer and the second type conductive layer, wherein the light emitting layer extends along a horizontal level from a top edge of the first type conductive layer and from a bottom edge of the second type conductive layer, such that an edge of

38

the light emitting layer does not contact the top edge of the first type conductive layer and the bottom edge of the second type conductive layer, and the bottom edge of the second type conductive layer is aligned with the top edge of the first type conductive layer, the light emitting layer is continuously formed on the micro-LED chip, and

the micro-LED chip further comprises a metal layer formed on the light emitting layer between adjacent micro-LEDs.

21. A micro-LED chip including multiple micro-LEDs, wherein,

at least one micro-LED comprises:

a first type conductive layer;

a second type conductive layer stacked on the first type conductive layer; and

a light emitting layer formed between the first type conductive layer and the second type conductive layer, wherein the light emitting layer extends along a horizontal level from a top edge of the first type conductive layer and from a bottom edge of the second type conductive layer, such that an edge of the light emitting layer does not contact the top edge of the first type conductive layer and the bottom edge of the second type conductive layer, and the bottom edge of the second type conductive layer is aligned with the top edge of the first type conductive layer,

the multiple micro-LEDs share the light emitting layer, and

the micro-LED chip further comprises a metal layer formed on the light emitting layer between adjacent micro-LEDs.

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